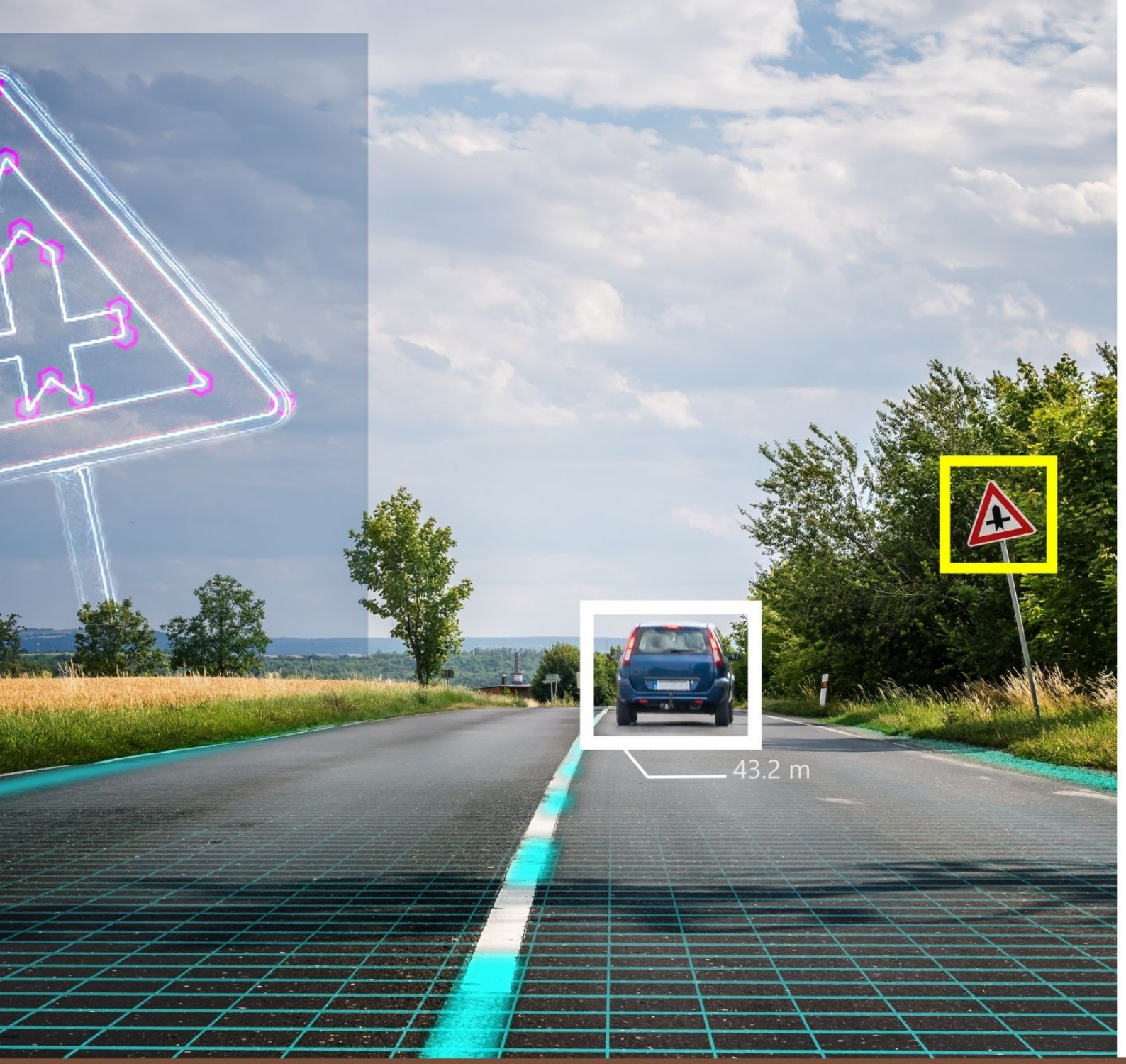


COMPUTER VISION

Prof. Jayanta Mukhopadhyay
Computer Science and Engineering
IIT Kharagpur

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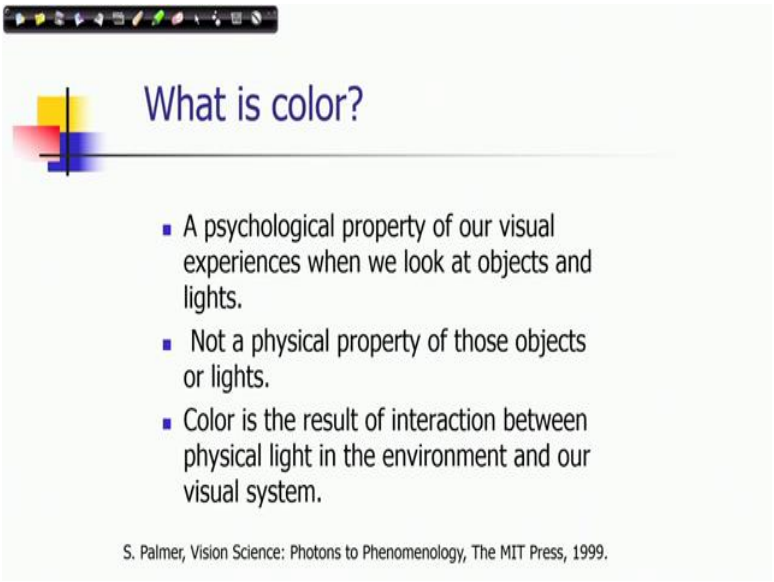
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Computer Vision
Prof. Jayanta Mukhopadhyay
Department of Computer Science and Engineering
Indian Institute of Technology, Kharagpur

Lecture – 34
Color Fundamentals and Processing Part – I

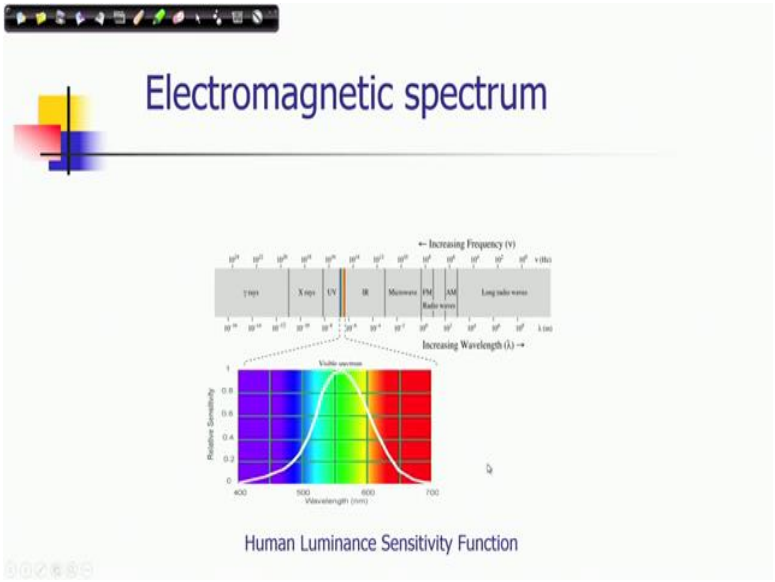
In this lecture, we will start a new topic on Color Fundamentals and Processing.

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So, let us try to understand what is meant by color and how do we perceive color. Color is a psychological property of our visual experiences when we look at objects and lights. It is not a physical property of those objects or lights. It is the result of an interaction between physical light in the environment and our visual system.

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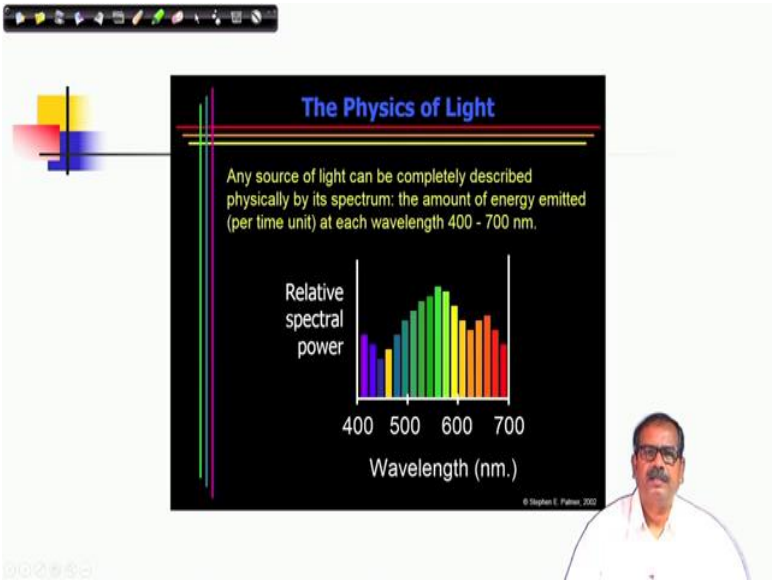


Consider the electromagnetic spectrum, the broad range of electromagnetic spectrum starting from a very long wavelength of radio waves, long radio waves to say gamma waves and you can find there is a very narrow interval in this particular spectrum at the higher end of spectrum frequency and that is that consists of the visible spectrum. Those are the frequencies of lights or colors what we perceive.

And, it has been shown here you know pictorially that how these wavelengths correspond to different colors. You can find that in our school textbook when we learnt about these phenomena of color in different optical bands. So, there are seven particular colors of rainbow that was distinguished and you can find out that from violet to red that famous acronym of VIBGYOR you can identify those spots in this particular picture.

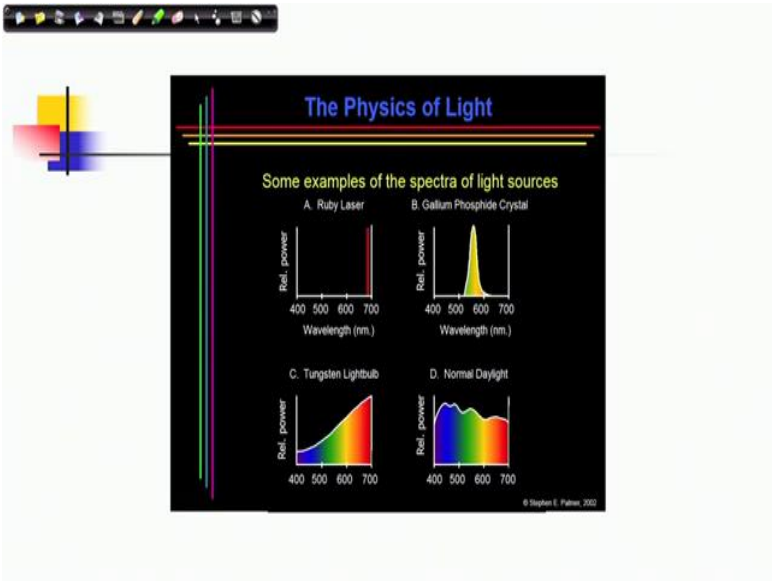
So, interestingly our sensitivity to this part of spectrum is very high and if we see the curve of this luminance sensitivity function which is a function of the wavelength and we can find that it is maximally sensitive when in the green zone that is what it is shown here and later as we move towards the either towards higher frequency range of violet or violet zone and lower frequency range of infrared zone there this luminance sensitive function slowly gets decayed.

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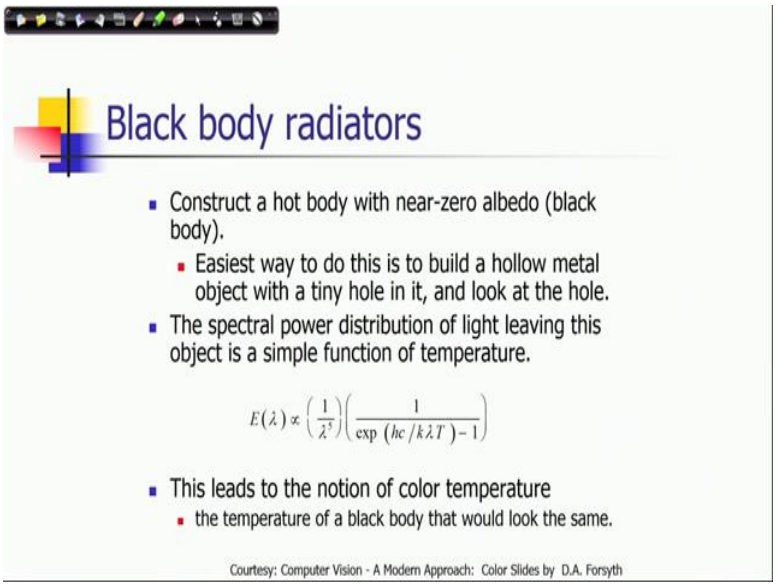
The thing is that the light which is causing the sensation that may not be also having uniform energy distributions over all the wavelengths. So, there is a relative spectral power we call it relative spectral power, it is energy per unit time that is what we need to you know consider here and for example, this could be a signature of a light source so, where you have a relative spectral power distributions over these wavelengths. So, this is how a source of light is described. It is the amount of energy emitted per time unit I mean that is the power at each wavelength from 400 to 700 nanometre and that is a visible band of these electromagnetic wavelengths.

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So, some examples of these light sources if you consider a ruby laser, it is a very pure electromagnetic wave radiator at the near about 700 nanometre which is the red in the red region, the color is red. Whereas these are other examples gallium phosphide crystal or if we consider a normal day light more or less this spectrum is uniform that you can see here.

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So, one of these particular characterisations of this luminance source is a blackbody radiators which is an ideal energy emitter in that sense and which has only observes from the environment and have its own energy spectrum.

So, one of the example of constructing a blackbody could be let us say first this is a hot body with near-zero albedo or that is why it is blackbody if it does not reflect the light from the environment then we cannot see it. So, that is the concept of blackness in this case. And the easiest way to do this is to build a hollow metal object with a tiny hole in it and we can look at this hole. So, the inside it becomes an ideal blackbody radiator.

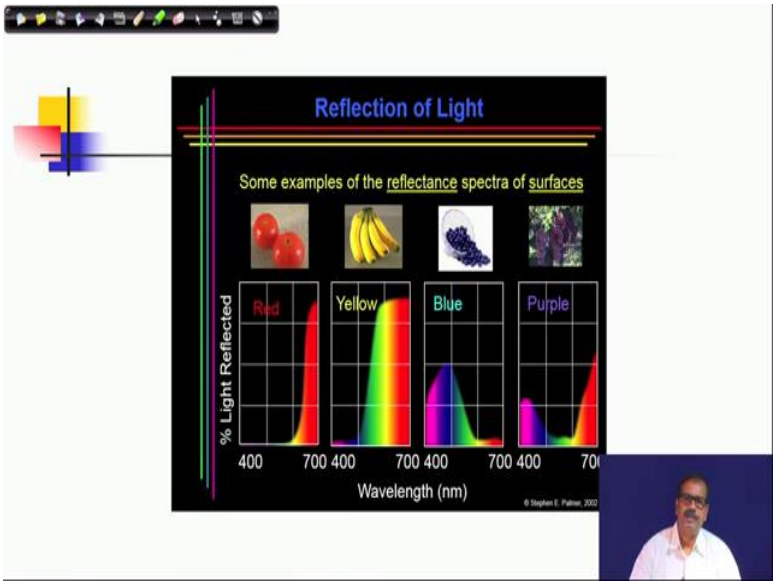
And the spectral power distribution of a blackbody radiator it is a simple function of temperature. So, which is radiating and this is relationships of energy imitated in power wavelength energy power wavelength that you can see here. And, this is proportional to this factor where you know that this h is that you know Planck's constant and you can find and k is the Boltzmann constant; those are not defined here. So, it just shows that the kind of functional relationships between the temperature and the wavelength and the distribution of energy with respect to wavelengths given a temperature.

$$E(\lambda) = \frac{1}{\lambda^2}\left(\frac{1}{\exp\left(\frac{he}{k\lambda T}\right) - 1}\right)$$

So, this leads to the notion of a color temperature because if a light source has own spectral power distributions or if I say energy emission distributions over the wavelengths and then which one is closely resembling to this blackbody radiations of a particular temperature, we consider that source has equivalent to having a blackbody source of that color temperature. So, we can tag any luminant source by any color temperature, from there itself we can get its spectral power distribution.

So, this is the temperature of a black body that would look this same when you are considering the notion of color temperature.

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Let us now consider that phenomena of reflection in the environment because this reflection is the major phenomena by which you know we sense the environment we see the objects that we have discussed from the very beginning itself in our very first slide we considered reflected lights projection of 3-dimensional scene points to a 2-dimensional plane.

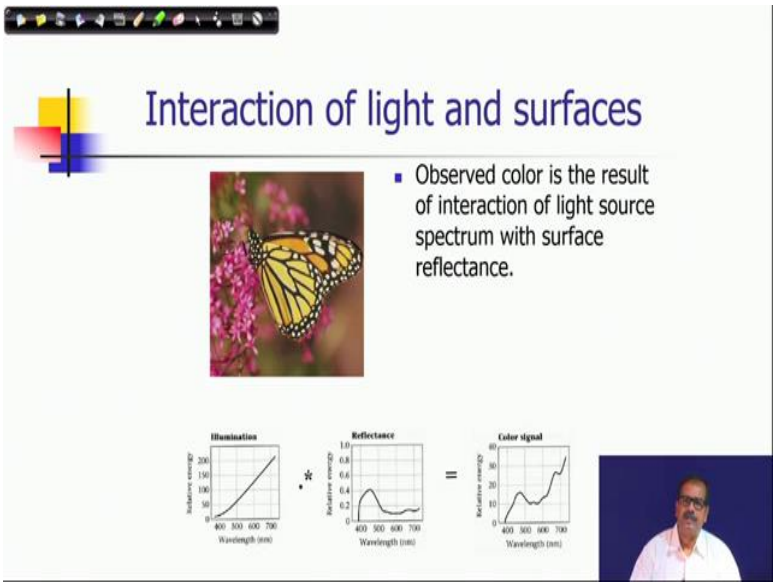
So, in our visual system also we are getting reflections of from a 3-dimensional scene and it is basically falling into our retina of eyes that I will be coming you will be discussing in my next slides, but the nature of that reflected energy that that also depends upon the kind

of surfaces you have. So, every surface absorbs a part of its incident energy and also reflects and in different wavelengths it performs this job in different differently. So, if I consider the percentage of light reflected across wavelengths that also shows the material property of that surface with respect to this particular task.

For example, you can see the picture of tomato here which is of red surface so, it is expected the energy what is reflected from the surface points they will have high content of rate wavelengths. So, higher wavelengths in the optical range those are the part of that energy of the reflected wave. For banana which is yellow it the spectrum looks like this. This is a percentage of light reflected in different wavelengths and it has high reflectance value of within this zone and particularly the yellow zone is very prominent including the red zone also as you can see here.

And, this are the blueberries and naturally this is expected that the reflection in the blue range of the visible spectrum, those will be high and this is another object which is a purple color. And, there we can see that the reflectance in this interval is very slow whereas, in this particular zone in the blue zone and also in the red zone is quite high, that is interesting that you have the purple reflections, but still you have sufficient component of red zones and blue zones. There are explanations of these behaviours of our visual response.

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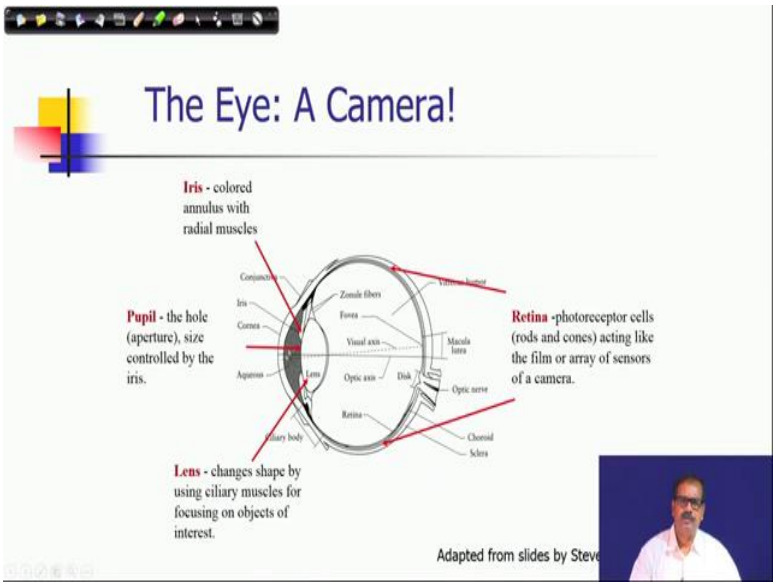


So, finally, as we can see that it is the interactions of light and also the material property of the surface, light source and the material property of the surface that results into our

visual sensations. So, that is what we observe as a colour. So, if I consider a light source, the reflected energy distributions across wavelengths in a light source and also they consider the reflectance values or percentage of reflectance reflected energy from surface points and at the surfaces at different wavelengths.

So, for every wavelength the energy what we receive in our visual system that is simply product of these two factors that is the power of illuminations or reflect the relative energy of this particular wavelengths energy of this wave for any wavelength and also considered the corresponding reflectance value of that wavelength you multiply, and then you get the relative energy of that particular wavelength that figure you will get in the received signal. That is how the received signal or received sensation is received sensory signal if I say exciting signal that is how it is characterized in this way.

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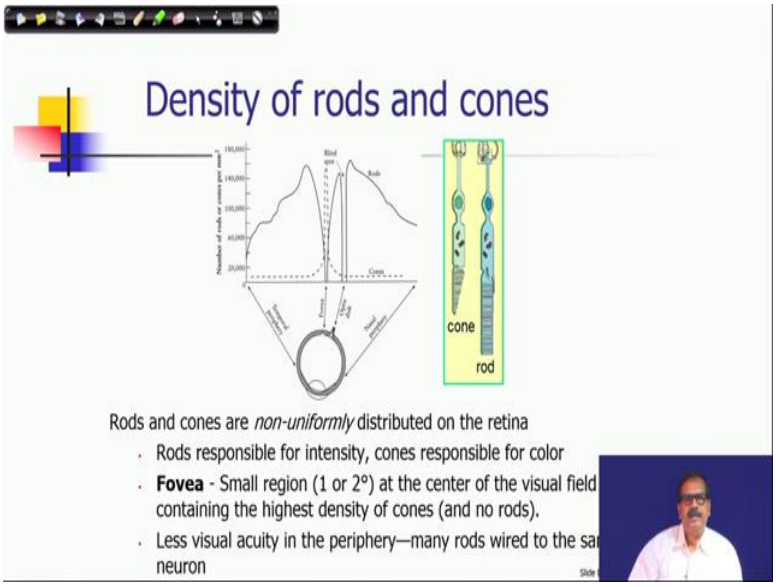


So, now let us consider the function of our eye in our visual system, now it is acting like a camera as you can see in its contraction. Let us say there is a lens and that project see 3-dimensional points in a environment to a 2-dimensional plane, it is not a plane I should say it is like a spherical screen why are these points are projected semi-spherical screen and that is called retina where these points are projected and sensation is carried out from that point through our nervous system to our brain to understand to which gives all the perception of this color and also we can understand the you have seen what is there in front of us.

So, the components in this part which has very direct analogy with the function of different components of a camera you can see lenses this part. Similarly, the retina which were there is a screen kind of thing or which acts like an image plane and there are photoreceptor cells. So, it carries the sensations which transduces this energy into the electrical signal form. So, the retina consists of this photoreceptor cells and from there the sensation is carried through our nervous system to brain.

And, there are different kinds of auto receptor cells in our retina they are called rods and cones. So, I will give more details in next slides. Here there is also an iris which is in colored analyst with radial muscles and there is this pupil the hole or aperture and apart like an aperture of a camera through which the optical lights the energy is transmitted through the lens or lights are coming to the system entering into the system and this is like an aperture of a lens and this its size could be controlled by the iris.

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So, about the rods and cones there are two types of receptors in the retina that I mentioned the shape of cone and rod we can see here, their cells basically their cells and they are attached with the corresponding optical nerves which goes to our brain through optical through the nervous system there is a pathway through which the sensation is carried out it is carried towards them. And, this is attached to these particular retinal surface. This is an attachment surface what is there and the shape of this attachment as relate to the

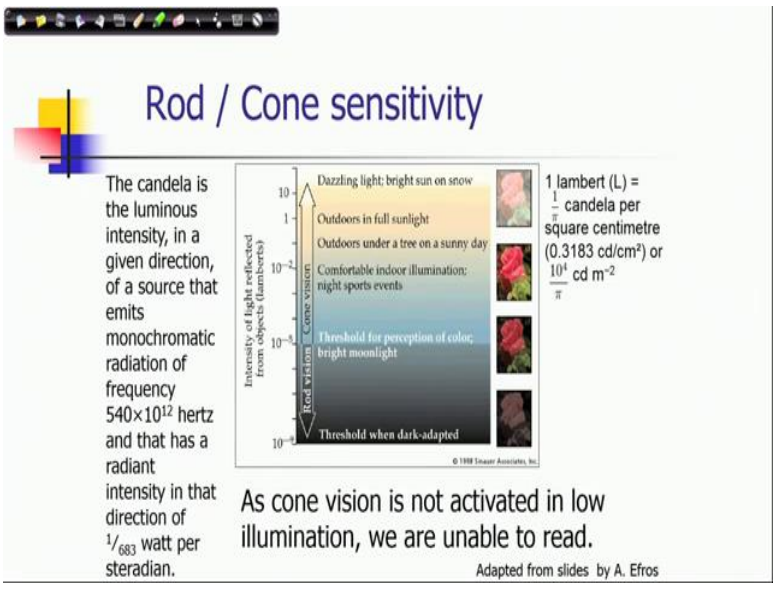
nomenclature of these kind of cells this is since this is a conical shape we call it cone and this is a cylindrical shape we call it rod.

Now, the distribution of these rods and cones is not uniform over this retina. So, you can see that the cones are highly concentrated over this. So, this is the distribution of cone and cones are highly concentrated over the retinal part. So, let me just point out. So, this is what you see is a distribution of cone roughly, this is the distribution of cone that is a curve what is shown here and the outside the curves they are the distribution of rods and it is very much concentrated on a particular region and which is a very small high visual acuity region and which is called fovea.

So, this region of the retina is called fovea and here the degree that is formed in this region is a very narrow degree. It forms a very narrow degree about 1 to 2 degree top range. So, the function of rod circuit is responsible for intensity and cones it is responsible for color and the fovea as I mentioned it is a small region that is 1 or 2 degree at the center of visual field containing the highest density of cones and rod.

So, this is how the structure is there and in the periphery you have less visual acuity. There are many rods where to the same neuron. So, that is the kind of structure.

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So, let us try to understand how rod and cone they are responsible they function they are responsible for sensing the light and their sensitivity we can see it varies. Rod vision is

sensitive to low illuminated environment whereas, cone vision is more sensitive in a very in a bright or highly illuminated environment.

So, you can see from this particular variations it is shown here that this is their zone which is for the rod vision zone and this is a zone which is for cone vision zone. And, this is the scale which is showing the intensity of lights reflected from objects and follows the lambert, you know the unit is a lambert unit. So, if I say what is a lambert, 1 lambert it is basically the definition is in this way. It is $1/\pi$ candela per square centimetre; candela is an again unit of energy radiation.

So, it is the luminous intensity in a given direction of a source that emits monochromatic radiation of a frequency of this 540 terahertz frequency and that has a radiant intensity in the direction of a particular power per steradian that is being emitted that is 1/683 watt per steradian. So, if this is a kind of a intensity, this is a kind of intensity or energy emitted form this is the amount of energy emitted per steradian at that frequency and from a source then that unity is you know one candela in that given directions that is and if it is 1 by candela per square centimetre, then the intensity of that you know surface at that 0.1 lambert. So, that is the definition.

And, for very high intensity object points we will be able to perceive through cone whereas, low intensity visions or through rod. So, visual acuity is more active in the cone sensitive zone and that is why we cannot read we are unable to read in low illumination because rod vision is not visual activity of rod vision is less.

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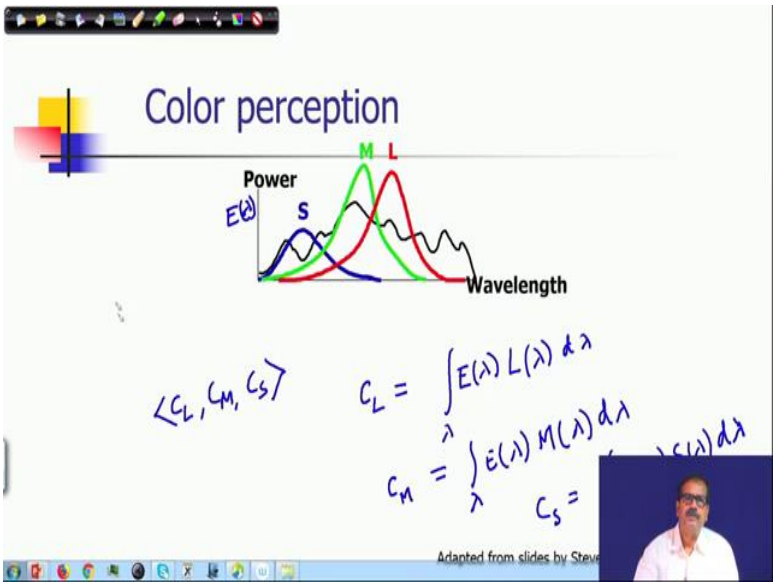


So, now let us understand that with this background the physiology of color vision. So, as we have seen there are three kinds of cones so, in our visual system and you can see that there are three responses like this is a one kind of wavelength this is one kind of response which is shown here as the S and or short wavelength response and that is the response in the you know blue zone. There are medium wavelength response which has been shown here as M and there are long wavelength response. So, there are three types of cones. So, it is cones are also there are three categories and they have different types of wavelength you know responses.

And, so, the ratio of this long wavelength, medium wavelength and short wavelength cones they also widely vary. As you can see that more number of you know long wavelength cones are in the red zone that is the ratio is 10, whereas in the medium wavelength it is 5 and short wavelength it is 1 and almost no S cones in the center of the fovea. So, in fact, there is a nice picture which shows the distribution of different types of cones by showing the colored dots in this particular two-dimensional point.

So, if we flatten the retinal surface near the fovea particularly and we can see in the centrals part hardly there are blue cones or a shorter wavelength cones; mostly it is red and the green cones or medium wavelength cones that we can see in this picture.

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So, just to model this color perception we can consider this particular you know simple mathematical you know explanation or mathematical model, that let us consider the incident energy in our retina a having is relative you know spectral power distributions as shown here across the wavelengths. And, then you have the same you know sensation is received by three different categories of cones. So, we will be explaining the color perceptions only using the cone vision. So, we are not considering rod vision in perceiving colors.

So, the color perception is a phenomena due to our sensations through cone cells and there the same sensation is you know same energy is received by different kinds of cones, but they have different you know different effects or responses because they are wavelength response they are wavelength or there they are response of to their wavelengths are difference. For example, this is the long wavelength cells, this is the filter response for long wavelength cells and this is the filter response for medium wavelength cells and this is a filter response for short wavelength sense.

So, as we have considered the simple model that what we can do if I consider this received energy is can be shown as see $E(\lambda)$ then in our long wavelength rest filters in the long wavelength cones the received energy can be modelled as $E(\lambda)$ $L(\lambda)$. So, what is being finally, received the output of this output from the long wavelength cone is $E(\lambda)$ $L(\lambda)$. So, it is an integration over λ and that should be the response from long wavelength cones. So,

$$C_L = \int_{\lambda} E(\lambda)\, L(\lambda) d\lambda$$

$$C_M = \int_{\lambda} E(\lambda)\, M(\lambda) d\lambda$$

$$C_S = \int_{\lambda} E(\lambda)\, S(\lambda) d\lambda$$

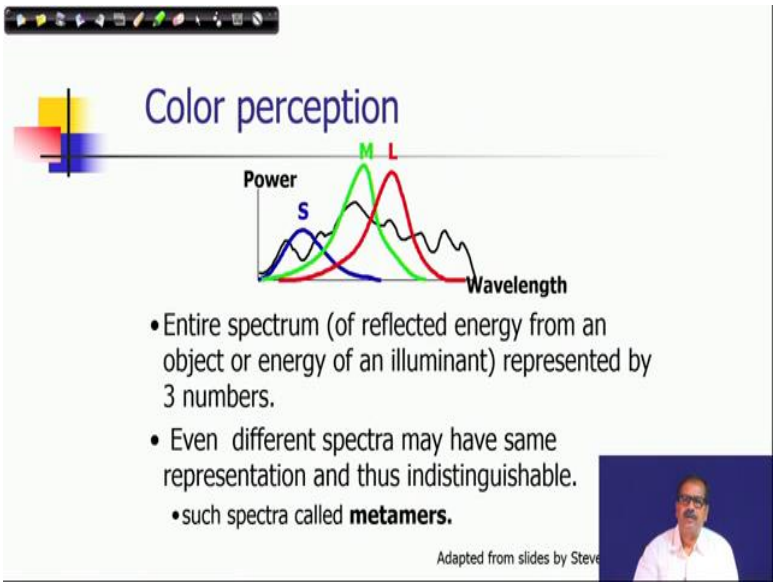
So, this is what we will be obtaining from a long wavelength cone; mathematically this is the operation that has been carried out there.

Similarly, for the medium wavelength cone we can write it as E(λ). So, this is a response this is the output of the of the filter with spectrum frequency spectrum and say M(λ)or wavelength spectrum as E(λ) that is a filter response and then this is output we get from the medium wavelength cone and for short wavelength cone in the same way we can write it as E(λ) S(λ).

So, now you can see that given a particular energy particular or particular energy from the environment which is received by all these cones we get we can finally, this can be coded into three factors one from the long wavelength cones another from the medium wavelength cones, another from the short wavelength cones. And, those sensations are carried to our in our brain through the nervous systems and those are perceived as a color.

So, finally, color perceptions can be you know considered into this from that it is the effect of these sensations which is giving the ultimate you know our which is resulting into the perception of the color of that particular light and this actually leads to the trichromatic theory of color which I will be describing next.

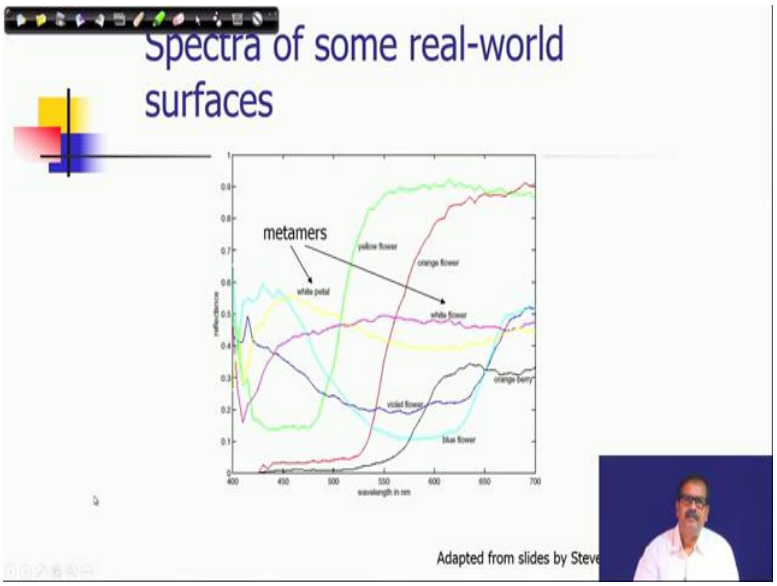
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So, just to summarize once again, in this color perception we have entire spectrum and entire spectrum of reflected energy from an object or energy of an illuminate that could be represented by three numbers, as I explained how these numbers could be mathematically computed given all these facts. And, then the it also know tells us that even if we have different spectra it can have the same representation because finally, as you can see there are it could lead to the same set of triplets by doing that product and sign integrations over accumulations of this product of the corresponding response and also the power over the wavelengths that would give you the same vectors.

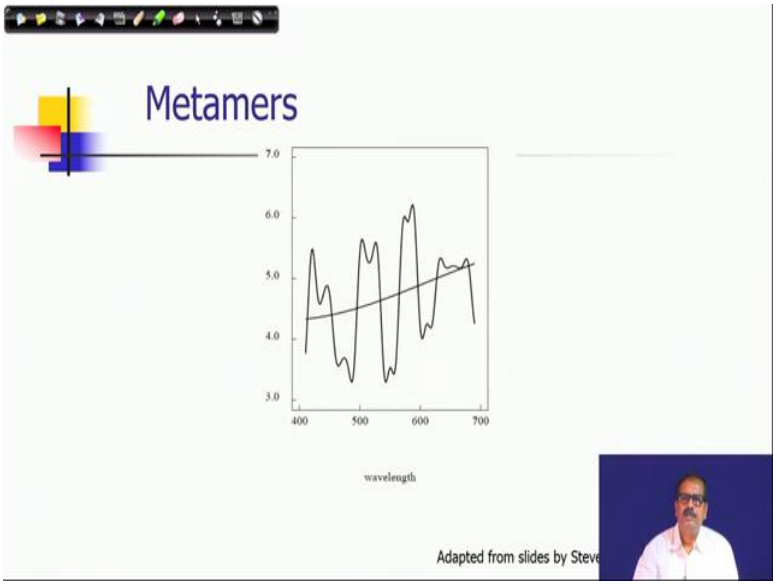
So, it becomes in the indistinguishable and such spectra they are called metamers.

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So, there are some examples of this metamers we can see here. Consider this spectra or the relative power spectra for different kinds of reflected waves. See this is a white petal spectra which has been shown here by this particular color and another in this pink curve this is a this is a white flower and we our sensation of white petal and white flower is similar it is a white sensation. And, but as you can see there is it there is a lot of variations over the distribution of the relative power over the wavelengths in these two different sources, but still we perceive them as white and this is an example of metamer.

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There could be another example that you know you can see that this is a widely varying relative spectral, relative power over the wavelengths whereas, there is a smooth variations. But, final effect of integrations of the product of our cone response and with these you know energy distributions, with these power distributions of a particular at a particular wavelength final effect would be would may be same and that is how that it could be also another pairs of metamers.

So, with this let me stop at this point for this lecture we will continue this discussion of color perception. Thank you very much for your attention.

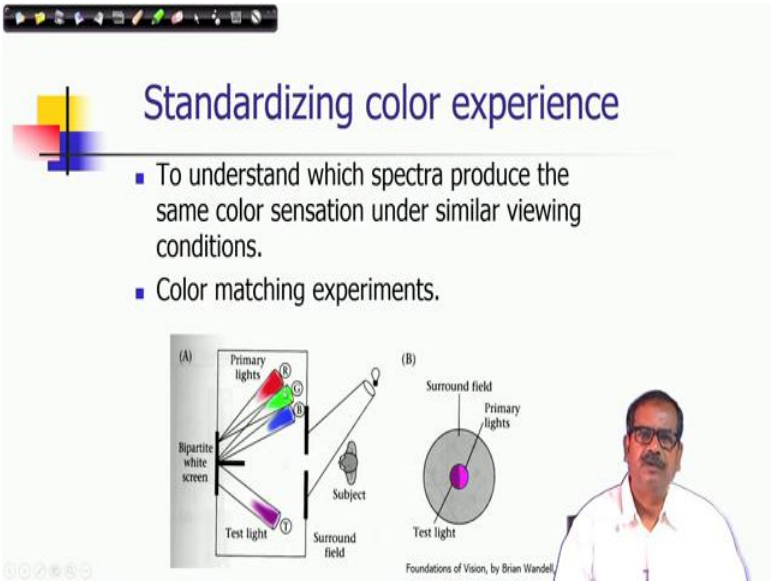
Keywords: Electromagnetic spectrum, black-body radiators, color perception, rods and cones.

Computer Vision
Prof. Jayanta Mukhopadhyay
Department of Computer Science and Engineering
Indian Institute of Technology, Kharagpur

Lecture - 35
Color Fundamentals and Processing Part – II

We are discussing about fundamentals of color perception and in the previous lecture we discussed how this perception is a result of interaction of the energy which is being received in our retina, and also the response of the corresponding retinal cells. And this could be factored into three vectors or into three factors, which can identify a particular characteristics of an incident energy or characteristics of the spectral distribution of the incident light in our visual system.

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So, that is how we experience color. And to standardize this experience; that means, to calibrate with respect to sources and then understand color perception under any kind of color source or any feature or response from any arbitrary color sources that would be our objective next. So, for that , we need to do color matching experiments.

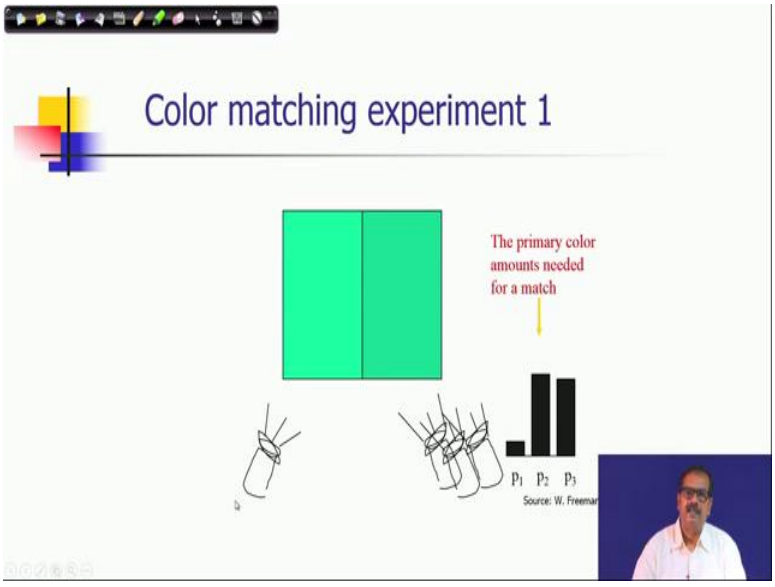
So, let me discuss what is meant by color matching and how these experiments are carried out. So, here is an example of a setup for performing color matching what you can see that, there are three primary sources of lights and of three different wavelengths and there are three pure wavelength sources of lights and we call them primary as we

have seen that a color could be considered as a finally. The color is represented by a combination of three different color components into red, green and blue zone of the spectrum. So, we have chosen such primary color sources in that way and consider a test light which needs to be calibrated with as a factors of these three primary sources, that test light is T.

And if I display this test light, this is once again a say Pure monochromatic color source of a particular wavelength. And if I display it on a screen and the screen is designed in this way there is a circular zone and here in that screen and there is of course, a surrounding field to provide you the contrast and the test light is projected here and this part is the combination of this red, green, blue, that is projected here.

So, you can vary the relative you know strength of this energy sources and you can note down that what in what proportion they are mixed and they are superimposed here. So, that proportion itself could represent the same test light as a factor of these primary colors. So, this is known as color matching. So, you have to vary the proportion in such a way that these two colors there should not be any distinction between them.

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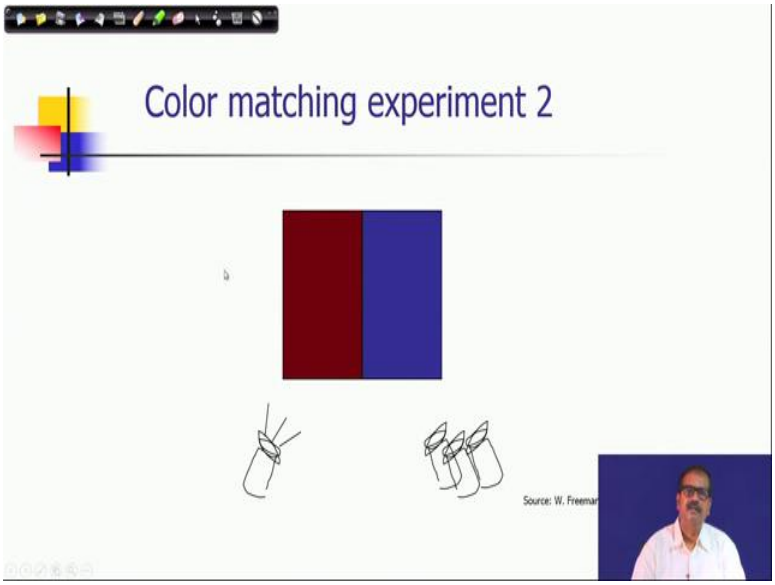


So, we can describe an hypothetical color matching experiment diagrammatically once again it is showing the same setup what I described earlier. So, this is that test light and if I project it for example, it produces a color like this. consider there are three primary sources of red, green, blue and a certain combination produces this. Say this is the

combination of these primary sources, it shows the relative proportion of the amount of energy emitted by these primary sources and they are superimposed here and that is the color.

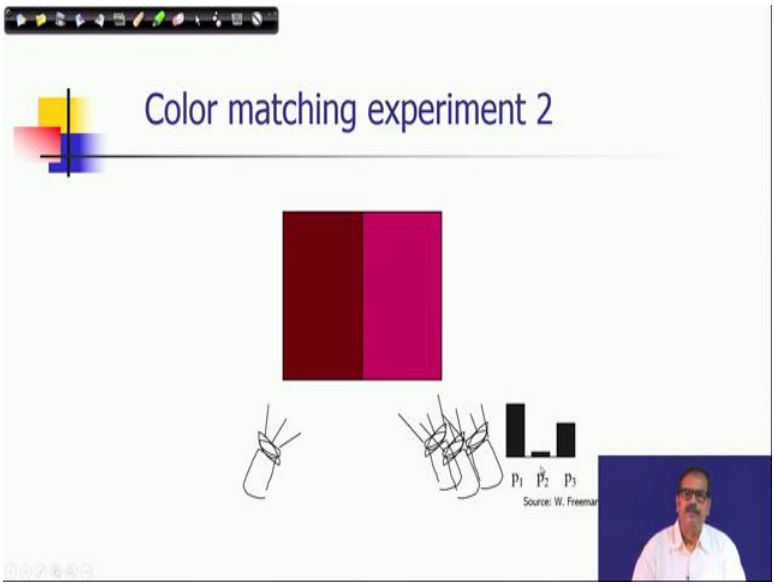
Now, if I vary one of them for example, p_2 I could see that this color is going near to this throughout visual sensation toward visual observation and further increase may cause exact matching of this two. So, then we say the, relative distribution of these three primary colors, is representing this particular color. That is how we have standardized the representation of this color with respect to these primaries through this experiment. And so, this is the primary color amounts needed for a match. So, consider repeating this experiment for every kind of light sources of different wavelengths.

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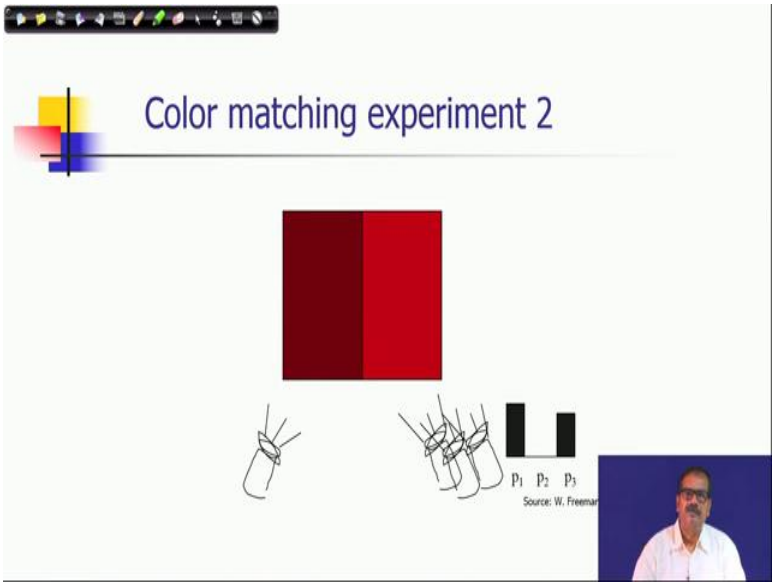


There is another scenario that also I should explain here and this is we are considering a type two scenario of this experiment. Here you can see that this is a color of the test light of a particular wavelength and again this is the color produced by the superposition of different primary colors, in some certain proportions.

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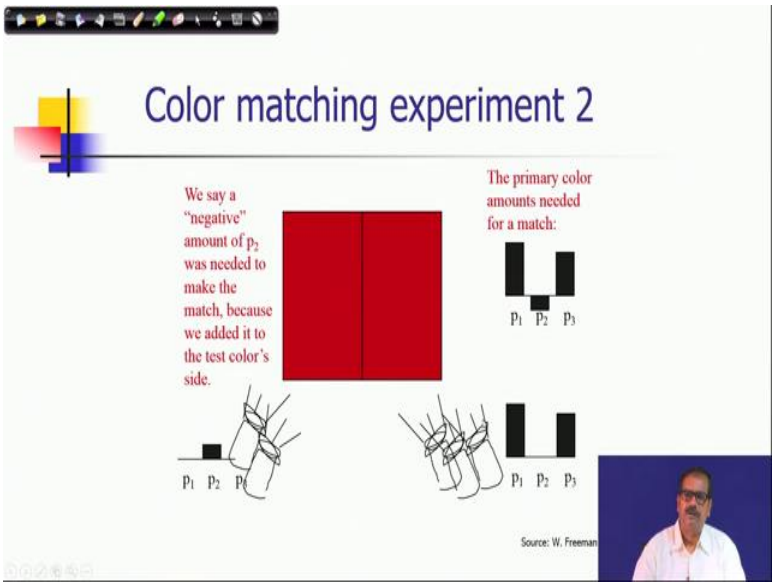


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Consider this is a proportion which is producing, as you can see that the primary p_2 is a very small amount and in fact, if we further reduce it we will find that this color becoming nearer to this one, but still it is not the same now in fact, what we can do there. Consider you add instead of projecting primary from here now, it is add primary from this part.

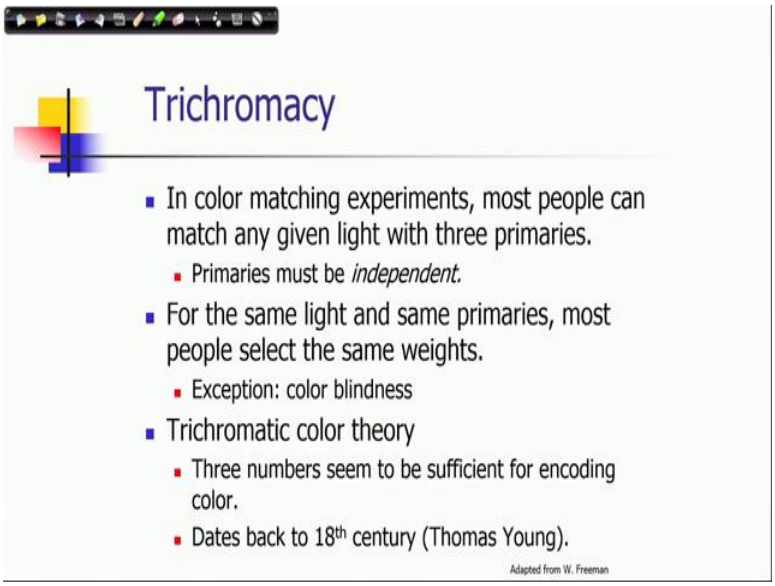
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Which means, say if I add a primary here then we can see that these two colors they become same, which means that as if you are subtracting the p_2 from these combinations p_1 and p_3 these proportions. And then you are subtracting p_2 with this proportion from this combination, then it is producing the same sensation.

So, we say negative amount of p_2 as needed to make this match and this is the another kind of experiment where you see that it is not only additive superposition, but also even the negative matching, negative additions from the other side. theoretically you can get the equivalent sensation in this way and you can obtain a figure obtaining proportions like this. So, you can represent in this in this fashion that this p_2 in it needed to be subtracted from p_1 and p_3. So, in the color matching experiments you note down all such proportions including their addition or subtraction for every wavelengths you can get a chart art for different colors for, different wavelengths and that would give you the color check.

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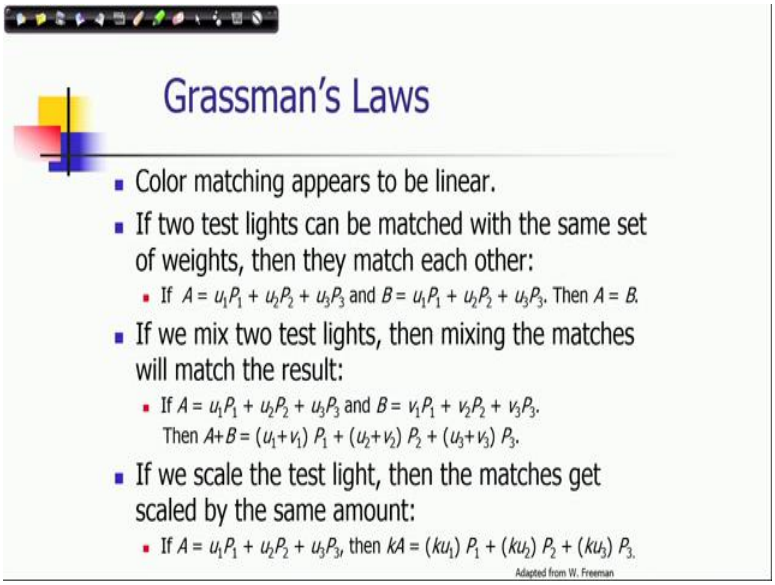
So, by standardizing this chart, we can have a trichromatic representation of any color which means using three primary colors and this feature of color representation is known as trichromacy.

So, it is the fact that most people can match any given light with three primaries and primaries must be independent which means that, none of the ; two other color with addition or subtraction whatever may be that with any proportion that should not be able to produce the particular color. So, that is how the independent, it is this definition is the analogy is with that linear independence relationships among the vectors. And for the same light and same primaries most people select the same weights that is also observed experimentally.

So, that is how the uniqueness of color representation across our perceptual systems of the varying human visual perceptual system so, that unique representation is established through this kind of empirical studies. Of course, there would be exceptions when there are color blindness. So, there would be different kinds of perceptions for that.

So, we are talking about normal vision and there it is an uniquely represented given primary with that same proportion to represent a color. And this gives you the trichromatic color theory, which means that three numbers seem to be sufficient for encoding color, it is a very old theory it dates back to 18th century it is proposed by Thomas Young.

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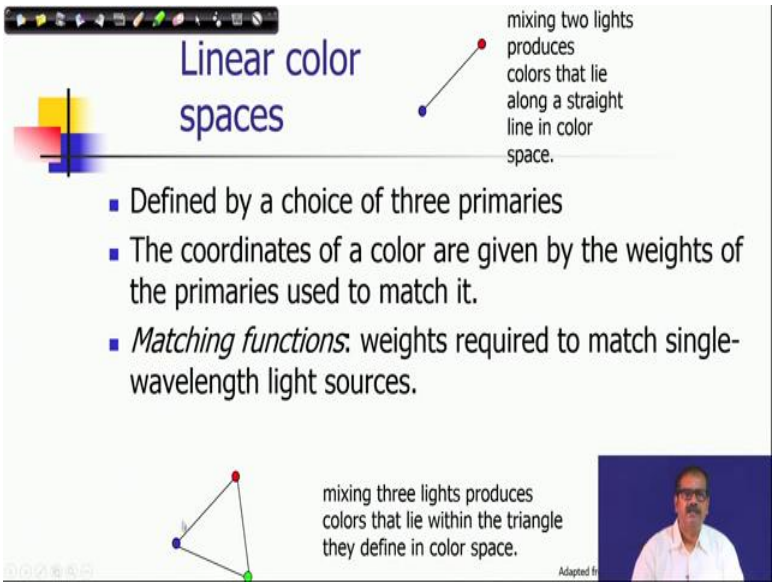


And as we can see that there are a lot of similarities with respect to the linear space and also the color space. So, Grassman proposed this laws of color matching using this property of linearity. So, it is considered that it is a linear property and which means that if two test lights can be matched with the same set of weights, then they match each other. Say you consider there are three primaries p_1, p_2, p_3 and this is the amount to which is needed to produce A color a say u_1, u_2, u_3 there is a representation. $A = u_1P_1 + u_2P_2 + u_3P_3$.

And if there is another color which is also represented by the same values same proportions and same values actually u_1, u_2, u_3 then these two colors must be same. And if we mix two test lights and mixing the matches will match the result which means once again you get two colors A and B and having their different proportions of primaries and if I superimpose, the color what you get that would be represented by this factor. So, their property of linearity is also that property holds during the super positions of two colors also.

this property also tells you that if we scale the color scale the test light, then also the matches get scaled by the same amount.we see the similar kind of color, but intensity varies with that color. So, these are the three features of Grassman and three laws of Grassman which which tells that know a color could be represented in a linear space.

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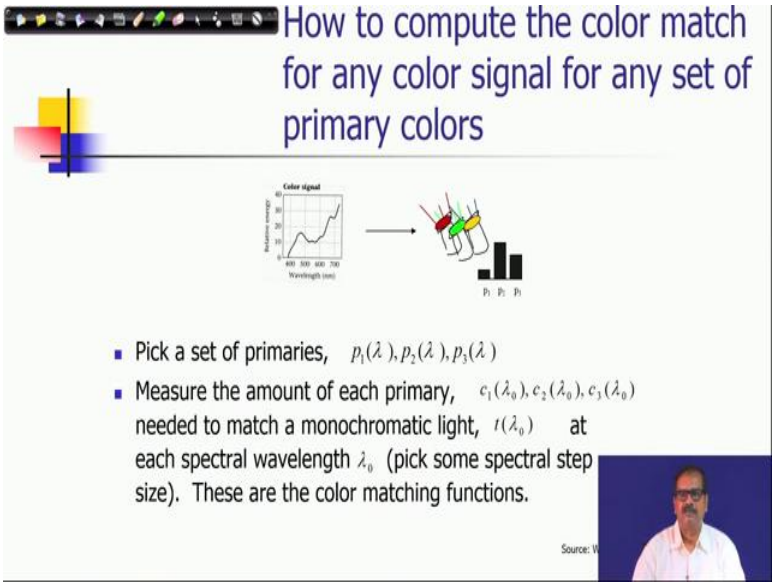
So, that is how the linear concept of linear color spaces they have been considered and it is defined by a choice of three primaries. So, there could be different color spaces given different sets of three primaries, because we will get different vectorial representations So, RGB is one common choice, but there could be different other primary choices.

So, the coordinates of colors are given by the weights of the primaries used to match it that is the representation and we need a matching functions; that means, we need these know standardization. So, a monochromatic light source with respect to this coordinate representation- three vector representation of a color.

And then we can apply this linear superpositions principles suppose you would like to have a mixing of two colors. So, if we represent them by that three vectors and any color if you consider a linear combination of these two, they will be represented by a color which could be also represented as a point in this particular straight line.

And if I consider three primaries and any color any combination of this color can be represented by a point in this particular triangle. So, these are all positive weights or the some of the proportion sum of the weight should be equal to 1. it is a convex combination we are considering and then the point should lie in this case within this triangle.

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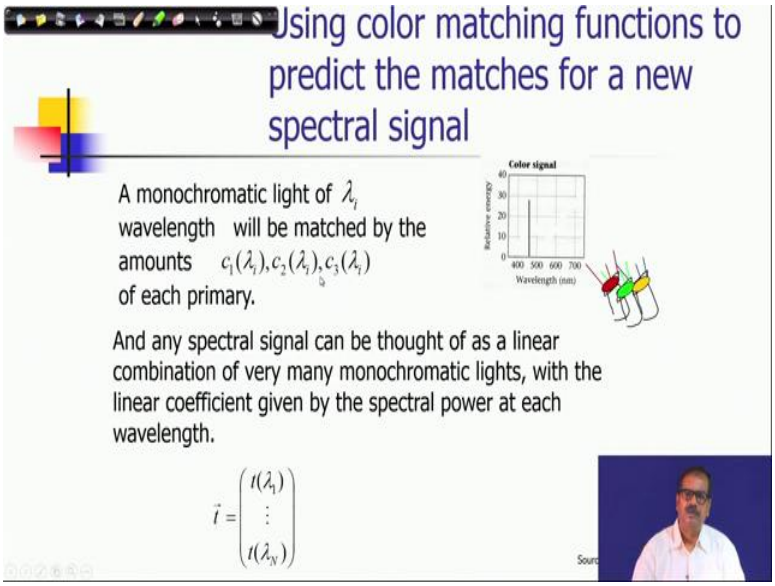


So, we have described this experimental techniques that how to compute the color match for any color signal for any set of primary colors. So, now we will be considering the representation of a color signal because as you see that so, far we have represented light of a monochromatic wavelength which means a pure color and then you have represented by the three vectors.

But again the light could be considered this color or the color signal could be considered as linear combinations of different wavelengths at different proportions. So, for each wavelength we can have its representation from the color chart; that means, this is the three vectorial representations say for a λ, this is the amount for primary p_1, amount for primary p_2 and amount for primary p_3. And so, we can measure the amount of each primary needed to match a monochromatic light.

So, in this way we can consider that whatever relative amount of energy is there in the spectrum of a light source, in the same proportions all these three vectors needed to be mixed and that would represent this color signal. So, color signal would be also finally, represented by a three vector form.

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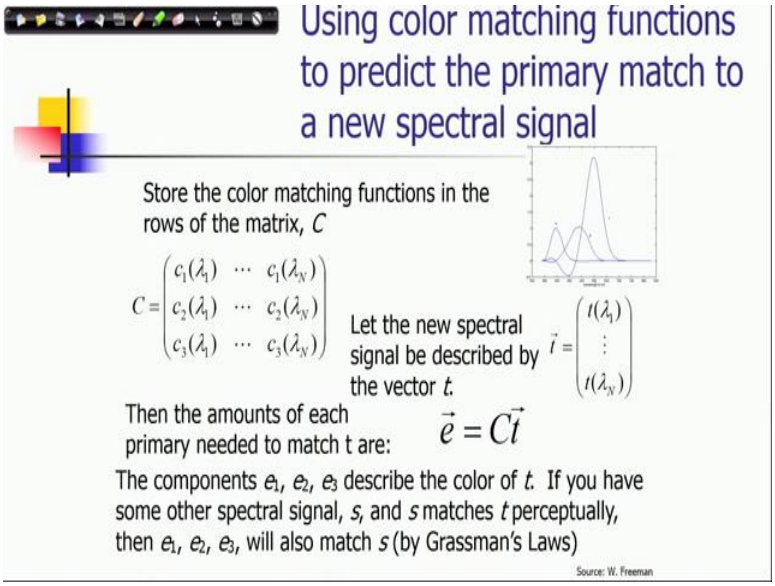


And using this linear superposition principle we can compute them, if we know the corresponding color responses are three vectorial standard representations of these wavelengths so, that is what we will be describing here. So, a monochromatic light of λ a wavelength, it could be matched by the amount say c_1, c_2 and c_3 for each primary and consider any spectral signal, it is thought of as a linear combination of so, many monochromatic lights and this linear combination is can be represented in terms of a vector.

So, this is the amount of the energy which is there in the wavelength λ_1 in that light source and consider there are wavelengths from λ_1 to λ_n. we can discretize the range of the wavelengths and also use the amount of energy for each wavelength which is present in that color signal. So, this is a representation of the color signal in terms of spectral power distribution. So, this is what is spectral power distribution and that is discretized in this form.

So, now, for each one, once again λ_1 is represented by the amount of primary color. So, it is normalized representation. So, if I multiply. So, each one should be multiplied by this $c_1\lambda_1$.

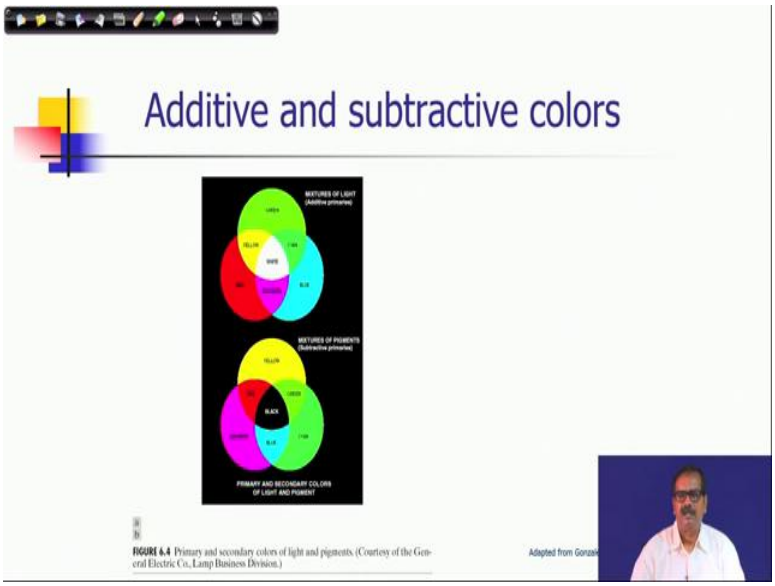
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So, we can consider this, say they store the color matching functions and then this is the representation of the color signal and now as I was mentioning you multiply this matrix with this particular combination. So, this matrix(C) is a $3 \times N$ as you can see and this ($\vec{t}$) is $N \times 1$.

So, finally, what you get? You get 3×1 vector for each energy, each component you will get for example, $t(\lambda_1)$ will give you $c_1(\lambda_1) \times t(\lambda_1)$ for the c_1 component part of it and then $c_2(\lambda_1) \times t(\lambda_1)$ for this c_2 component and $c_3(\lambda_1) \times t(\lambda_1)$ for the c_3 component. And for every wavelength you are finding out these components and then again add them component twice that is what it is simple matrix multiplication that is what it is explaining. So, in this way you can apply Grassman's law to get the representation of a color signal by three vectorial form or three components.

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So, we will be considering now production of or renditions of colors in the technology and you can see that now there are two different types of technologies available. As we have discussed that a color could be represented as a superposition of primary colors now this superposition does not mean always an additive mixing, there could be also subtractive component.

But the problem is that even in today's technology we cannot you know have both the properties either the principle should be on additive mixing, principle of the technology or it could be only on subtractive principles. So, there are two types of systems for color renditions either it is an additive system or it is a subtractive system.

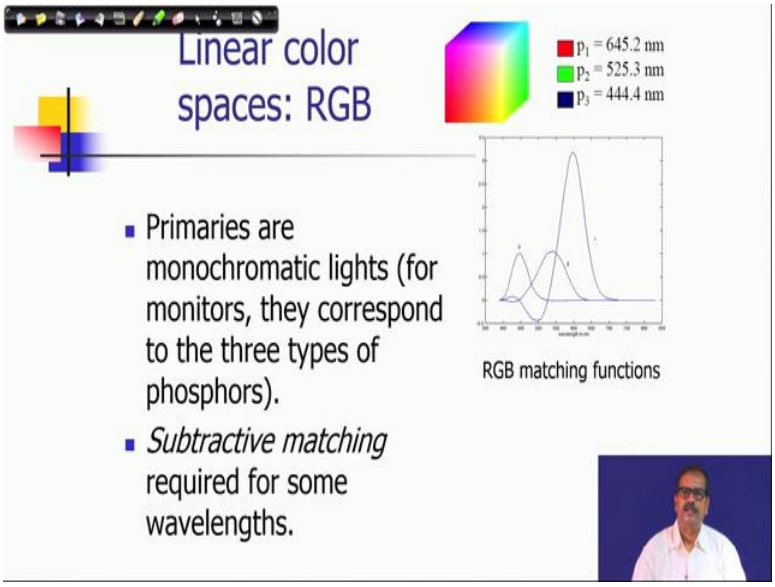
So, this slide particularly it is showing that this is a additive system, where the red, green, blue color, they the primary colors and a different proportions these colors when they are added all of them are positively added there is no subtraction, then you produce different kinds of colors like you can have yellow here, you can have cyan, you can have the magenta color here.

But on the other hand in the subtractive system primary colors could be magenta yellow and cyan, and which means that in the magenta the red color gets absorbed and then you have the sensation. So, it is absorption of red color, but which will produce a sensation of yellow, absorption on green which will produce the sensation of; of magenta and this is

a sensation of yellow. And the absorption of blue which will produce you the sensation of sand and using them you can produce different kinds of colors.

So, this principle is not you know subtractive principle by which these colors are produced. So, example of this kind of system is the printing system. So, when you print colors on white pages, we use a subtractive principle whereas, when we you display colors in a computer monitor or any display screen or any cathode ray tube using just normal televisions. So, there we work with the principles of additive superposition.

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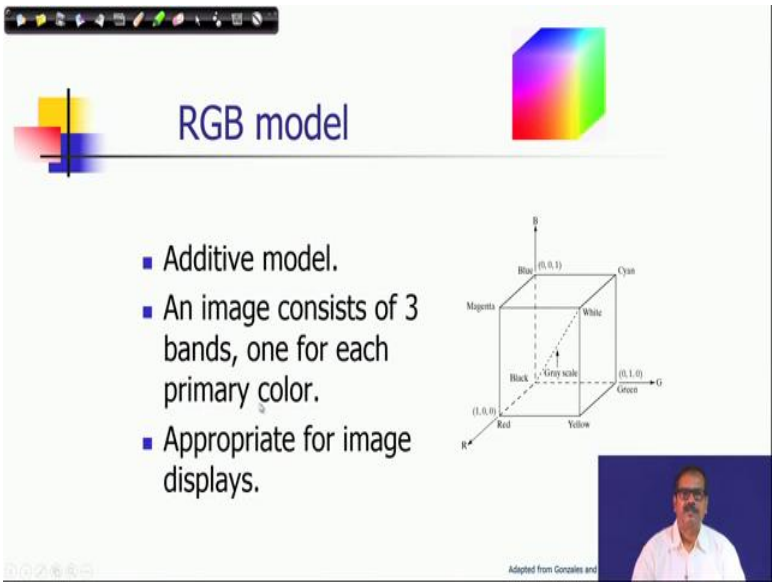
And we can talk about know different colored spaces linear color spaces, the common color space is an RGB color space because when we captured the color information in cameras, this is the space in which we capture the color information of a point. Because we use red filter, green filter and blue filter in the camera when the optical energy is incident on a lens before that we apply those filters. So, you get only the energy in the red zone or green zone or blue zone of the spectrum accordingly, and then these three components are independently captured for the same light source for the same incident energy and that is how a color is represented.

And that space is a natural red, green, blue space or RGB space and in the RGB matching function if I show it, over as a function of wavelength which has been displayed here, you can see also the corresponding wavelength which is representing red, green and blue which is shown here.

So, one of the thing you should observe here that it is not always additive mixing for every wavelength. For some wavelength for example, you consider this red matching response for some wavelengths it is subtracting. So, that is why in the RGB color space every color cannot be produced by only additive color.

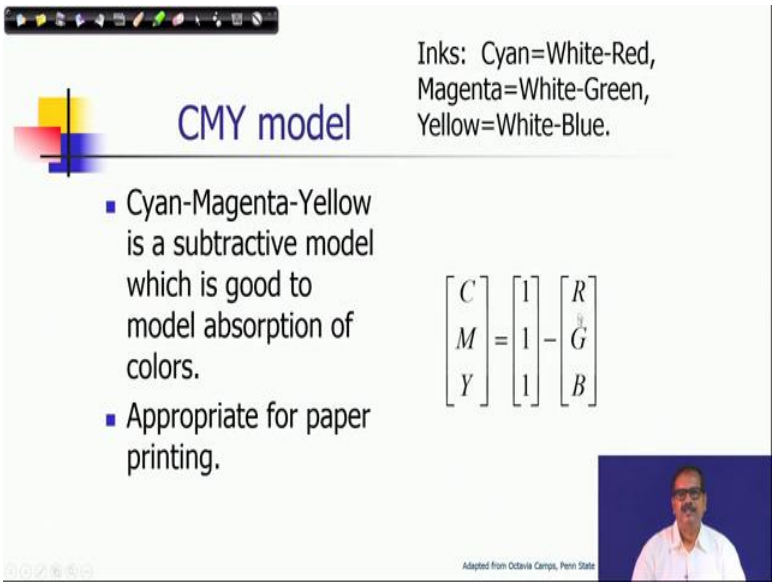
So, that is one kind of property and that is the reason why every display additive display cannot produce all sorts of colors, there are set of sets of colors which cannot be rendered, because they require subtraction and which is not possible by the technology what is giving that perception or producing that color for our perception. So, this is the thing we should note that for some wavelengths you record subtractive matching.

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So, just to summarize an RGB model could be represented in this normalized form where you can consider 1 is the highest intensity values and 0 is the lowest intensity value and then you can represent every vector within that normalization. And it is an additive model an image consists of three bands one for each primary color and it is appropriate for image displays.

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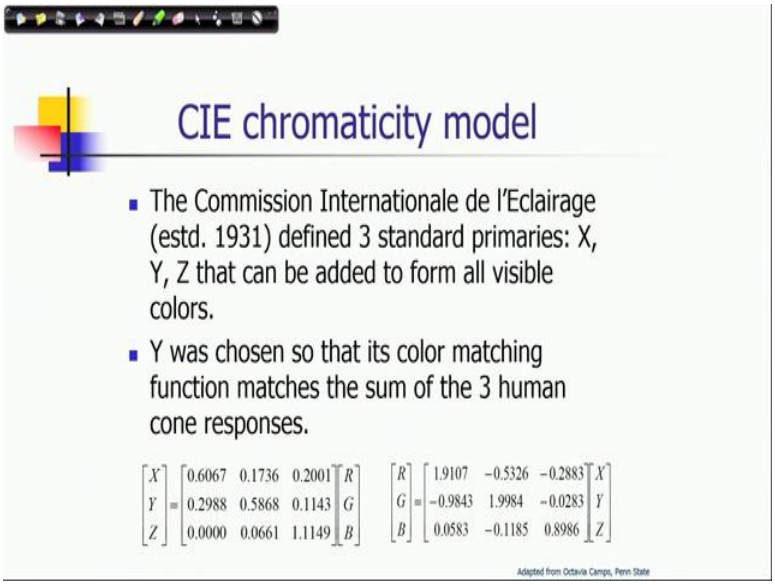


Whereas the other subtractive model the primary colors are cyan, magenta and yellow and can see that they are actually complementary colors of red in this way the complementation is explained here.

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = 1 - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

So, you can see also. If I represent the red, green, blue primary colors those components, if I normalize them within the range 0 and 1 corresponding cyan, magenta and yellow also can be represented from there for that color in the subtractive model. Because in the subtractive model what you are doing. If I produce this amount of cyan means this amount of red will be absorbed in this particular treatment.

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And this is appropriate for paper print. There is a another standard color space and which has been provided by the chromaticity model and which has been provided by a standard body of international body of color which is known as here that commission internationally de I'Eclairage, I mean this is in the French pronunciation my pronunciation is not correct.

So, we call it in short CIE, I will refer it to that. So, international body of color and this is established in 1931 and it defines three standard primary colors the objective is that in this space you do not require any subtraction, you can produce all color using addition, but the thing is that it is a hypothetical space. So, you cannot have light sources which will produce this kind of primary colors. But, mathematically we considered that these are the three primaries, three hypothetical primary colors. And in this case Y component was chosen such that the color matching function matches the some of the three human code responses.

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 0.6067 & 0.1736 & 0.2001 \\ 0.2988 & 0.5868 & 0.1143 \\ 0.0000 & 0.0661 & 1.1149 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

So, the example is that, it is a, linear transformation of any vector in RGB space that gives you X Y Z now this is a primary color. So, X is this combination of RGB will produce this X and this combination of RGB will produce Y and this combination of

RGB will produce Z as you can see that this factor is more than 1. So, it is not possible to produce because we have normalized all RGB component to 0 to 1.

So, this primary color is not possible to produce all these primary colors not possible to produce physically, but mathematically this transformation is defined. And now the advantage of this transformation is that : use the same matching functions with the matching experiments whatever you have with RGB, if I transform all those matching functions using this transformation you will find all of them becomes positive.

All the RGB components, even some of them is negative in the matching function due to our experiments, but after transformation you will find all of them becomes positive in this case. You should note that this transformation is an invertible transformation and as you can transform an RGB vector to XYZ space. So, we call this space as CIE XYZ space or simply XYZ color space, you can again transform it back to RGB space.

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 1.9107 & -0.5326 & -0.2883 \\ -0.9843 & 1.9984 & -0.0283 \\ 0.0583 & -0.1185 & 0.8986 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

So, inverse transformation is shown in the above fashion, and you can see that this is the matrix and your corresponding XYZ component is shown here and if I multiply this with RGB, if I multiply this with XYZ, then you can get the corresponding RGB component. So, let me give a break here for this in this lecture. So, we will continue this topic of discussing about color fundamentals and later on processing with color images in our subsequent lectures.

Thank you very much for your attention.

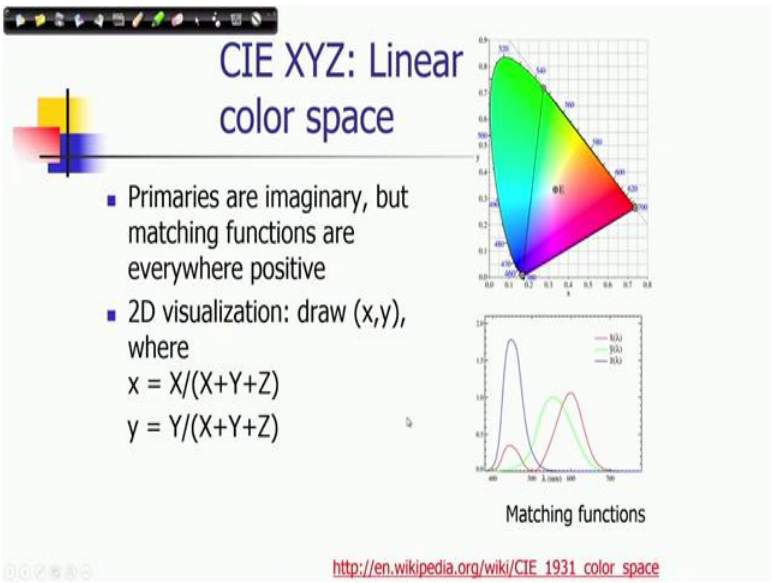
Computer Vision
Prof. Jayanta Mukhopadhyay
Department of Computer Science and Engineering
Indian Institute of Technology, Kharagpur

Lecture - 36
Color Fundamentals and Processing (Part III)

We are discussing about trichromatic representation of a color according to our perception or three-vectorial form of a color. In the previous lecture, we discussed how color could be represented in its natural space of RGB, natural in the sense that we captured the information of color into these three components, red, green, blue but the problem is that not all possible colors could be produced by using these three components in an additive mixture of those components. So, for a convenient representation of color, where some of these vectors even all these vectors could be represented by positive numbers in their components.

There is an hypothetical color space which has been designed by an international body of color; and in acronym it is called CIE, and that space is called XYZ space and there is a transformation which has been proposed by that body which has been suggested by that body or recommended by that body. Using that transformation, you can convert any color represented in RGB space to that XYZ space and since it is a linear transformation, so this is also a linear color space.

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So, we call this space as CIE XYZ color space and here as I mentioned primaries are imaginary, but matching functions are everywhere positive that is the advantage we have. So, if I again plot those matching functions in terms of color representation of color in across the wavelengths, then we can see actually all these values are positive values here and so this is the for, so this is representing $X(\lambda)$), this is $Y(\lambda)$, and this is $Z(\lambda)$.

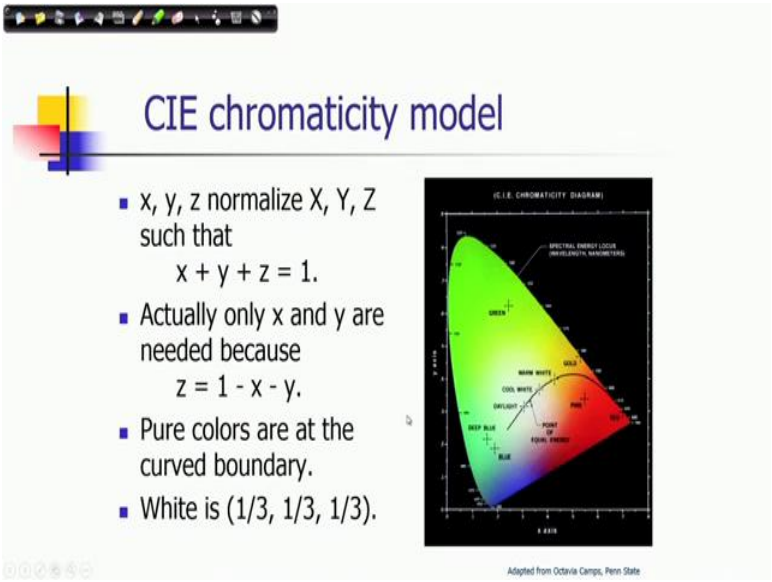
Now, one of the representation of color, you can consider also a two-dimensional representation because the part of the color representation as we can see using Grassman's law that if I scale the vector still you will get the same sensation of the color, only its intensity changes. So, if I can separate out this intensity from the other quantities, then also we can represent a color.

So, with respect to intensity or normalize the color representation with respect to intensity, then this normalized representation once again can be represented in a two-dimensional space, so that is a motivations of representing color in a two-dimensional space and as you can see that this normalization of XYZ component happens in this way, so this is normalized x and this is normalized y.

So, here I am showing you a normalized representation of all colors which has been converted to their respective x, y, z coordinate given these wavelengths, and they can be represented in a two-dimensional space. We call this space as normalized x, y space and a point is representing color here pictorially, even the color dots are representing the corresponding coordinates.

So, this phase is called CIE XY chromatic space, because here the intensity information is missing here, only we are interested on the relative representation of two components which is sufficient to identify a color uniquely in this particular space. So, this is what is a normalized representation.

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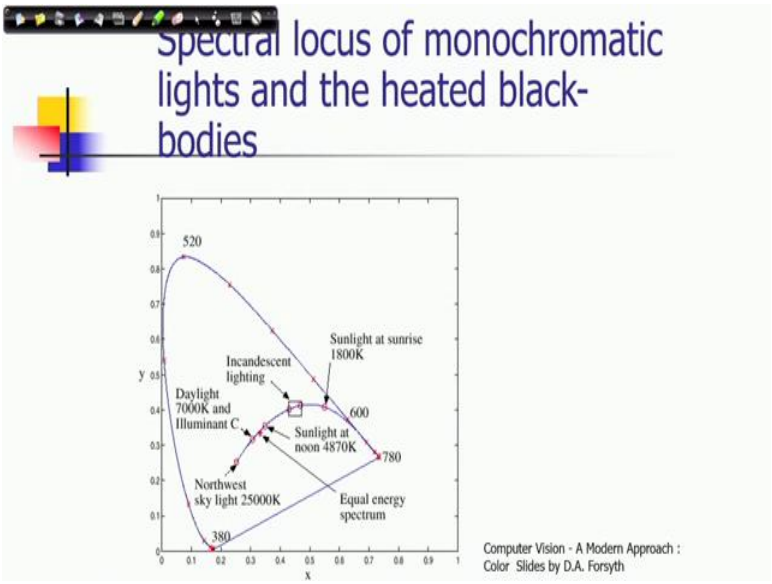
So, to understand this particular representation, we follow a model. we call it CIE chromaticity model. As I mentioned that it is a normalized x, y, z and since the sum of them is equal to 1, so it is sufficient to represent any point using x and y and in this model, you can see the pure color is in the boundaries. So, this boundary represents pure colors. In fact, this is the red zone of having 700 nanometer and as you proceed through this boundary wavelength is decreasing which means frequency of the energy is increasing or electromagnetic wave is increasing and red, green and this is a blue zone and wavelength is near about 360 nanometer. So, these are the pure representation of color.

In between it has other components of this color, I mean we will be again defining them. Actually the white point means you have all the components with equal magnitude and this point can be represented by $\left(\frac{1}{3},\frac{1}{3},\frac{1}{3}\right)$ in the three dimensional form but in a two-dimensional form it is simply $\left(\frac{1}{3},\frac{1}{3}\right)$.

In this diagram, we are also showing a curve which is interesting it is showing the particular color in our visual perceptions there, as if their color remains almost same, only the purity of the color varies and this is the curve, this is a locus of same chromatic sensation in the sensor, no, I should say the pure color what is being shown here in the

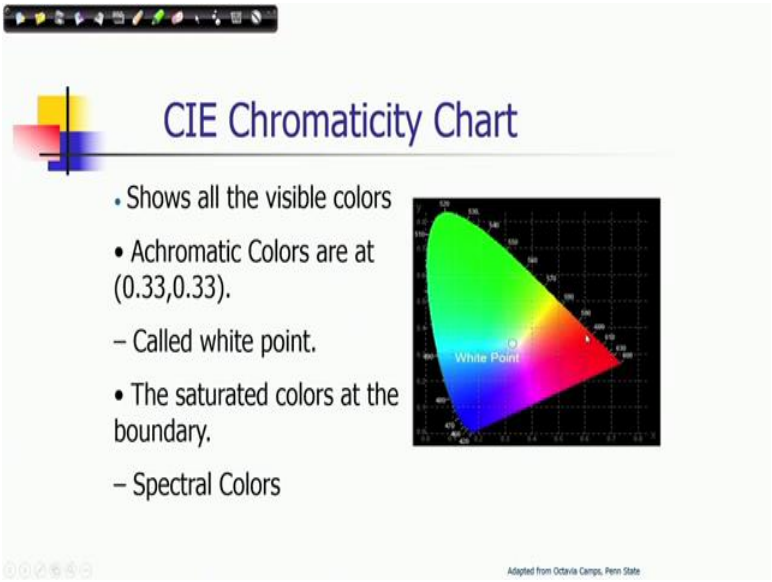
periphery of this curve. That sensation still carries over this curve. even if the point moves that sensation still carries. its whiteness of the color differs.

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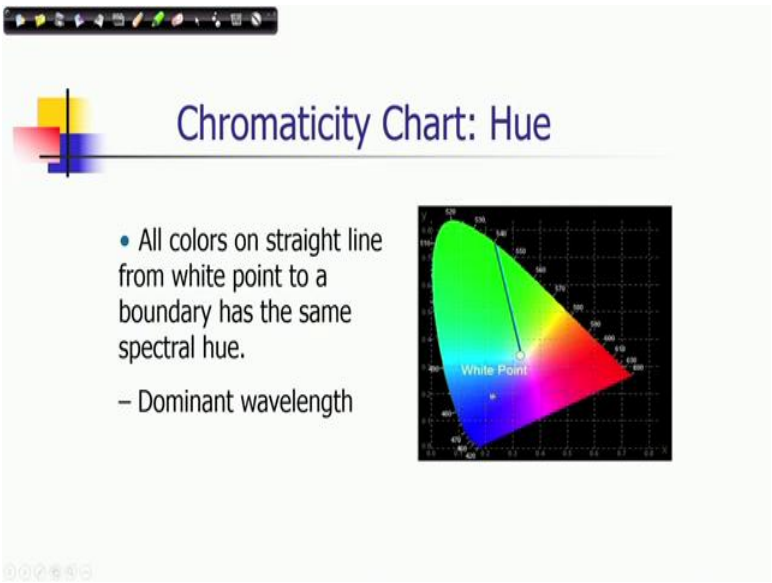
So, this is called the spectral locus of monochromatic lights. So, it is showing here that how different light source of different wavelengths that can be represented here also and you can see that. if I consider the representation of sunlight even the sun which is at noon which is equivalent to a blackbody radiator of 4870 Kelvin and which is almost a white light source but when it goes in sunrise or even sunset, it is becoming more reddish and you can see the (Refer Time: 06:55) trend captured in this particular curve.

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So, CIE chromaticity chart can provide you a simple representation of color using two particular components which is called saturation and hue; I will be defining them. So, any color which is achromatic means which does not have in the sense of our white sensation that is what is achromatic color, and that is represented by as I mentioned it should be $\left(\frac{1}{3},\frac{1}{3}\right)$ and this is a simple representation whereas, the saturated colors means pure colors they are at the boundary and there is a spectral colors.

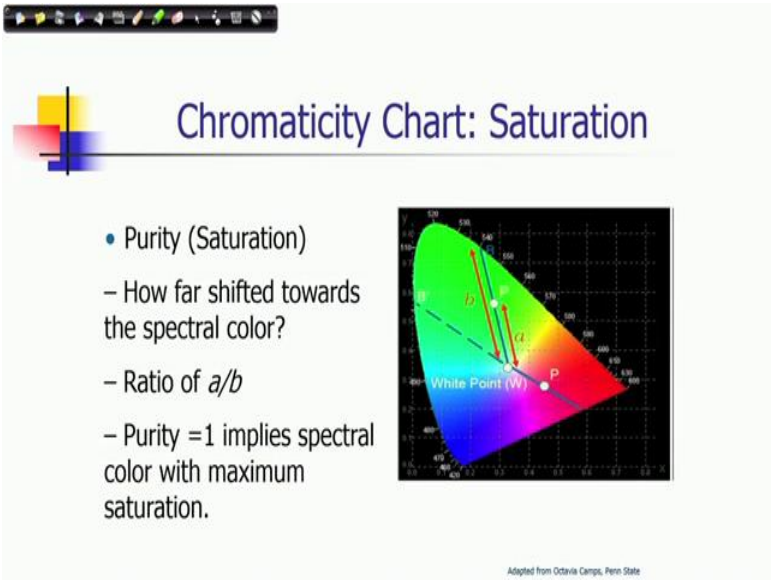
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And so if I consider a radial point, a straight line from this white point connecting to any point in the periphery, now this is a simple model where it is representing the same color represented by this wavelength whereas, the whiteness of the color varies as you move towards periphery the colored is becoming more pure which means whiteness is lessened as it as it moves inward it whiteness is going to increase. So, the this is called saturation.

So, less white means less whiteness in the color means it is more saturated. So, the presentation of hue in this form it is giving the directions, it is remain, it is represented by the I should say a direction from the white point which is directing to a particular point in the periphery. So, we can represent it by an angle also with respect to a reference direction.

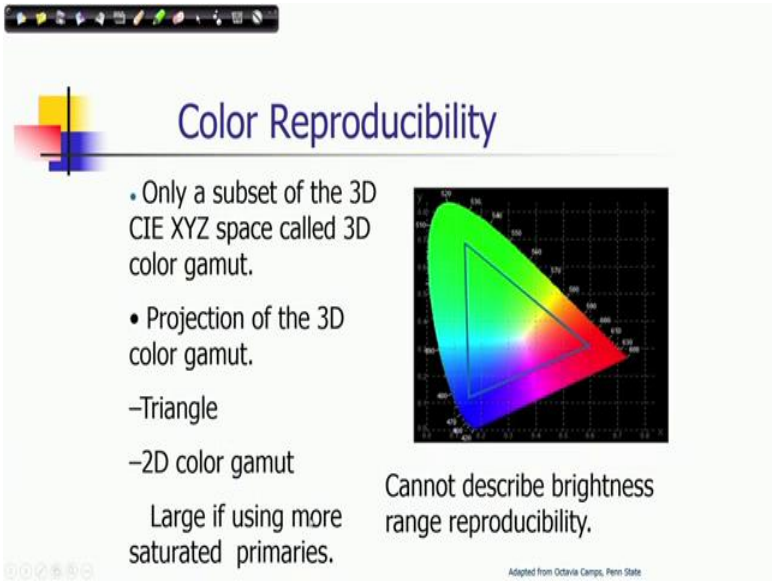
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Whereas for hue it is the relative proportion of these two lengths how close it is to white point, and what is the total length of this particular radial segment which is touching to the periphery of the chromaticity chart. So, how far it is shifted towards the spectral color? So, this ratio of $\frac{a}{b}$ it is giving the saturation as you see that it is as it is moving more outward this is going to be increased.

So, it in this scale particularly saturate varies from 0 to 1 in this definition. So, this is it this definition may vary also this is one sort of simple representation. So, periodic well to 1 implies spectral color with maximum saturation.

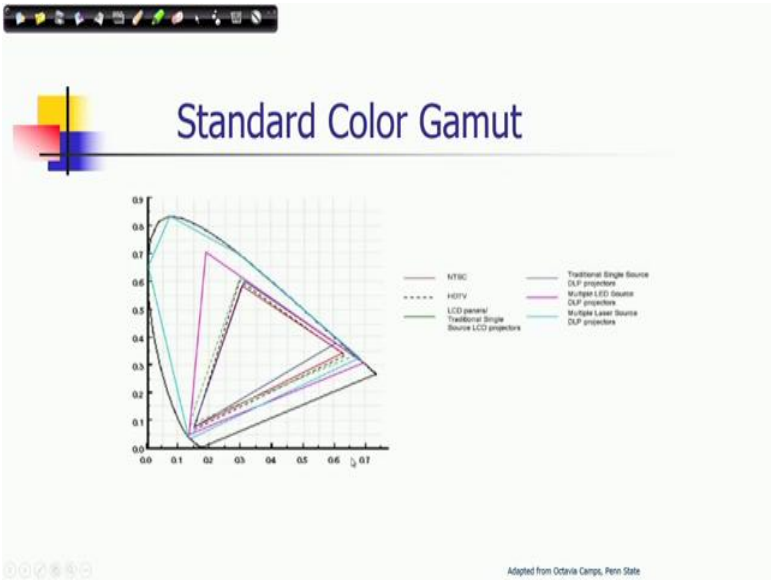
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So, as I was mentioning that color reproducibility depends upon the choice of primaries because once you choose a set of primaries then in the XY chromaticity chart itself, we can represent them as three points. Now, these forms a triangle, so any linear combination of these primaries that would be any linear combination means additive combination, that should be represented by a point within the interior of this triangle, any point should lie within the interior of the triangle or in the edges of this triangle. So, this is the limit of representation of color by these set of primaries, so that is what this particular fact is described here.

So, if I project the 3D color gamut, then you will get this triangle and we call this also a 2D color gamut. So, color gamut is a set up reproducible colors given a set of primaries. So, it could vary, this set could be large if I consider the primaries are more saturated, because if it is more saturated, it will be moving towards the periphery and you can see the area of this triangle will be increasing in that case.

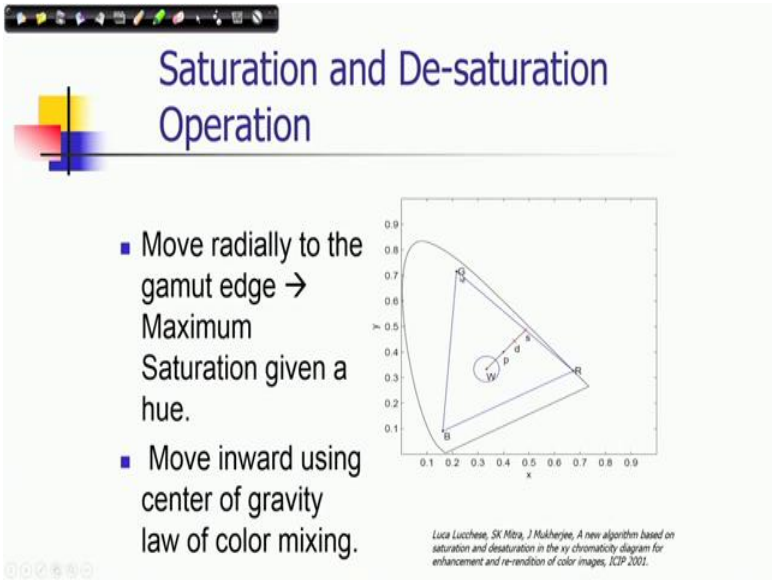
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So, there are various primaries are used and which increases the scope of color gamut for displaying colors, and in fact, there could be a some kinds of display or where you can have more number of primaries instead of rate, that means, you do not have an independent set of primaries, you can have some dependent set of colors also whose combinations are also considered and in that way you can increase the range of color reproducibility.

One such example has been shown here. So, this is a multiple laser source DLP projections where you can have multiple primary. So, it is given by the we should not call them primary color in that sense, we should call those color sources whose linear combination is producing the other colors and, so these are showing you these vertices are showing you this number of colors.

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So, one of the interesting operations from the processings point of view is called saturation and desaturation operation. So, let us understand this particular operation. As we can see that given a color image, we can represent the color of any pixel we can get them into this color gamut. We can represent by any point which will lie within this color gamut and when we consider the primary of the corresponding color system is given by this points here, which is shown as RGB here.

So, if I consider say p is a true color, which is the color of the source image. And then using our model, if I draw a vector or draw a line connecting the white point which is at the point $\left(\frac{1}{3},\frac{1}{3}\right)$ of this color space that is a coordinate or you can see here (0.33, 0.33) is a coordinate point. So, if I know connect these two points W, p and extend it, when it intersects the edge of the color gamut, so that is the maximum possible reproduction of a color of this same hue; and in the same direction, this is a maximally saturated color in this representation.

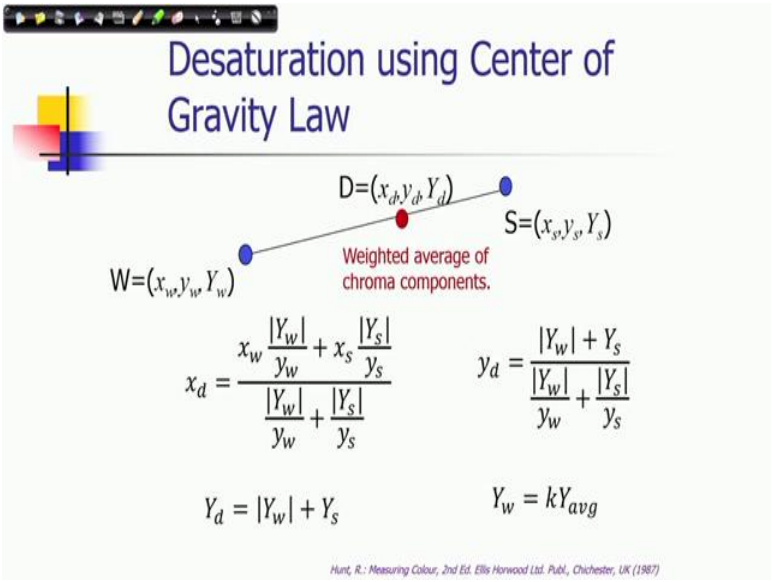
So, we can compute this point and then again you can convert it back to RGB, and you get the corresponding RGB component. In that way, in this way, you can get the maximum saturations. However, if I again any point in this straight line itself can represent a color having the same hue that means the color corresponding to the wavelength which is represented by this vector.

So, there is another operation called de-saturation that means saturation may not give you always pleasing sensation. So, it is not true that more saturated colors and more pleasing there in our perception. So, we feel comfortable with the right balance of colors their saturations as we are accustomed by perceiving different phenomena of nature. So, sometimes it is required even to reduce the saturation and move inwards towards the white (Refer Time: 14:21) point following the same vectorial direction or along the same line, so that is called de-saturation operation.

So, in this diagram, what it is shown that say p is the point which is a color of the pixel and if I connect W, p and extend it, and when it intersects with this edge B, G, then it is a maximal saturation point; and if I move inward also it is the de-saturation point d and this circle is showing here. I explain the algorithms of saturation de saturation, usually the color within this circle most of them are like whitish. So, it is very difficult, it is not visually pleasing if we disturb these points.

So, sometimes for the processing of saturation and de-saturation, we leave out these points which are very close to white. So, some empirically chosen threshold we can model it as a circle around that I mean it is a threshold radius around that point W and we can leave out from this processing.

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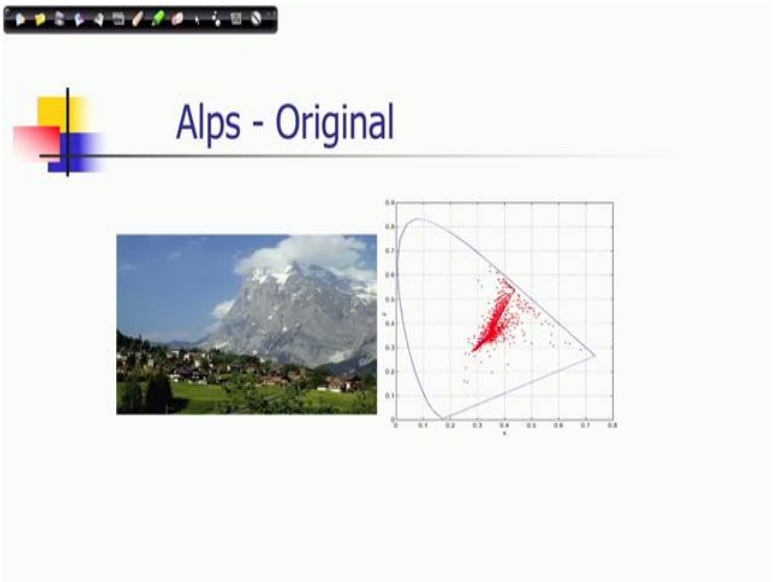
So, this operation of saturation desaturation that is used for enhancing color images. So, we will describe this algorithm before that. let me explain the desaturation a process,

there is a work which proposed a particular intuitive way of obtaining the weighted combination. So, giving these two points W and S color point in the x y chromaticity space, you can get a desaturation point between them because as you see that this is desaturated with respect to this S by following a property called center of gravity law, which means that it is a weighted combination and weights are chosen in such a way. So, these are the weights, so $\frac{|Y_w|}{y_w}$. So, I will explain this notations or symbols.

So, this is the weight. If multiply with the corresponding thing you know values and take the weighted average, you get x_d, similarly you get y_d. You note that , this value is nothing but multiplication of this weight into y_w and this value is multiplication of this weight, weight y_s.

So, you get this weighted combination of x_d, y_d whereas Y_d is the intensity part is defined as the sum of these two values ($Y_d = |Y_w| + Y_S$) and the intensity of whiteness can be also estimated as the average of Y of the image a fraction of it k. So, empirically which you can choose, for example, you can choose k equal to 1 also.

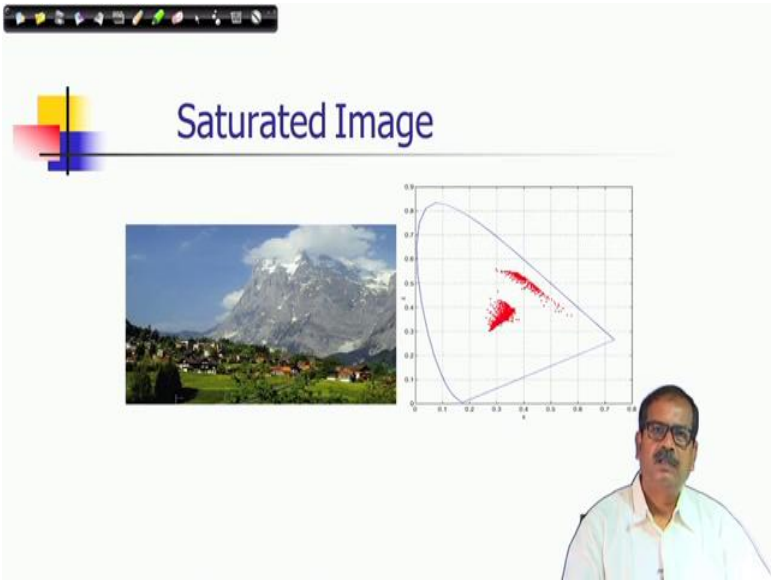
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So, one example of saturation process has been shown here. There is an image where we named this image Alps because this is the image of a particular mountain and speak of Alps and this is the original image. And in the original image, if I plot the colors of the

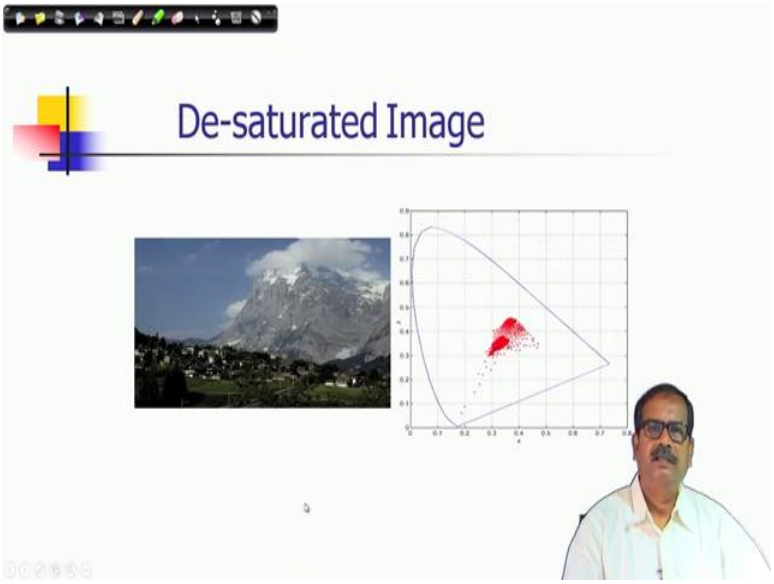
pixels in the x y chromaticity chart you can see this is a distribution and you can see from this distribution itself, the triangular gamut is perceptible because all these points they lie within the triangles of these primaries which have been assumed to some values considering the properties of the display.

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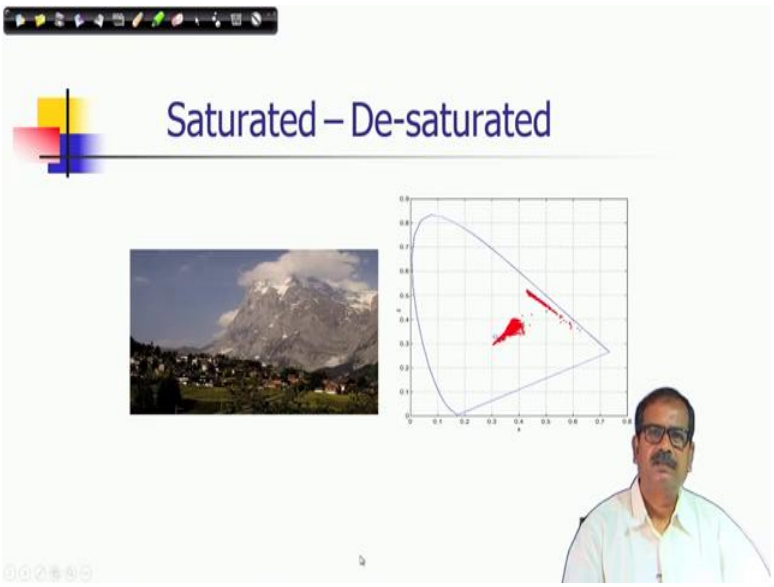
So, if I saturate and if I leave out those points as I mentioned without processing and saturate other points, we can find that they are lying at the edges, and then this is a saturated images. You can see some of the reddish points are if I look at the previous one. So, this point is becoming more reddish in the saturated images.

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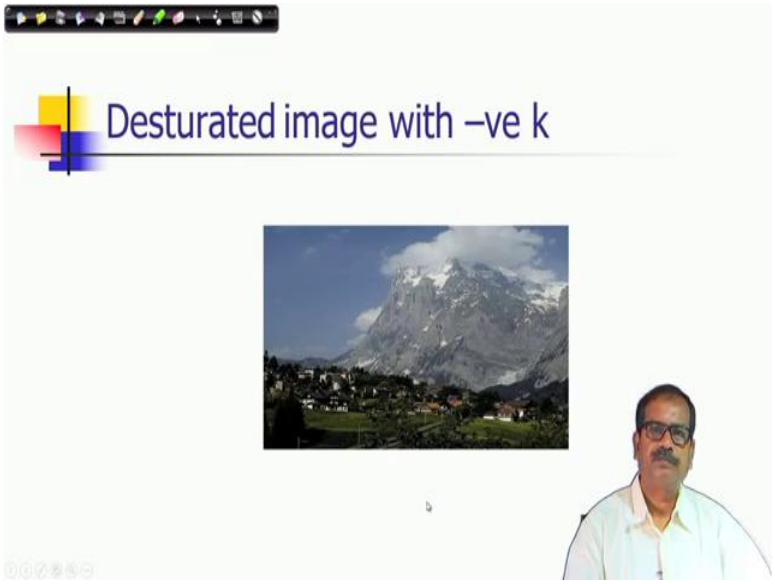
And following desaturation operations using that central center of gravity law, you get a desaturated image like this and where the chromaticity chart, the distribution of color is showing in this form.

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So, that is just desaturated, and saturated, desaturated, after saturation then you performing desaturation, then it looks like this.

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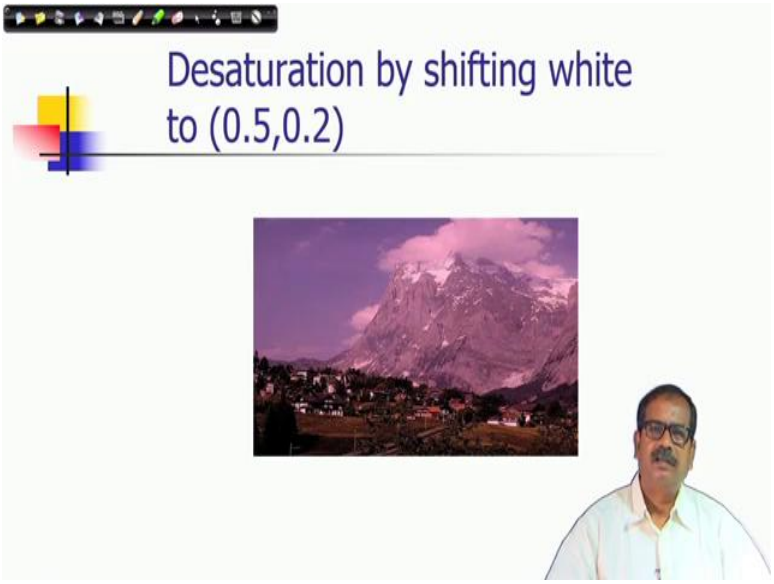


$Y_w = kY_{avg}$

There are some variations you can make. So, if I use the negative value of k as the you know intensity at the white point that you remember kY_{avg} . let me show you the value of the effect what is defined by k. So, this is what is k. So, it is a multiplication factor with the Y_{avg} value and that is giving Y_w and this is how the intensity at this point is determined.

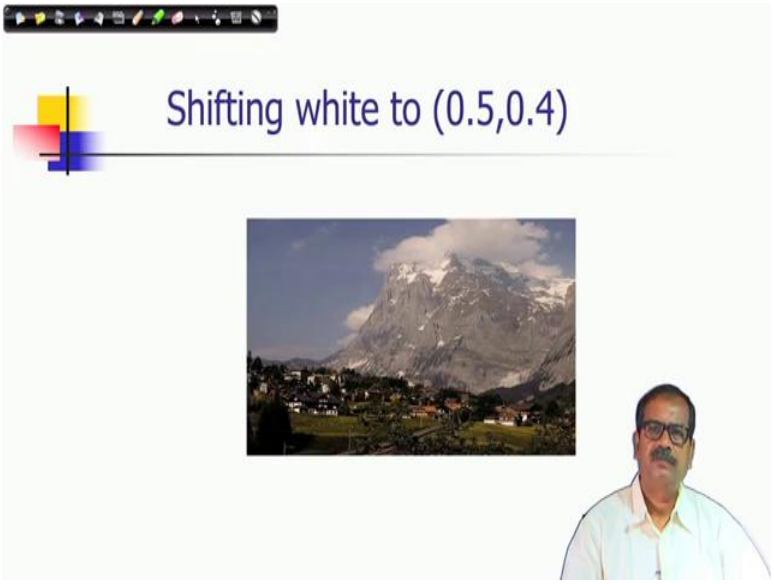
So, this component of course, this component is negative means you are not taking the modulus in that sense. So, you are subtracting it in fact. So, if I use that negative k, so you have to modify that definition, it should not take modulus and then what we get, we get these kind of images. This is the effect.

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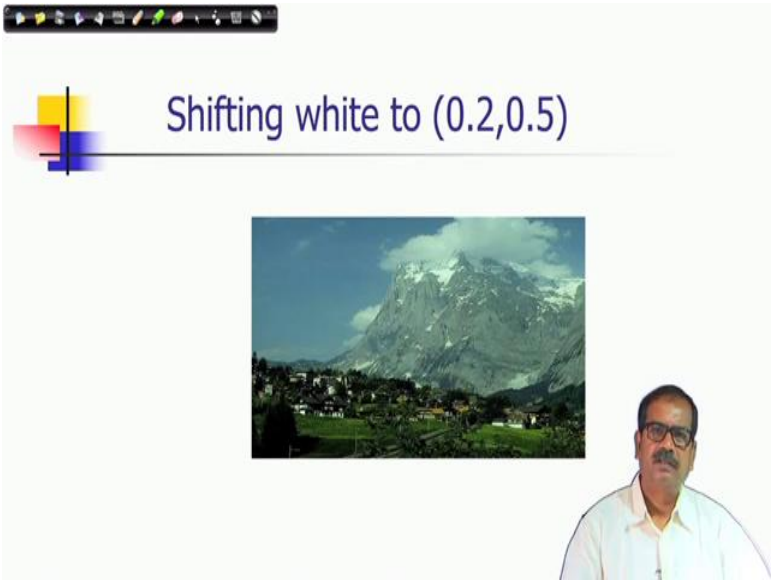
If I shift the white instead of $\left(\frac{1}{3},\frac{1}{3}\right)$ as the white point, the same process we carry out same computation you carry out by considering your white point is (0.5, 0.2), which is towards say reddish zone or and then now you can see that the effect, effect of reddish.

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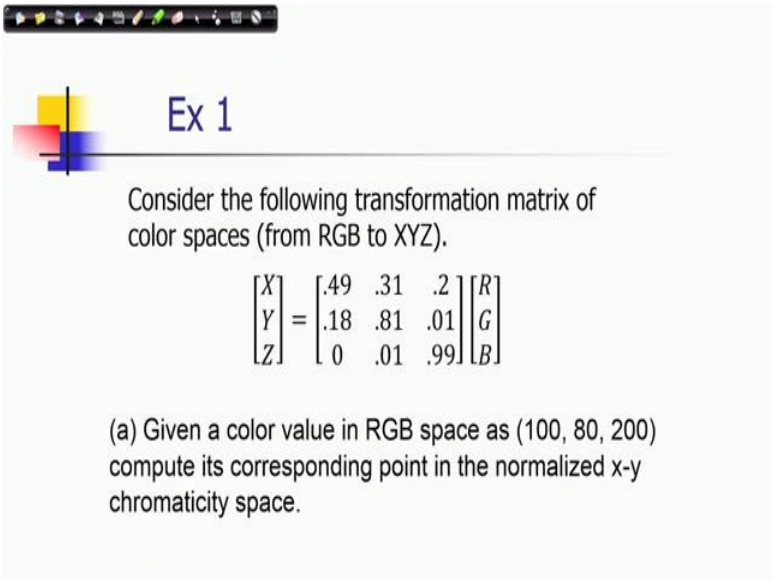
Similarly, if I shift white towards greenish zone, then this is the effect or towards bluish zone this is the effect.

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So, there are different effects those are possible by varying this kind of points.

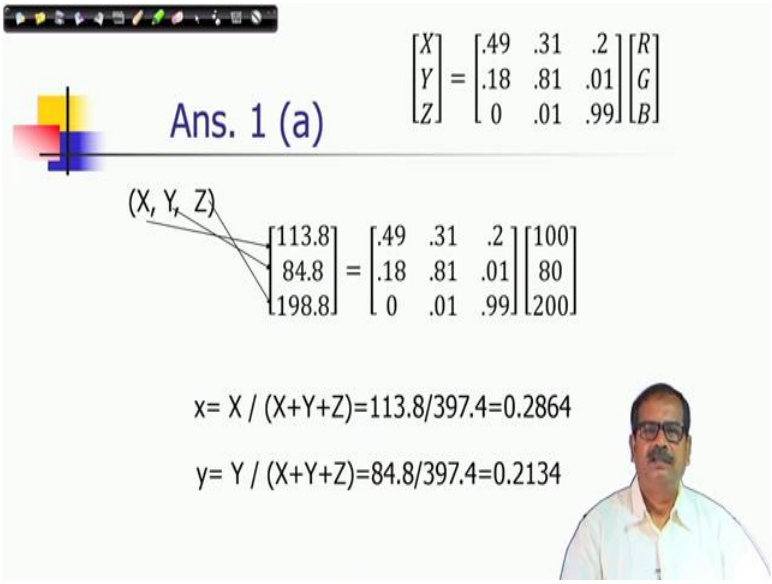
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So, this is one example of processing with the color images by using the normalized CIE chromaticity chart and let me discuss one exercise how do we compute this chromaticity point, it will show you those computational steps, so if you can solve this particular problem. So, in this case the problem says statement says in this way that considered the following transformation matrix.

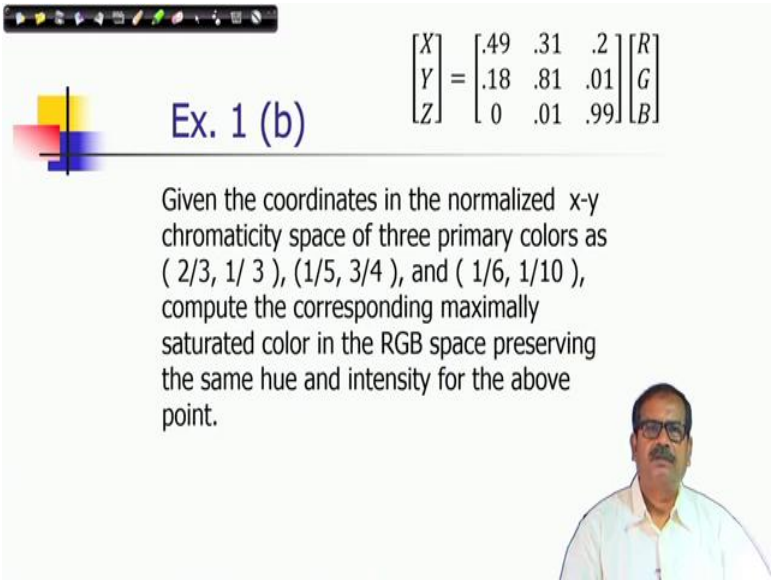
So, this is the RGB to XYZ transformation matrix, given any RGB color you can convert it to XYZ. So, first given a color value in RGB space this is the three vector representation (100, 80, 200) compute its corresponding point in the normalized x-y chromaticity space. So, this is what we would like to solve.

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And this is very straightforward what we have to do, you have to simply multiply the RGB vector with the transformation matrix, and then you will get corresponding XYZ vectors and then you have to normalize with the definition. So, we know that the normalized, so these are the XYZ components and this is a normalized X which has to be; which has to be divided by the sum of these components. So, you get a value say 0.2864 and y you get a value similarly 0.2134. So, this is a part of the assignment and of that problem.

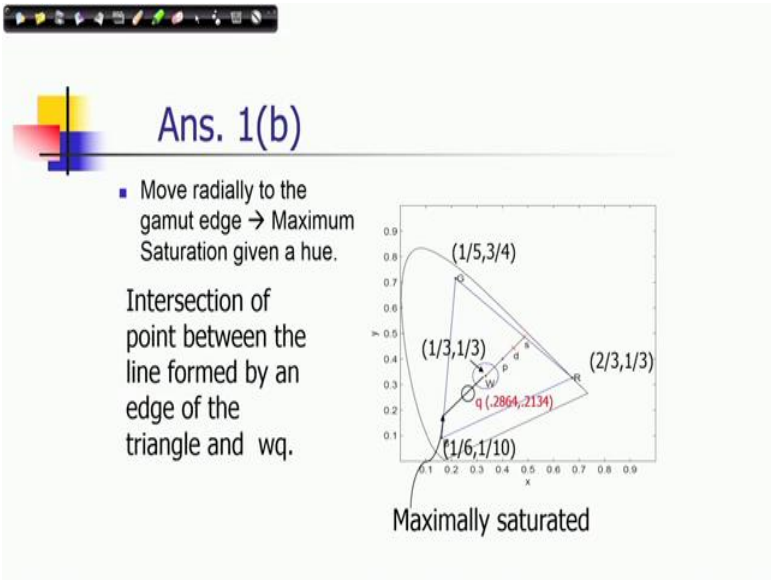
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The next what you need to find out that we need to find out a maximal saturated point of that particular color. So, where you get the points of the no vertices of that color that triangular gamut in this way. So, let me read out once again given the coordinates in the normalized x-y chromaticity space of three primary colors as these $\left(\frac{2}{3},\frac{1}{3}\right)$, so note it is it should be for R; $\left(\frac{1}{5},\frac{3}{4}\right)$ it should be for green; and $\left(\frac{1}{6},\frac{1}{10}\right)$ it should be for blue.

So, in our representation, we will be using in our discussion using those three notations. So, what we need to compute the corresponding maximally saturated color in the RGB space preserving the same hue and intensity for the above point.

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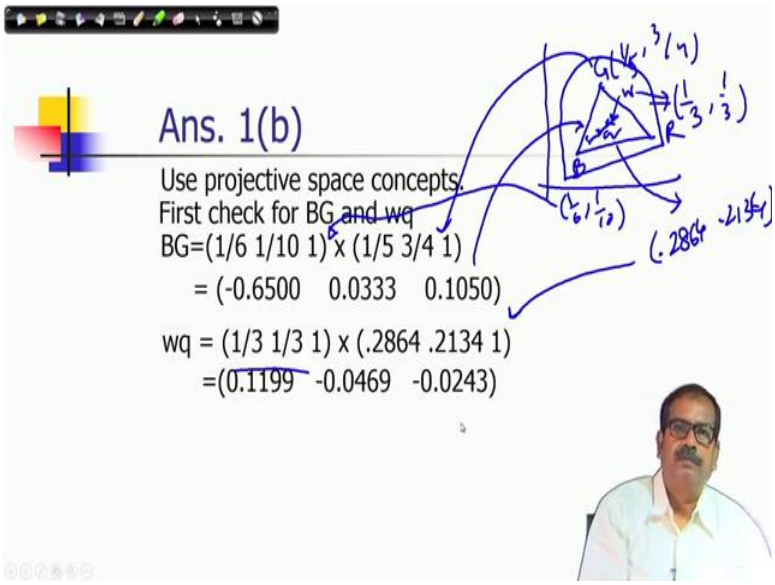
So, if I perform this operation, so once again this figure is explaining what should be done but consider that you have a point, so, this is the red point, this is a green and this is a blue in the chromaticity chart. So, this is how the triangular gamut is specified, and consider this as a white point $\left(\frac{1}{3},\frac{1}{3}\right)$ and this is the point you have computed already of that color vector in the x-y chromaticity chart and the thing is that you need to find out the maximally saturated point of connecting and say w, q, and which is connecting the triangular edge.

Now, it may happen that it is difficult to say the exact position of this point, it may happen this point is here, and it is intersecting this one and actually the intersection of this edge is, outside the gamma triangle. So, you need to consider all these options or even it can intersect in this direction this edge and now you have to find out that in which direction this point is there, that means, you have to consider the corresponding angle and then only choose that direction for the maximally saturated one.

So, there are several options. Now, in our case we will be considering only intersection with these two because by looking at the coordinate positions we can identify that these are the two possible candidates of edges, and we will be considering and the solving this in that fashion. So, we have to find out the intersection of point between the line formed by an edge of the triangle and wq. So, let us proceed with this computation.

You should note this maximally saturated as I mentioned that point or whatever the intersection is there, then you have to decide before declaring it as maximally saturated.

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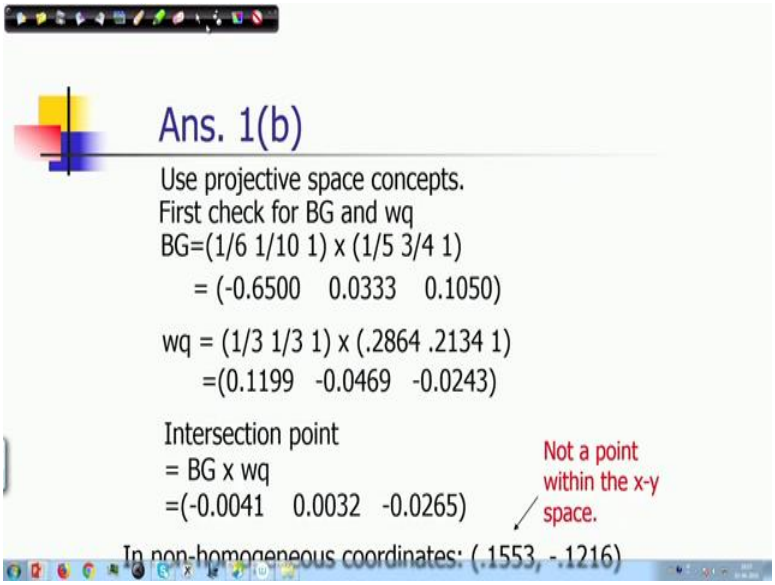
So, for computing this intersection, we will be using projective space concepts. We have already studied that how you can compute the intersection of lines in a two-dimensional coordinate space, if I extend it into the projective space representation. So, first check for intersection of BG and wq. So, note here once again let me draw. we have a CIE chromaticity chart, and then there is a gamut triangle and say this is R, this is G and this is B, and this is w and say this is your q.

So, we are trying to find out intersection of this edge with respect to the corresponding triangular edges. So, the BG these computations could be computed as with respect to BG the intersection could be computed as the cross product of these two vectors. As you can see that this point is represented as this point is represented as $\left(\frac{1}{6},\frac{1}{10}\right)$. So, in the homogeneous coordinate system, this is $\left(\frac{1}{6},\frac{1}{10}\right)$, and then G is $\left(\frac{1}{5},\frac{3}{4}\right)$. So, this is the representation. So, if I take cross product, this is giving you the equation of this line BG.

And similarly I will get the equation of wq. So, this is the equation of w q. So, this is w which is $\left(\frac{1}{3},\frac{1}{3}\right)$, so this is the homogeneous representation and q is (0.2864, 0.2134) as

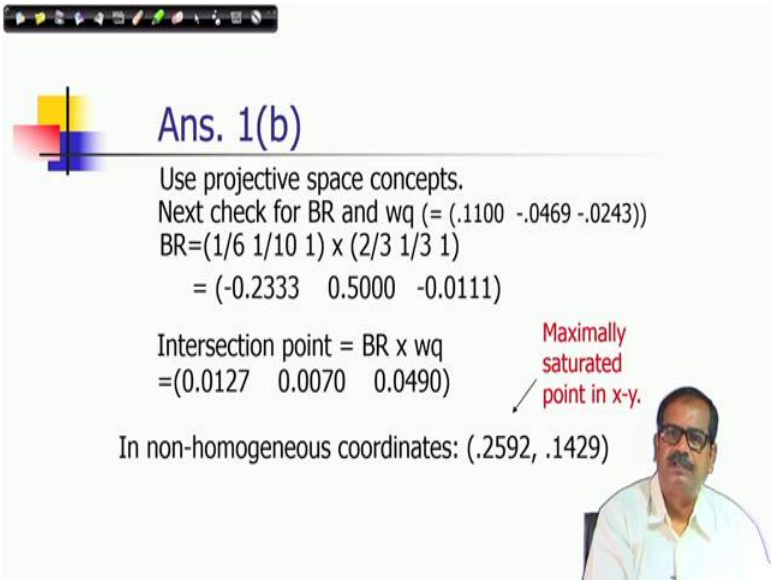
we have computed earlier. So, this is a homogeneous representation. So, if I take the cross product of these two, then we will get this particular representation and value which is representing the straight line wq.

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So, now intersection of BG and wq will give you the point, this is the point, again this is homogeneous coordinate representation. So, in the non homogeneous form, this is the point. Now, you observe that there is a negative and as you know that all the points are to be positive and there is a range between 0 to 1, it should lie within that range. So, this point is not within the x-y chromaticity space.

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So, we should not consider these points. So, now, we should consider the other edge BR, and find out the intersection with wq. So, wq we have already computed which has been shown here and now BR is given by this again in the same way we are computing the cross product of these two points in the homogeneous representation; one is representing B; the other one is representing R and then you get this is a straight line in the projective space.

So, now, the cross product of these two lines BR and wq that would give you the intersection point and this is the intersection point you will get. In the homogeneous coordinate representation in the non homogeneous coordinate representation it is (0.2592 , 0.1429). So, finally, this should be your answer, so it is a maximally saturated point in x y. So, this is an example by which you can see that how this point maximally saturated point could be computed using x-y chromaticity chart given any point in the RGB color space.

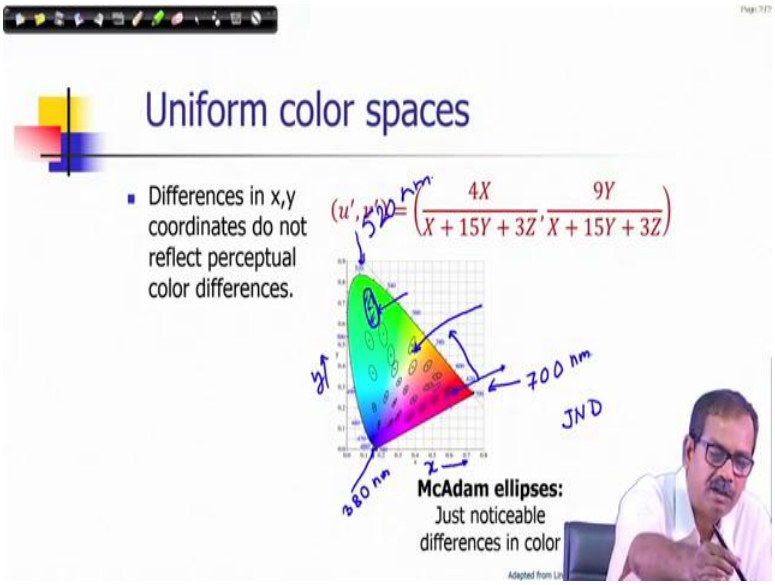
So, with this, let me stop here. We will continue this discussion of you know Color Fundamentals and Processing in our subsequent lectures.

Thank you very much for your attention.

Computer Vision
Prof. Jayanta Mukhopadhyay
Department of Computer Science and Engineering
Indian Institute of Technology, Kharagpur

Lecture - 37
Color Fundamentals and Processing (Part IV)

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We continue our discussion on Color Fundamentals and Processing of color images. In the last few lectures I have discussed how colors could be represented as a trichromatic vector. And, how the perceptual factors of color in terms of chromatic experience in terms of hue and saturation could be obtained from these calibrated chart as proposed by an international body of colors or in acronym CIE. There is CIE chromaticity chart from where you can represent a color in a x y normalized x y chromatics space.

And as you can see here also in this particular figure that the colours are shown in a semi elliptic region and where the peripheral regions peripheral curve is showing different wavelengths. For example, the wavelengths starting from, see this point is 700 nanometre and this point is 520 nanometre and this point is 380 nanometre. And, across this periphery as you traverse from this point and come to this point, wavelengths are decreasing in this order. And different wavelengths, they correspond to a particular color, a pure color that is sensed by us when that electromagnetic wave of that wavelength is received the energy from those wavelengths.

But, in the chromaticity space any color when they are represented as a superposition of three primary colors particularly when we capture in terms of red green and blue, then we can convert those color points into a two dimensional chromaticity space that we have discussed that is x y chromaticity space. So, for example, this is the x axis of that space, this is x axis of that space and this is y axis and any particular color at this point it has been shown also.

what is the color of that pixel that is how this you know visualization has been created this graphics have been created, where it is also showing the colour. So, you can see there if there are large regions of having almost similar color sensation and the minimum the region where color are almost indistinguishable they can be approximated by ellipses.

As you can see one example of this ellipse is shown here a larger ellipse, here in this region you can see this ellipse I am drawing it just for your convince. So, centring this point all the color covered by this point all the chromatic point they will generate the same sensation. Now, these ellipses are known as McAdam ellipses and these ellipses are also called just noticeable differences in color ellipses or JND ellipses this is a very common acronym what is used.

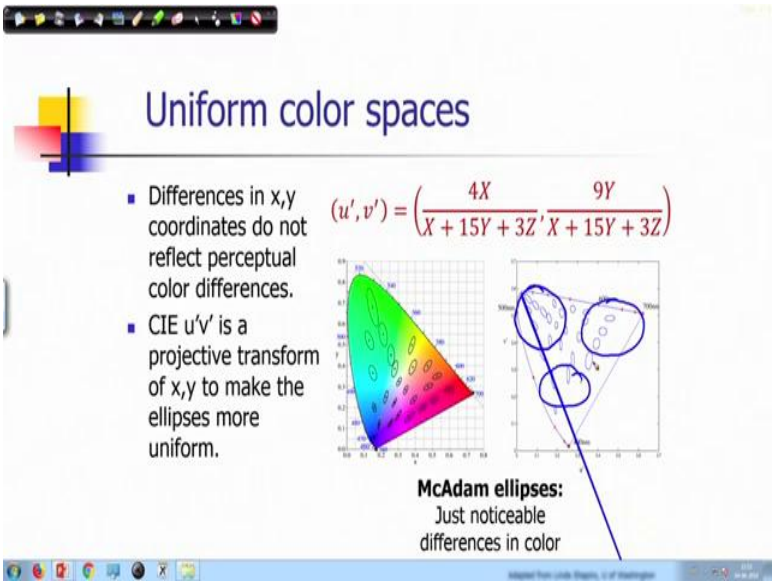
So, if your points so within these JND ellipses so all the perception of color would be almost similar in our human visual system. Now the problem here in thisx y space is that the size of this ellipse is non-uniform, you can see the size of this ellipse at this location is quite large. Whereas, near to the red region, this size is quite small very small ellipses are shown here, so that is a problem with presentation of color in an x y chromaticity space. So, there is a proposition another proposition for another color space, where you can try, you can make this ellipse.

Size of this ellipse is more or less uniform. it will not be exactly uniform in every place. But we try to minimize the variation of this sizes and this color space is called uniform color space, the uniformity the uniform that adjective comes from the point of view that we are trying to make those sizes uniform by making some transformation on those (x, y) point itself. So, so this is one example of this transformation. you can see that just to make it clear let me rub these points.

So, you can see that the transformation I have displayed here. so this space is called u' v' or UV in short that is a uniform color space. Again it is a chromaticity space, space

of representing only the chromatic components and the kind of transformation from the x y z coordinates of the color representation of the color has been shown here.

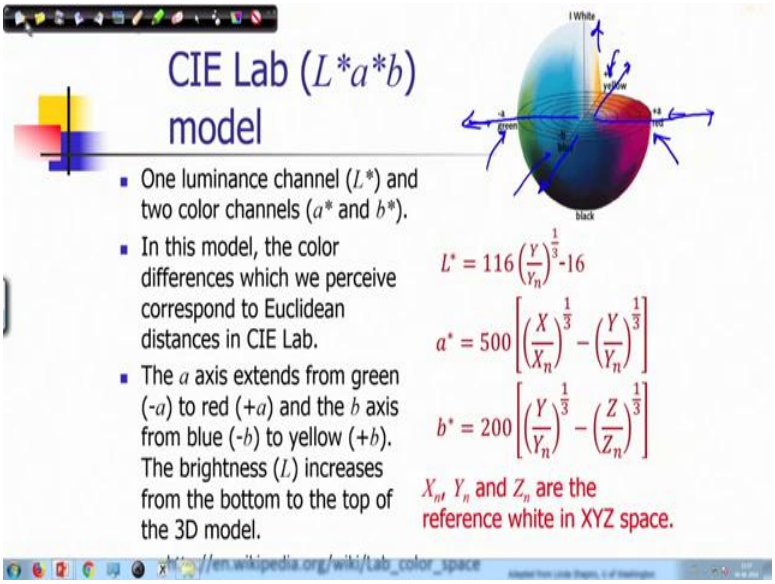
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So, the point to note here that it is a projective transformation of (x, y) to make the ellipses more uniform, say you consider the this diagram. Here you can see that the differences of sizes of ellipses in this zone which corresponds to the green zone in this space and in the red zones or in the blue zones, they are reduced grammatically. So, this space is no more suitable when we are trying to compare colors in terms of defining certain distances.

Though it is ellipse it is not circle, circle so Euclidean distance is not very appropriate in that sense in this space. But still you can consider the neighbourhoods are sizes are almost at least sizes are similar. So, they would be more appropriate or they would be more effective for distinguishing colours when they are represented in this space.

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So, let us continue this discussion on color representation. let us see what are the spaces are there. So, one of them with respect to the discrimination of colours there is another very effective space that is proposed also by the international color body CIE and this space is called Lab space or we define this space also $L^*a^*b^*$ space. Here the distances would be more effective Euclidean distance would be more, as the distinguishable the ellipses. If I convert in the "ab" space or that is "ab" chromatic space, so those will be more circular in that shape.

So, in this space we have one luminance channel and the two color channels a and b, I am referring them a and b those sometimes here shown as a^*b^*. As a transformation is also shown here you can see the transformation in this part. So, we have $L^*a^*b^*$ and you can see that is a non-linear transformation which is made using once again they are made using x y z representation of the colour. And note that all this x y z values they are normalized by the respective values of reference white.

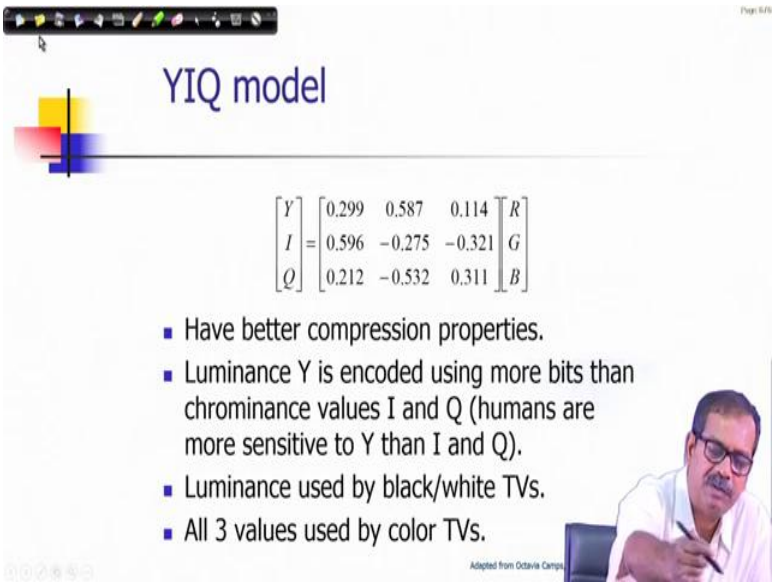
So for example, reference white is represented as X_n, Y_n, Z_n, usually the white is that white color and we know that corresponds in the chromaticity chart of x y chromaticity chart the point is nearly $\left(\frac{1}{3}, \frac{1}{3}\right)$ in the x y coordinate. Almost all the components of x y z would be equal, but they may not be exactly equal. so this is a kind of calibration you need to do with respect your system.

So, find out what is the definition of white in your system and it is a kind of white balancing operations in the color representation. So, once you get this values and then you can normalize x y z coordinates, as $\frac{X}{X_n}, \frac{Y}{Y_n}, \frac{Z}{Z_n}$ and then again use this function to derive $L^*a^*b^*$. And so in this model the color differences which we perceive it correspond to Euclidean distances in CIE lab. So, it is very effective and the problem of distinguishing even in the uniform color space that becomes much reduced. I mean we can we can mitigate that problem to a great extent if we use $L^*a^*b^*$ for this purpose.

Just note that the a axis in the case it extends from it is shown in the figure also. So, this is the a axis this is the negative part, this is the negative direction and this is the positive direction. So, $-a$ is green so this is a green portion these are red, similarly for b axis this is a $+b$ and this is actually luminous. So, $+b$ is showing here and this is $-b$, so this is yellow this part this is yellow and this one is blue.

So, blue to yellow and green to red that is a kind of variation of hue in this space along this axis principal axis and along the you know L axis which is shown here as a vertical line it is a intensity what is being varying alone that axis.

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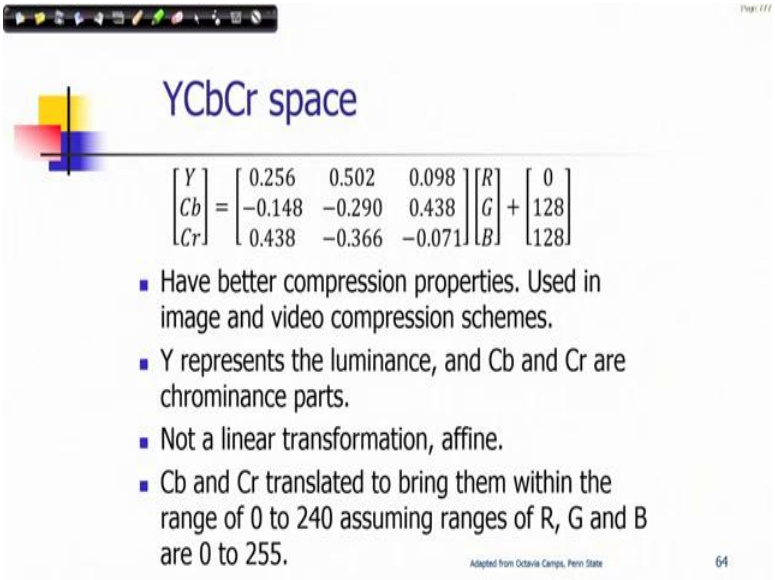
There is another color space which is called YIQ color space and it is respective transformation has been shown here it is a linear transformation, it has the property of better compressibility of information. So, in usually the chromatic component of I and Q

they require less bandwidth. So, to represent the color variations. So, if you consider any radio signal color radio signal and if the color is represented into the three forms other than convert them from RGB to this form, then the component of I and Q in that signal will require less bandwidth.

And that is why this particular conversion is used in television signalling color television signalling where this conversion has been made and we can also used in the digital television signals. So, so it is a luminance y is encoded using more bits because it requires more bandwidth. Where as a chrominance values I and Q they require less width. So, the idea is that humans are more sensitive to the intensity y. And in fact if you have black and white television signal, then you know you can just simply use the y and there is a signalling encoding.

So, that even the encoding of I and Q will not affect the receiver of black and white TV. But that is the different point which we are not going to elaborate here. So, the fact is that luminance is used by this black and white TVs and all three values are used by colored TV. So, this YIQ is also a very good model for sending information by factorizing information into these three channels.

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The space YCbCr space we might have already mentioned this space when we discussed about image transformation. Actually this is a space this is the color space used for image compression in JPEG standard and in many other standards subsequent standards

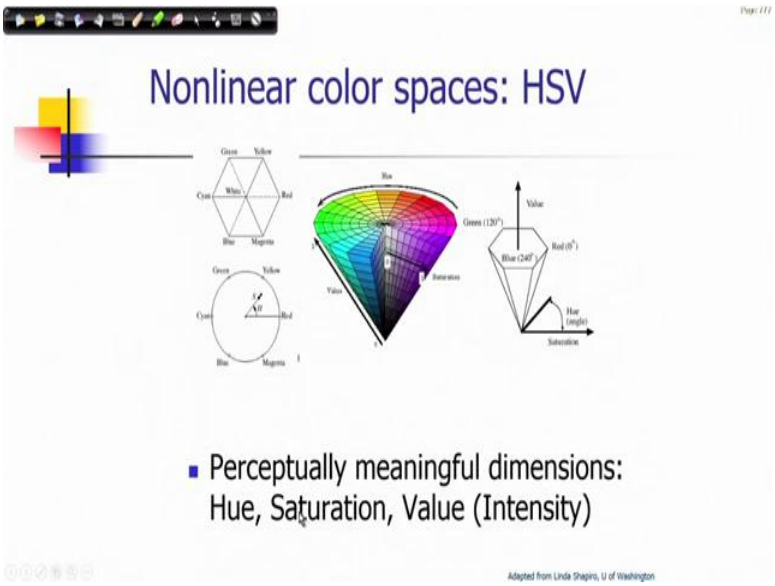
also Vario standards. So, you can see the transformation here. there is a linear part of it and then there is an translatory part of it which is which makes this transformation as an affine transformation.

And just to note that it is also having a better compression properties, because of that it is used in different compression schemes and here Y represents the luminance and Cb and Cr they represents chrominance parts. So, Cb is called complementary blue and Cr is called complementary red that is a full term. But it is not a linear transform it is an affine transform because, there is a additive component in the transformation, translatory component or additive component at the end like this particular component which makes this transformation affine.

Otherwise you know if I leave aside this part and consider modified Cb as $Cb - 128$ and Cr as $Cr - 128$ then you have a linear transformation from RGB to that. So, that is also something used in color representation. So, this translation is done with this motivation that these values without this translation it could it is ranging in a negative zone.

So, for this inconvenience of representation in digital storage and I am making all those values positive. So, that you can represent then you know in unsigned white color representation or pixel representation. Assuming if red green blue the values they varied from 0 to 255 you can see the values of Cv and Cr of the transformation will vary from 0 to 240. So, that is one of the motivations why we make this kind of affine transformation.

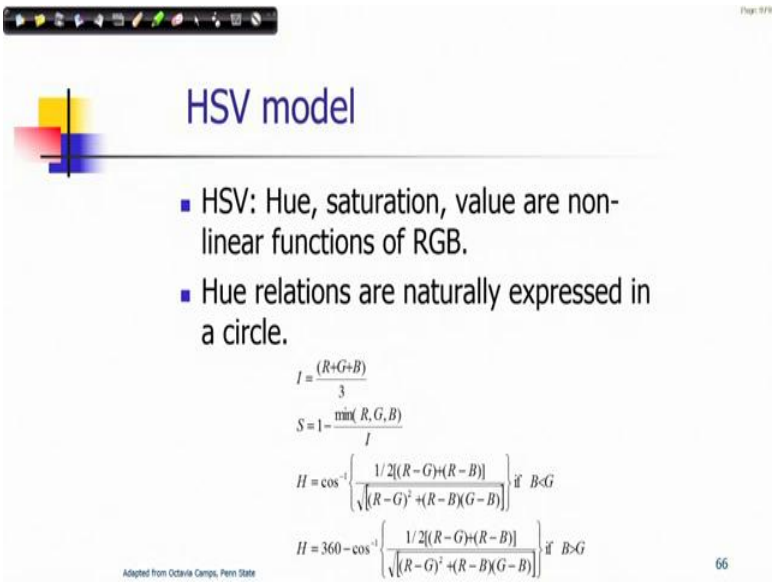
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There are other non-linear color spaces where again this hue and saturations they could be factored out from the achromatic information's like intensity. So, this is showing one such example, where you can see it is a conical shaped representation of a space and at the the intensity is varying as we move towards this and as it as move towards the vertical directions.

Whereas spatially in the horizontal directions you have the direction of saturation and the hue is represented by the angle in this. So, such a kind of polar representations what has been shown here. And, depending upon angle you can see .so your reference direction is red and as you increase your angle it is if there is a transition from y green cyan blue magenta to red as you can find out in this circle representation or in this hexagonal representation.

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So, we can elaborate this transformation further the mathematical form that we will see. So, the interpretation here is that HSV stands for hue saturation value, value stands for intensity . So, Hue relations are naturally expressed in a circle as I shown as I have shown here and these are the expressions. So, how the intensity which is a value represented here by I which is just a average of red green blue values and then saturation expression is also very simple it is one minus of fraction of intensity and the minimum value.

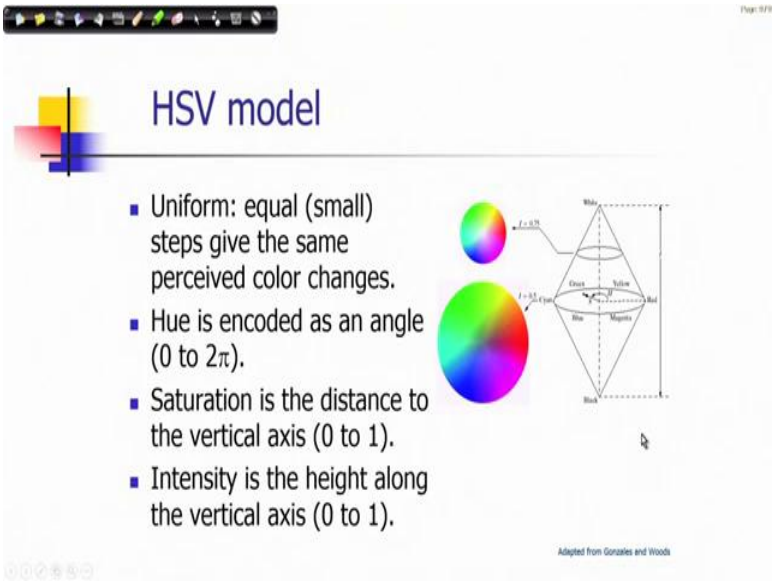
$$I = \frac{(R+G+B)}{3} \qquad S = 1 - \frac{\min(R,G,B)}{I}$$

So, fraction defined as a as a ratio of minimum of RGB to the intensity value and it is a one minus of that value saturation and where as hue is also expresses angle as has been interpreted in that space.

$$H = \cos^{-1}\left\{\frac{1/2[(R-G)+(R-B)]}{\sqrt{[(R-G)^2+(R-B)(G-B)]}}\right\} ifB < G$$

$$H = 360 - \cos^{-1}\left\{\frac{1/2[(R-G)+(R-B)]}{\sqrt{[(R-G)^2+(R-B)(G-B)]}}\right\} ifB > G$$

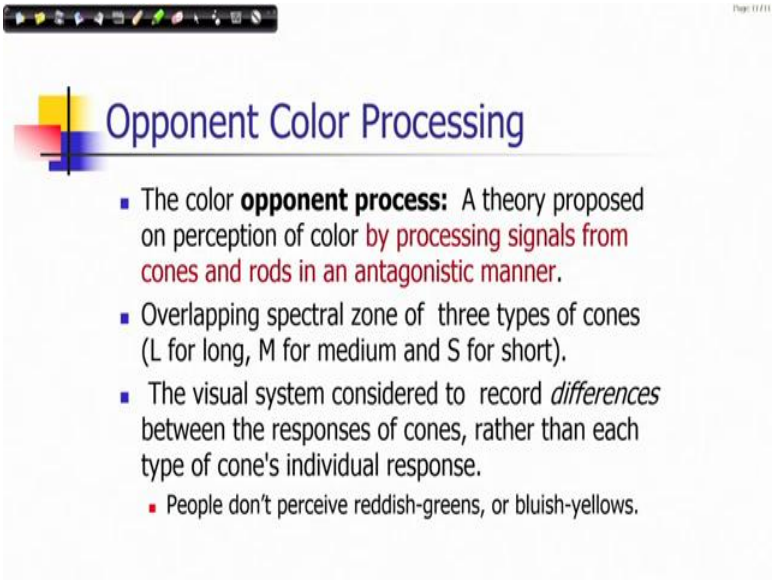
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So, the implication is that HSV model it is uniform that is it is equal states it gives a same perceived color changes. So, it has that uniform representations of non distinguishable colours that kind of neighbouring color, non distinguishable color that shape also here it is a quite uniform it is a three dimensional representation. So, it is if I say that is a kind of circular ball around a point, so it has that effect.

But though you should note the the conical regions it is a double conical region what is being represented as it is shown in this part of this diagram and around this point you take small spheres spherical neighbourhood and that would be uniform here. And saturation is the distance to the vertical axis that is 0 to 1 and intensity is a light along the vertex. So, these are all normalized representation. So, ranges of saturation intensity and Hue is an angle, so it is it ranges from 0 to 2π .

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The other interesting color space it has again a theory of opponent color processes opponent processing of colours that is going on in our visual system. it is supported by that theory. this space has been proposed. what it does that you know the theory says that by processing signals from cones and rods in an antagonistic manners.

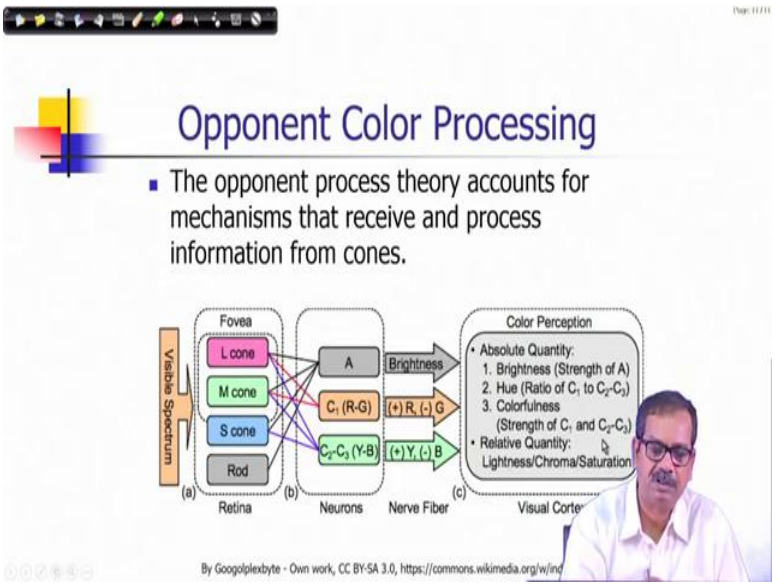
So, they act in an antagonistic manner. So, rods are kind of inhibiting when cones are very active or rods are active then cones are inhibiting. So, that is an antagonistic way they act. However, so let us consider the representation and in the overlapping spectral zone of three types of cones that we has been considered. We know that three types of cones we discussed that long wavelength zone cones that is if that is a spectral response. If they response move to the longer wavelength that is called L medium wavelengths called M and short wavelengths call S.

So, L M and S types of cones are there in our visual system and it consider the differences between the responses of cones, rather than each type of cones individual

responses. So, when we sense a color signal when the received energy in terms of three stimulations of red green blue in three primary forms. It is not red green blue directly being processed for the interpretation of the color, rather the differences of red and green and say blue and also red green part we will see later on what is the model of this difference in the next slides.

So, because a when the fact it is supported by this kind of psycho visual phenomena, that it is hardly there is any perception of reddish green we cannot talk about reddish green or bluish yellows in our color perception. So, from there this opponent color processing theories are developed.

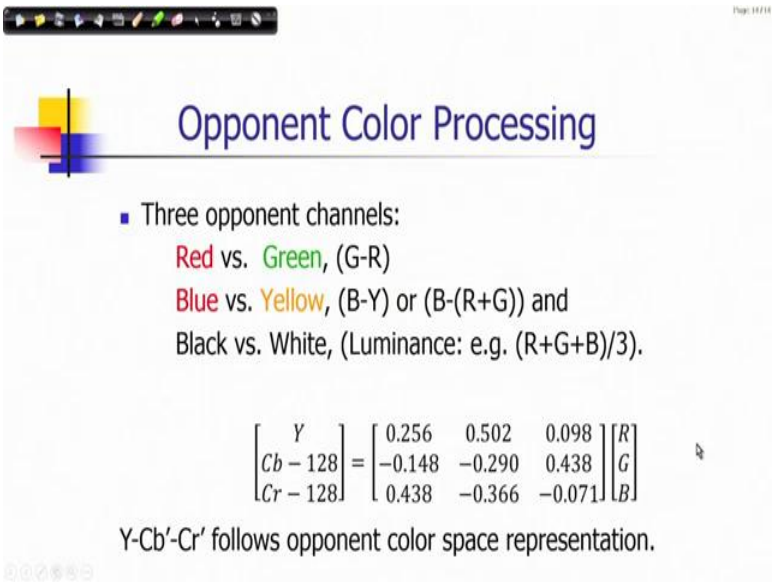
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So, you can see that what kind of representation is there. So, this is a L cone and M cone S cone and you see that L and M they are participating in an antagonistic manner with providing you C1 which is $R-G$ that is a component. So, green is opposing the red you know excitations and S cone it is $Y-B$ where y is basically represented by $R+G$.

So, that is what is opposing so and all the three cones and the rod the participate in producing the brightness value. So, you have now this three factors of brightness this is the a chromatic part and $R-G$ and $Y-B$ as the chromatic part. So, this is how this color has been represented and mathematically we can model in this fashion.

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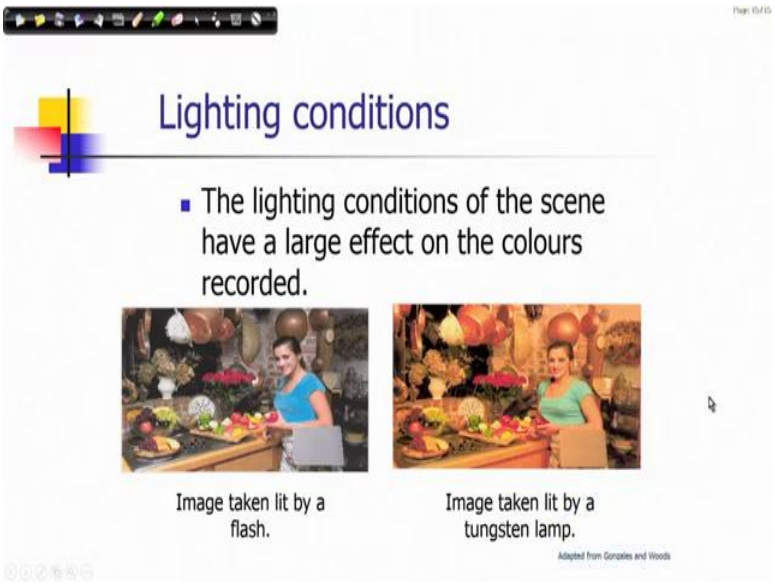


So, we have three opponent channels like red versus green is represented as a $G-R$ and blue versus yellow is represented as $B-Y$, Y is representing yellow or $B-Y$ is represented as a additive composition of red and green or $R+G$ and where is luminance is black versus white $\frac{R+G+B}{3}$ there is a representation. So, this is if you note the that Y Cb Cr conversion where you consider on the linear part, you will find that this is somewhat following this principle because you can see this is red green blue.

So, Y part is representing particularly this luminance which is a weighted average of red green blue. Whereas, the Cb complementary blue part you can see it is representing this component it is blue from the blue you are subtracting addition of red and green. So, that is a complementary blue representation complement representation and Cr this part it is the red. So, red is , you are subtracting major component is green and also some amount of blue is also subtracting in this model.

So, we can see that Y CbCr model in fact that proposition also tends their motivated by this representation of opponent color processing and that is used heavily in various applications. So, this is what is our Cb' prime Cr' is presented to show $Cb-128$ and $Cr-128$ in this slide.

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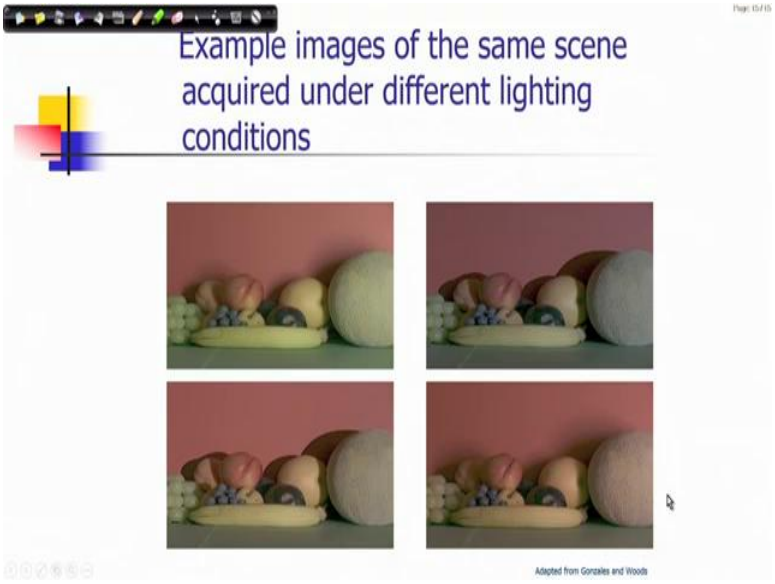


So, now we will be discussing about another phenomena which is observed when we perceive color. It is the variation of intensities, our environmental illumination and how do you perceive colors. You can see when we take photographs or when we take images in varying illuminations the component red green blue they widely vary.

As you can see the super positions of those primary colors they reproduce you know different kinds of image colors in the images, though they are of the same scene. For example, in this case this image is taken by a flash light and this image image is taken by a tungsten lamp and you can see the differences. The interesting you know property is that as a human being we can adopt this illumination variation.

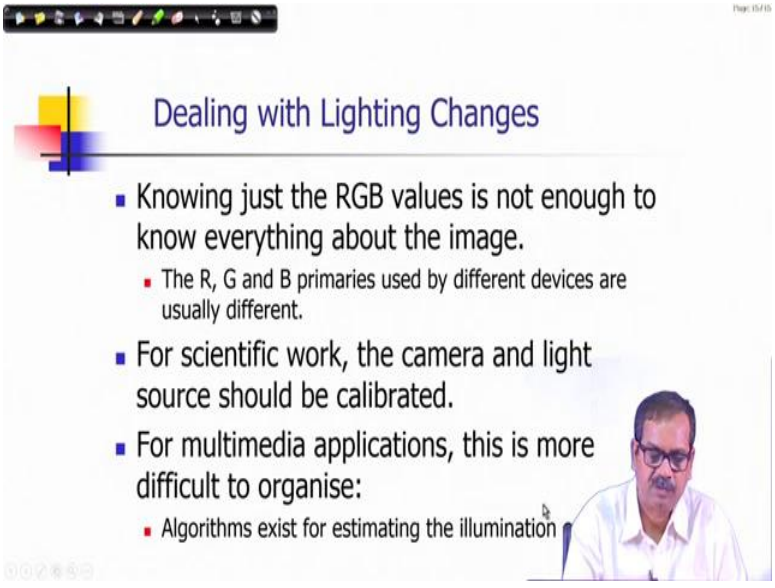
Even we can understand true red with largely varying illuminations, that means that does not have much effect on our color perception. But when you take the photographs there is a large variation of this red green blue component. So, we will see that how we can add these phenomena while processing and how actually it affects our understanding of color when we do the processing. So, this is the some of that lighting conditions of the scene have a large effect on the colors recorded.

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This is another example that same scene has been captured in different illuminations different lighting conditions. But, when you record this intensity values or record the red, green, blue components; there is white variation. But as a human being we do not see much difference we can adopt to these variations.

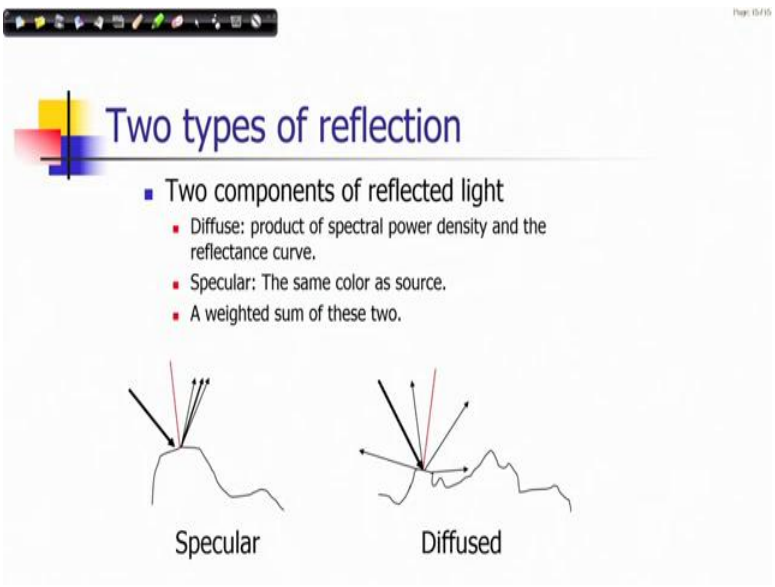
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So, so that is what knowing just the RGB value is not enough to know everything about the image. That is a challenge when we interpret a color pixel. And the RGB primary is used by different devices are usually different that is one consideration. So, for scientific

work we need to calibrate the camera and light source and for multimedia application this is more difficult to organize this factor has to be addressed there. So, there exist algorithms for estimating the illumination of colour.

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So, this phenomenon is of adaptation of human to perception of color same color or true color under varying illumination is called color constancy. So, when we tried to address this computation, we call that is a computation of color constancy. When we try to achieve this objective. now to understand the once again that you know how the colours are perceived or colours are produced. What are the components are there when we record the corresponding color signal now we have to consider two types of reflection that is occurring on the object surfaces.

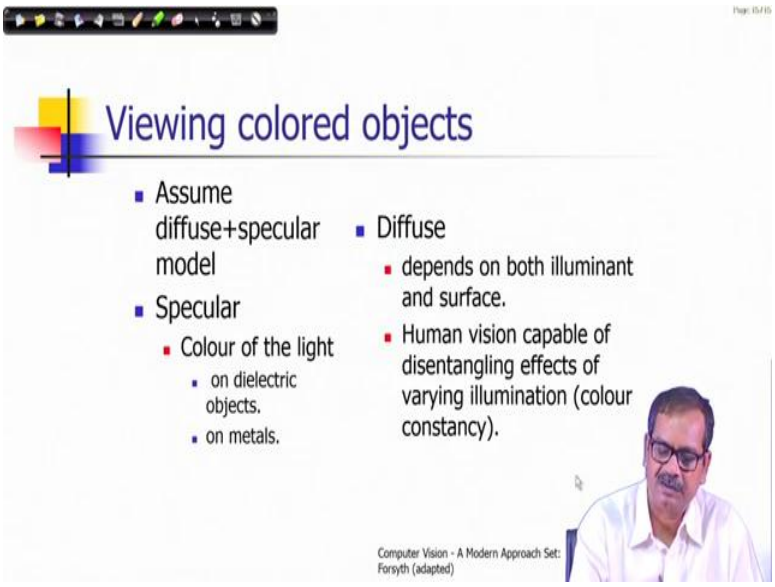
So, there are two components of reflected light, one component is a diffuse component and which is the product of spectral power density and reflectance curve as we have discussed and the other component is a specular component and this specular component is just like a reflection by a shining surface. So, you get almost a same color of the illuminant, it is acting those surface points are acting like a mirror point. So, it is actually providing you the color of the illumination or color of the light sources itself.

So, when the same light falls on a surface of diffused surface you get the color of the surface, when the same light falls on a specular surface or get the specular reflection you get the color of illumination, color of the light source. So, there are the two different

kinds of color we perceive from the same scene. So, in a diffuse reflection as you can see it has been shown that after reflection when the incident energy it is diffused, means it is reflected in all possible directions and this energy with equal energy in an ideal diffuse surface. And that amount of that energy reflected energy depends upon the angle of incidence in this case cosine of the angle of incidence it means proportional to that.

Whereas specular reflection its like a mirror light reflection. So, the reflected energy goes to a particular direction and if your senses are in those directions it will get those you know reflected energy. It is very narrow directions through with this specular reflection takes place almost following like a Newton law of reflection like a mirror reflections. So, when we perceive when we record color, record the intensity values or received energy at a point a weighted some of this two actually were recording.

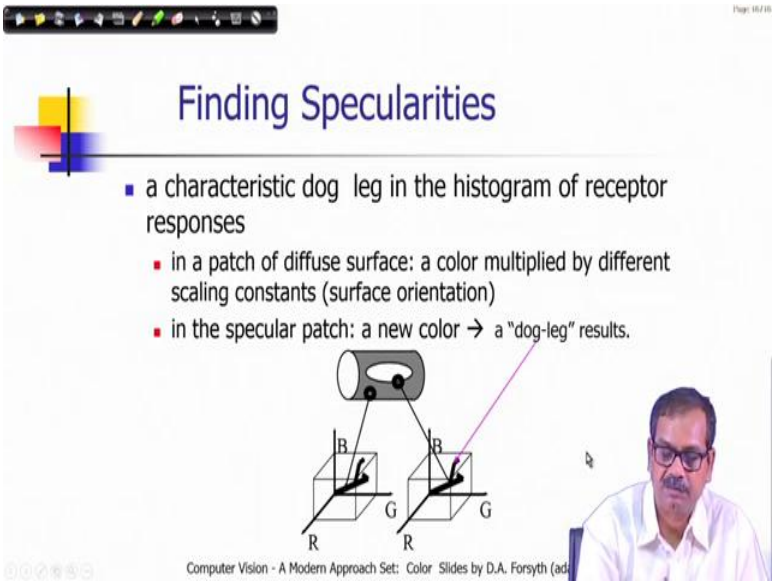
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So, let us assume this module diffuse + specular model. So, as I mentioned it is a kind of summarization of the things what I described in the previous slides. So, far it is case specular reflection you perceive color of the light you record color the light on dielectric objects on metals you can get this kind of reflection. Whereas per diffuse reflection it depends on both illuminant and surface. it is the product of the intensity is received from the illuminant and also the reflectance property of the surface, reflection coefficient of the surface.

So, the fact what I stated also that human vision is capable of adjusting the perception of color in varying illumination and this particular phenomena is known as color constancy.

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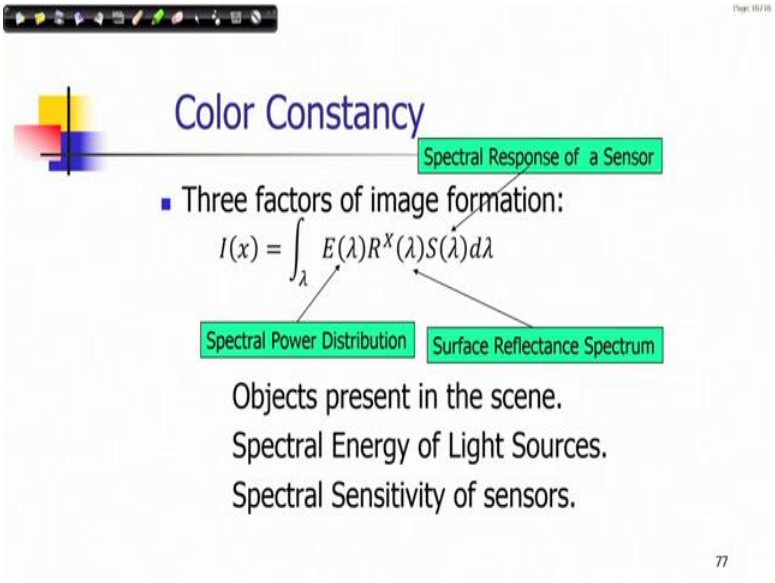
So, we would like to compute or we would like to represent the colors in in terms of illumination in variance representation of color. So, we will have to have this that is what is the color computation of color constancy. Now, we can observe if we plot the points this color points in a RGB space, we can observed a kind of structure a dog legged it is structure as a in the histogram of receptor responses.

So, in RGB space or if I just plot these points in the RGB space we will find out the histogram of receptor or dog legged structure in that plot itself. So, you can see here in this picture we are describing, the presentation of this colours. considered a diffuse surface and so all colours produced by this. diffuse surface. So, this is a diffuse surface all colours they would be more or less of having the similar hue and saturations. So, their color vector would be similar, but only they could be scaled by intensities amount.

So, they will have a in the RGB space, they will have representation of a particular directions, they lie on a linear segment again connecting the origin. But on top of that if there is a shiny surface that is a specular reflection, then this colours will be related to the color of the illuminate. it will give directions along this particular surface and since it is a super position. So, you will get shifted.

So, you will find another directions for some values we will find another points. In the space on top of that and this structure is a kind of dog legged structure. So, we observe this particular structure when you plot this color in a RGB space. So, this is what in a patch of diffused surface, a color multiplied by different scaling constants that we get and the specular patch, a new color and as a result you get this dog legged in this diagram.

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So, now we discuss about computation of color constancy. So, we see that there are three factors of image formations or for producing the sensory information from the surface from the energies of the environment which is reflected from a surface point. So, these factors are: what are the objects present in the scene, then the spectral energy of light source, that is a spectral power distribution and then spectral sensitivity sensitivity of sensors.

So, we can represent this particular thing as you can see that $E(\lambda)$ is representing the distribution spectral power distribution where as $R^{X}(\lambda)$ is showing you the surface point object point x and that is a surface reflectance spectrum and $S(\lambda)$ is the spectral response of a sensor. So, it is the accumulated response over all the wavelengths over the spectrum that that would be the output from this particular sensor and that is what is

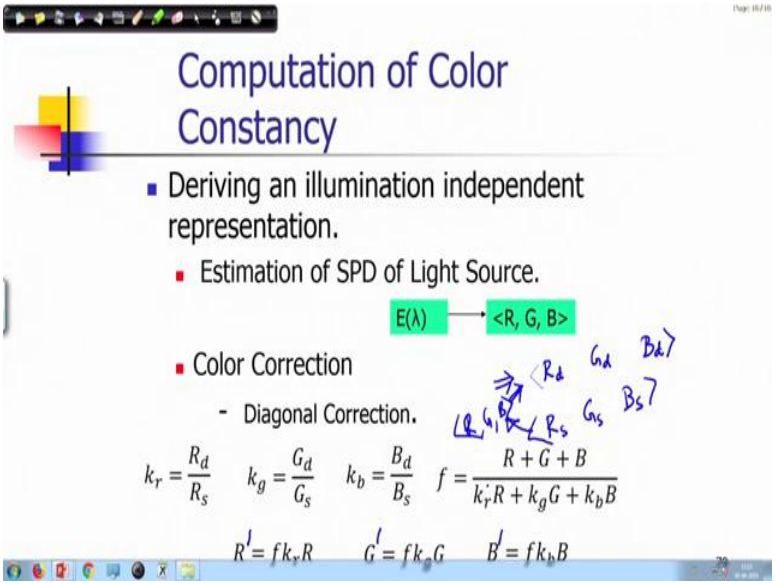
expressed as $I(x)$ So, if there are three different sensors are having three different types of spectral responses you will have three such components. $I(x) = \int_{\lambda} E(\lambda)R^{X}(\lambda)s(\lambda)d\lambda$

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So, that is of the color is represented, it is a same scene captured under different illumination and the problem of color constancy is that : can we transfer colours from one illumination to another.?

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So, this is the problem statement that we need to derive an illumination independent representation of color. And for that what is required? we required to estimate the spectral power density of light source that is the first thing and Then normalize the colors what you get in your as a response of sensors. you should normalize this colours with respect to the spectral power density (SPD) of the light sources.

So, the factor which will be doing that normalization that task is known as color correction and for estimation of spectral power density you need to compute the three vectorial representation of the color of the illuminant. So, there are one method is known as diagonal correction and so this diagonal correction is given in this form. If suppose you have a target color in the color invariant representation, you would like to transfer all the colors as if they are illuminated by this illuminator which has the color equivalent of representation of $R_d G_d B_d$

And, the actual color of the illuminator in the scene or color of the light is given by $R_s G_s B_s$. So, any color what you get say RGB any pixel which is obtained due to illumination how you can transfer you can modify it. So, that same color would be obtained from the illumination of this color vector. So, that is the correction you are going to do.

$$f = \frac{R+G+B}{k_r R + k_g G + k_b B} \quad k_r = \frac{R_d}{R_s} \quad k_g = \frac{G_d}{G_s} \quad k_b = \frac{B_d}{B_s}$$

So, first you find out the proportional factor of these spectral components of or primary components of this color vectors of target illuminator and source illuminator also you compute this factor f it is trying to normalize the intensity values. So, that you can have the same intensity value even after modifying and then your updated color value should be this. So, I should write here updated value.

$$R' = fk_r R \quad G' = fk_g G \quad B' = fk_b B$$

Let me put it R' that is the updated one which is $fk_r R$ then $G' = fk_g G$ and $B' = fk_b B$. So, this is how you can perform this kind of color correction. So, let me take a break here and we will continue this discussion in the next lecture.

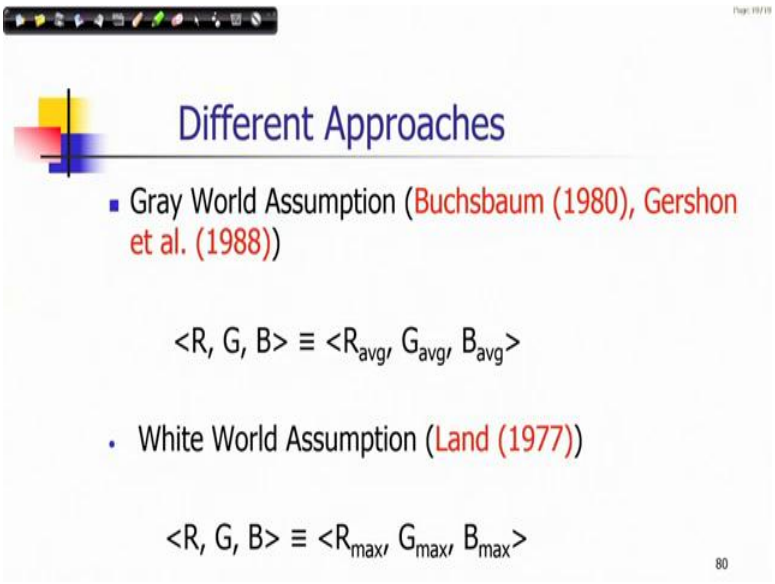
Thank you very much for your listening.

Computer Vision
Prof. Jayanta Mukhopadhyay
Department of Computer Science and Engineering
Indian Institute of Technology, Kharagpur

Lecture - 38
Color Fundamentals and Processing (Part V)

We are discussing about computation of color constancy and the objective is to get color representation which would be invariant to the color of the illuminant.

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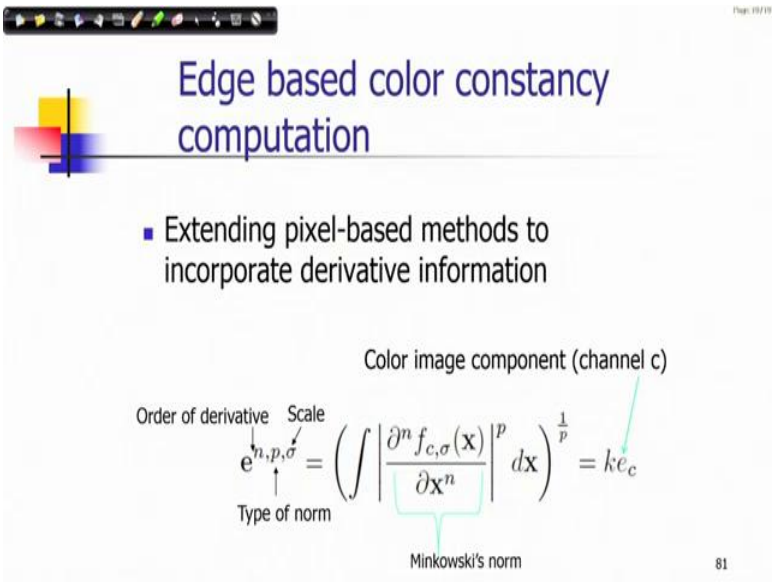
And for that there are two tasks that we have been identified we have identified; one task is to estimate the color of the illuminant. the next task is to transform the colors from the source color image to a target color image where there is a target illumination is desired I mean it should be color illuminated by a target illuminator. So, now, we will be discussing different approaches of estimating color of the illuminant.

So, you consider here that there is one technique which assumes that the world is gray. Which means that in the red green blue all these components if I average all the color then actually that color of the illuminant would be gray. So, this particular part we can describe in terms of red, green, blue.
If I take the average of the red green blue components this one is giving you the color of the illuminant. $\langle R, \quad G, \quad B \rangle \equiv \langle R_{avg}, \quad G_{avg}, \quad B_{avg} \rangle$

So, this is that types technique is a very simple technique you have to perform simply averaging of the color vectors and you get the estimation of the color of the illuminant. The other assumption is called white world assumption, so in this case the assumption is that the color of the illuminant will be given by the maximum of these components of red, green, blue. $\langle R, \quad G, \quad B \rangle \equiv \langle R_{\max}, \quad G_{\max}, \quad B_{\max} \rangle$

The color is almost like white and you just consider the maximum of their red green blue component. And here you need to just compute the maximum of red components of all the pixels, maximum of green components of all the pixels, and maximum of blue components of all the pixels.

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So, these two are very simple techniques, there are other techniques which use a more involved competitions and assumptions are bit different here. Here it is considered that the pixels which are lying on the edges, color of the illuminant they are more information of the color of the illuminant.

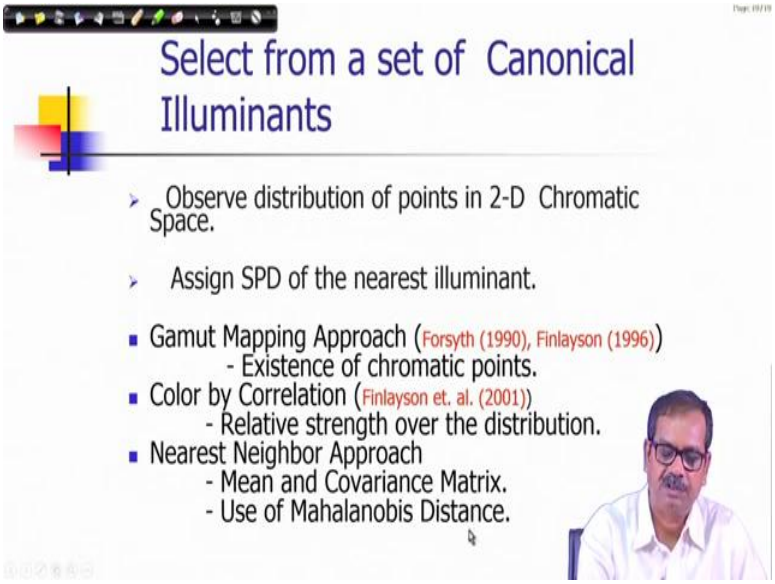
So, the idea is that when colors are reflected by the boundaries of objects, you get kind of a specularity and as we discussed that specular reflection contains the color of the illuminant. So, from there itself we are trying to estimate the color of the illuminant. So, one of these method that is shown here, it extends pixel based methods to incorporate derivative information.

That means, again you are trying to accumulate all such pixels intensity values or their derivatives of those intensity values. And then this aggregated form itself will give you the proportional color representations in the three vectorial (Refer Time: 03:25). So, you just consider these particular computations.

And here you can see that we are considering n-th order derivative of a intensity value which is represented by the function $f(x)$ and c is the channel I think I have. So, n is the order of the derivative, then p is a type of norm as you can see this is like an L^p norm. And this is also called Minkowski's norm in the continuous space. And σ is the scale; that means, it is smoothen by a Gaussian mask and then you are performing this derivative. And then you compute this particular competitions of L^p norm or Minkowski's norm. $e^{n,p,\sigma} = \left(\int \left| \frac{\partial^n f_{c,\sigma}(x)}{\partial x^n} \right|^p dx \right)^{\frac{1}{p}} = ke_c$

And for that particular channel see, so if you are doing it for red component for example, all the pixels then you will get a value which is proportional representation of the illuminant for that particular channel. And you carry out this competitions for every channel like green and blue you will get three such values and their proportional values will represent the color of the illuminant. And you can adjust your particular intensity of that illuminant and transfer the color using the color correction as I discussed in the in previous slides in the previous lecture.

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The other kind of approach could be that instead of computing the colors or accumulating the responses of the perceived specularities in pixel reflections. What we can do? We can consider something some kind of data driven approach; that means, we have some models of different canonical illuminants in terms of their RGB distribution. Distribution in a color space, it could be RGB space, it could be xy space.

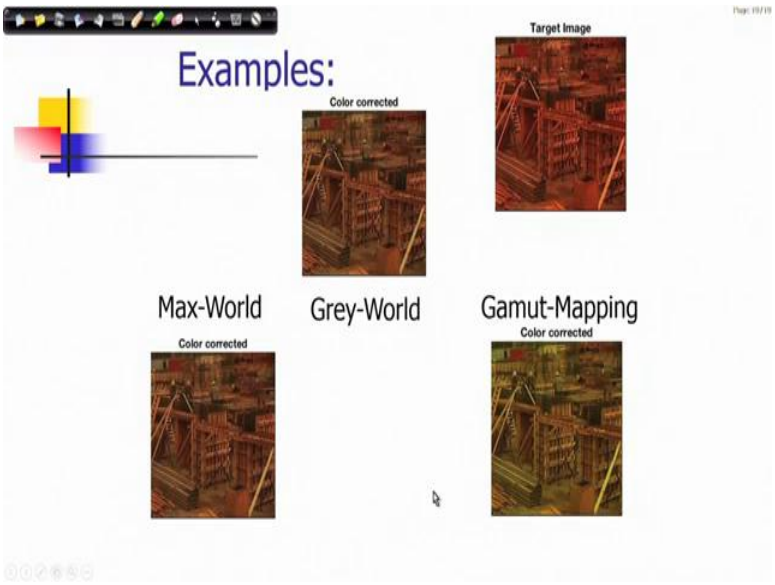
And then given an image we try to find out what kind of distribution is there in that space. So, the similarity between these two distributions can indicate that whether that corresponding scene has been illuminated by that given illuminant whose distribution is known. So, if I have a set of such illuminants we would try to consider the illumination with the nearest of them.

The closest approximation of the distribution what we get we try to attach that illumination for that scene in that way we can select is from a set of canonical illuminant. So, it is observe distribution of points in 2D chromatic space even you can do it in 3 D chromatic space also and; that means, including intensities; then you can assign the spectral power density of the nearest illuminant to them.

So, there are approaches like gamut mapping approaches, so as we have discussed that how xy chromaticity chart gives you a gamut. That means, the distribution of points in this particular space which should be lying in a triangle and from there you can compare the that distribution with the distribution of the canonical illuminants.

So, existence of chromatic points is important, color by correlation is another approach. So, it depends upon you know what kind of similarity, how do you compute this similarity, so it is relative strength over the distribution that is considered. You can use also nearest neighbour approaches like, you can use mean and covariance matrix of those distributions and some distance functions like mahalanobis distance function you can use. So, in this way you can select one of this canonical illuminants.

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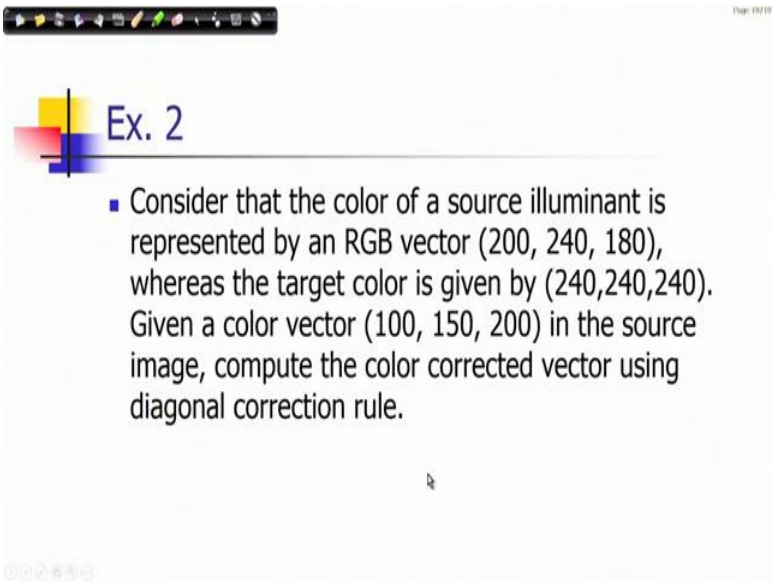


Some examples of these techniques, consider this is a target image and which means this image is illuminated as you can see there is a reddish hue in the illuminant. And that is why their color is more tending towards red. But, for if I perform color corrections and first you estimate the color of the illuminant using grey world assumption.

That means you consider averaging of the red component, green component and blue component use it as the color of this image. And if I use the target illuminant as simple white light like with the intensity value say (255, 255, 255). And then perform color correction following diagonal correction rule then you will get this image.

So, this is for grey world method, for max world method you get this kind of image and gamut mapping will give you this kind of image. I have not described in details gamut mapping you can go through some of the papers. And in our course or in this particular respect we will be considering simple approaches of estimating colors using say grey world assumption, max world assumption etc.

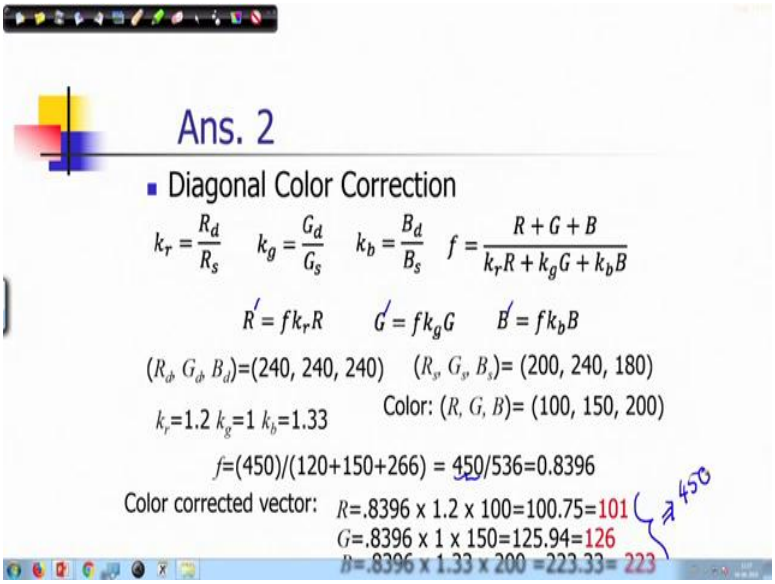
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So, let me discuss a problem here to let me solve a problem just to show you that how color corrections could be performed using source illumination and target illumination. So, let us consider this exercise and then we can understand the computational steps better. So, consider that color of resource illuminant is represented by an RGB vector (200, 240, 180), which means 200 is red component, 240 is a green component and 180 is the blue component.

Whereas the target color is given by (240, 240, 240) in my case as you can see in this example it is trying to get a white illuminant by with this combination that is the target. And given a color victor (100, 150, 200) in the source image compute the color corrected vector using diagonal correction rule, so this is a problem statement.

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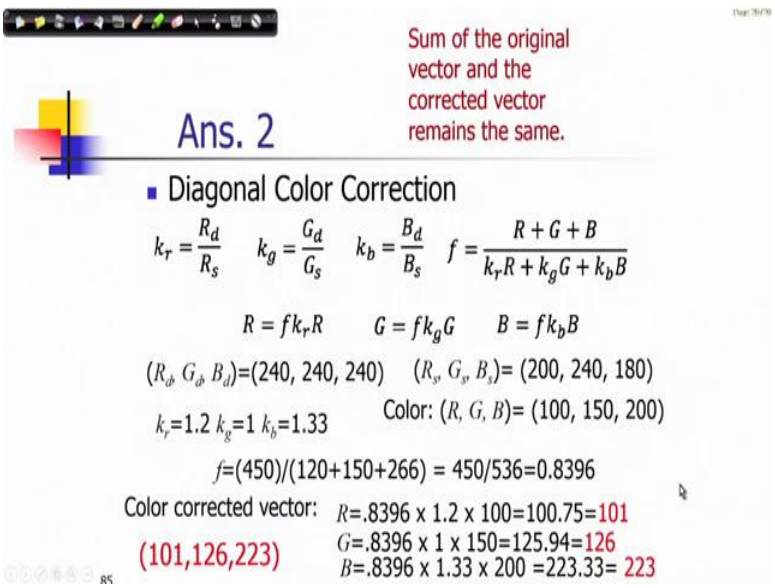
So, let me see how we can perform this competition it is quite simple and quite direct if you know this expressions as I mentioned. We need to first compute the proportional factors of target and source colors of the illuminants for each primary components like k_r, k_g, k_b you need to compute. And also you need to compute the corresponding factor for normalizing the colors. So, their intensity value remains still the same.

So, and then the expressions, so a modified colors are given in this form since they are modified I am using once again this notations. And if I perform this computations what is given you can see (R_d, G_d, B_d) is given and the component of source is also given. So, if you component this proportional factors by using this information those values are shown here. Then this color has to be converted into a form as if it is illuminated by the source given by (R_d, G_d, B_d) that means, (240, 240, 240).

So, the completion of f is this is the factor which is computed here and so color corrected vector as per these expressions R. You can see R becomes 101, it was earlier 100, now after modification, it is 101, green becomes 126 and blue becomes 223, so these are the modified color corrected values. So, you should note that the intensity in this particular computation if I add $(R+G+B)$ if I add these values and that is still coming the same which is actually with this value is 450.

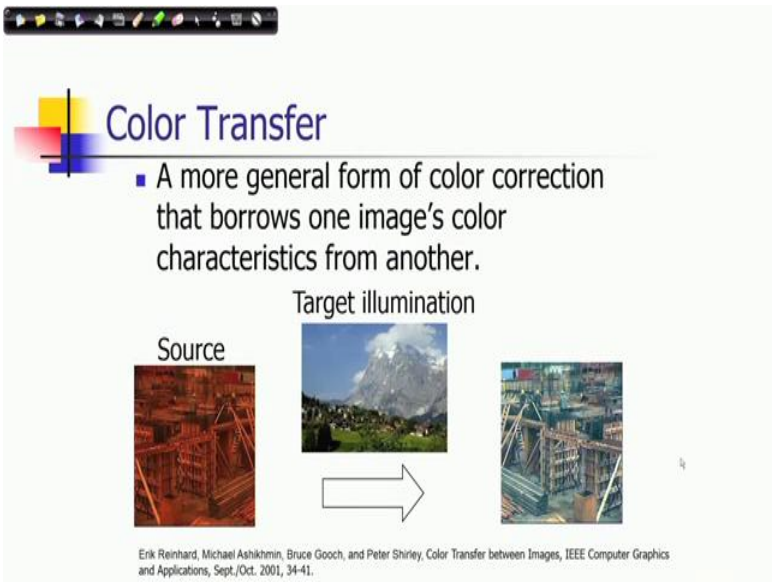
So, if I add 101, 126, 223, you will still get 450, so in this correction you are not changing the intensity values of the modified pixels. but what you are changing you are changing the relative proportion of three primary components because it is corrected for the target illuminant, so that is how we perform the color correction.

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So, this is the final answer in this problem and as I mentioned some of the original vector and corrected vector remains the same.

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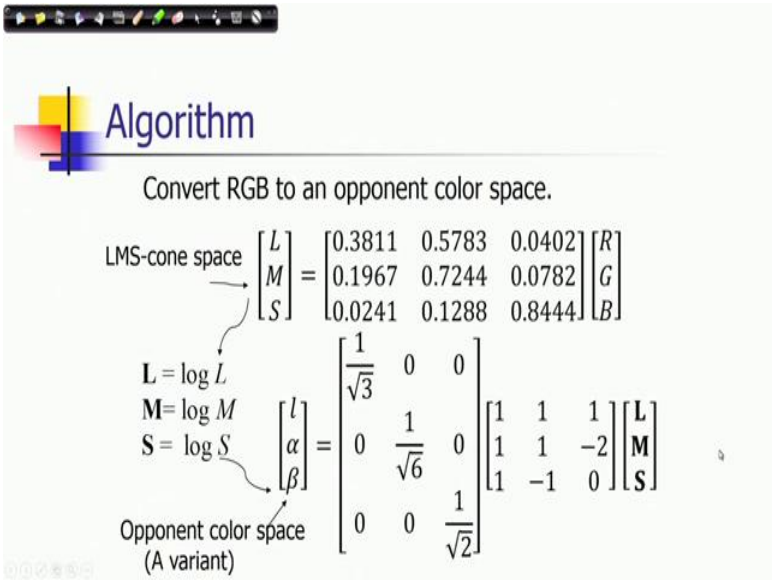
We will discuss about a more general form of color correction and that operation is called color transfer. In this operation what it does that there is an source image which has its own color distribution and there is another image which defines a target color distribution. So, the source image would like to borrow their target distribution in its own rendition.

So, an example can be cited here to explain this particular operation, consider this is an image which is that source image. And you can see that source image has an illumination which is a bit reddish illumination and all the objects inside it they are affected by that particular illumination.

Consider another image which is a completely different image having different types of objects, different scenario, different context, but it defines a an illumination or defines a target illumination, or target color distribution which is depicted by this target image. So, it is an image of alps, a peak of alps and, so this distribution should be transferred to the content to the objects of source image to the display of source image, so that it looks like this.

So, your image under this transferred illumination model or if I said with the transferred color characteristics that would look like this. So, here you can see the colors are quite different and distribution is quite akin to the target illumination of the as if they are illuminated by the illumination of target image.

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So, let us discuss about the processing of this particular operation. So, what we do here we process all this pixels in a different color space where we can separate out the chromatic components from luminance component. So, in this particular technique this is the way this processing has been done. first, RGB is the color space where the color values of the pixels are captured.

So, there convert it into another space which is called LMS cone space, so it is trying to model our retinal sensitivity of different cones. So, this model trying to provide as if the sensations which will be generated in our cones. And that is what it is trying to capture with this mathematical model, you can see it is a linear transformation.

And just to remind you that L stands for long wavelength, M stands for medium wavelength and S stands for short wavelength which means they are corresponding to red, green and blue zones of the spectral domain. Now, once you get this LMS value you further process it you take it to their log domain. Because, in some of the perceptual models it is considered that our perception is more it follows the logarithmic, it is proportional to the logarithmic value of the receipt sensory unit sensory single or receipt energy.

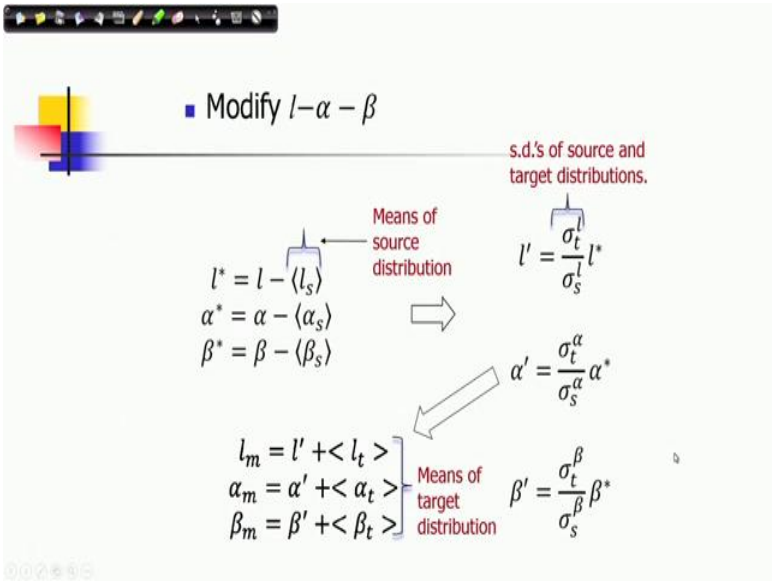
So, that kind of heuristics have been used here. we use the logarithmic values and use this logarithmic values to separate out the chroma components for the luminance components. So, you can see once again another transformation which is also a linear transformations in the logarithmic with the logarithmic values of the cone responses that we have considered in all model.

We call it LMS once again, but their font has been shown in a bold font and you can see the color space which has been given by $\begin{bmatrix} l \\ \alpha \\ \beta \end{bmatrix}$. These are the three components which is similar to an opponent color space model you can see particularly in this row. So, this is corresponding to the intensity component which is the addition of L, M and S and this one it is $(L + M - 2S)$, so L corresponds to red, M corresponds to green.

So, it is like yellow minus blue that kind of model opponent color model and this is $L - M$ which is red minus green opponent color model which we have already discussed

earlier. So, it is a variant of this particular operation processing this particular opponent color space, so we perform this kind of separation. Now, in the next ; we will be doing will be performing certain processing over the chromatic components. So, that you can transfer the distribution of colors in the target image to the source image, let us see how we can do it.

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So, for modifying the chromatic component and also the luminance component, first what we can do that, we subtract the mean of all those components from the source image. So, you can see here this symbols l_s , α_s , β_s they denote the corresponding l, α, β components of source image. And then you are taking their mean which is denoted by this angle bracket operations I mean this is the symbol just.

So, your subtracting those means and then up the subtracted values they are scaled up, so that the standard deviation of the target distribution can be transferred. I mean your source distribution also should follow the standard should be similar to this variances or standard divisions of the target distribution. So, you perform a proportional scaling- a scaling proportion in proportion to their ratios of standard deviations that could make these operations. $l' = \frac{\sigma_t^l}{\sigma_s^l} l^*$, $\alpha' = \frac{\sigma_t^\alpha}{\sigma_s^\alpha} \alpha^*$, $\beta' = \frac{\sigma_t^\beta}{\sigma_s^\beta} \beta^*$

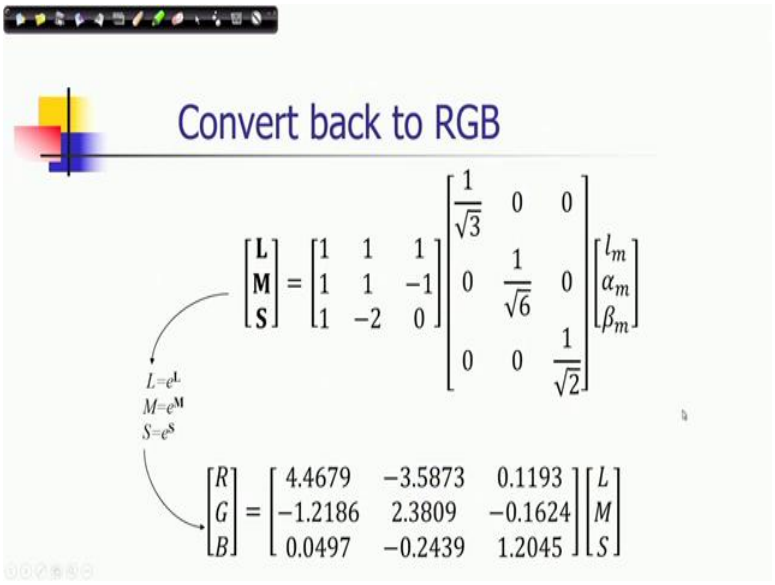
So, you can see here, this is the standard deviation of the luminance component of the target image. And this is a standard deviation of the source component of the luminance component of the source image. So, if I multiply the modified luminance value with this scale with this ratio or with this proportional factors. Then actually your making the standard deviation of the modified distribution is the same as the standard deviation of the target distribution.

So, that is what your performing by doing this particular operations and you are doing it for every component in the same way. So, the corresponding ratios of the standard deviations in those components in both target and source those are multiplied with the modified values. So, this can transferred the standard deviation operation also this is what it is just defines the corresponding symbols as I mention that they are the standard deviation of source and target distributions and defining those ratios.

So, once you have performed this then you have to also add the mean of the target distributions components which means for the luminance components, α component, and β component. So, in this way in the transferred domain, transferred color space or transferred colored values of source image in that distribution you have transferred the mean.

So, that distribution we have the mean which is equal to the mean of the target illumination, target distribution and also it will have the standard deviation of the target distributions. And that is in the $l-\alpha-\beta$ space. And now the rest of the job should be that you perform the inverse transformation to take all these components back to the RGB domain. So, for that what you should do; you should follow the inverse operations.

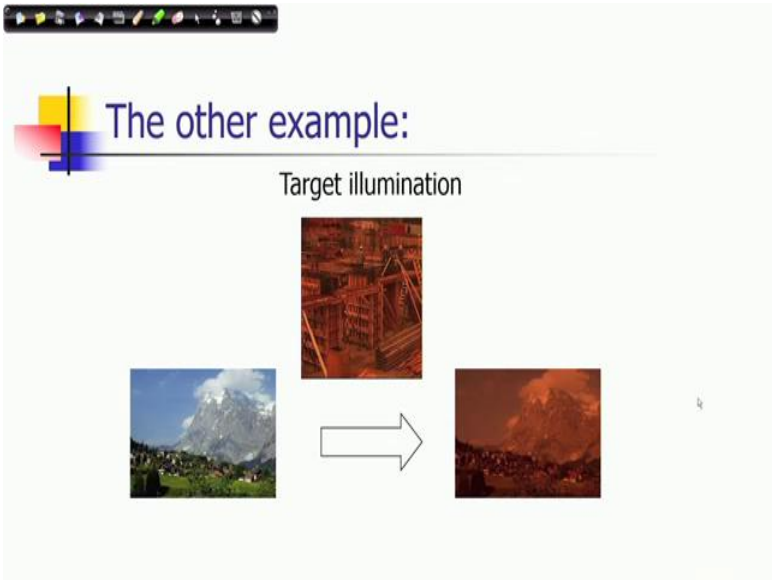
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That means you can see that after modifying the l, α, β components again you are perform you are taking those values into the this place **L M S.** But this is a logarithmic space of L M S cone color space, so this is a logarithmic of L M S cone color values. So, far from there again to you have to transfer to the original L M S cones color space.

So, you perform the exponentiation operations over those logarithmic values and perform once again the inverse operations to transfer from L M S to the R G B value. So, in this way you get the color transferred image from the source image using the characteristics or using the distribution of the target image. So, let me give you the other example of this operation.

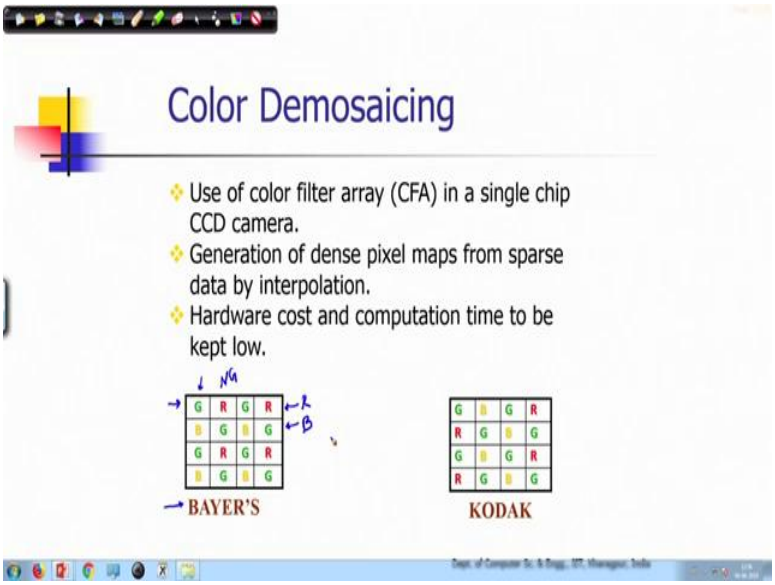
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In this case we will be considering we will show off the source and target images and we will see its effect which means your target illumination will be defined by the image which has that reddish illumination in the environment. And your source image will be the image which has been captured in broad day light in a every sunny day. And it is a once again its content it has been shown its a mountains peak which is very prominent and there is a valley, green valley which has been also displayed. So, under the color transfer operation let us see how it looks.

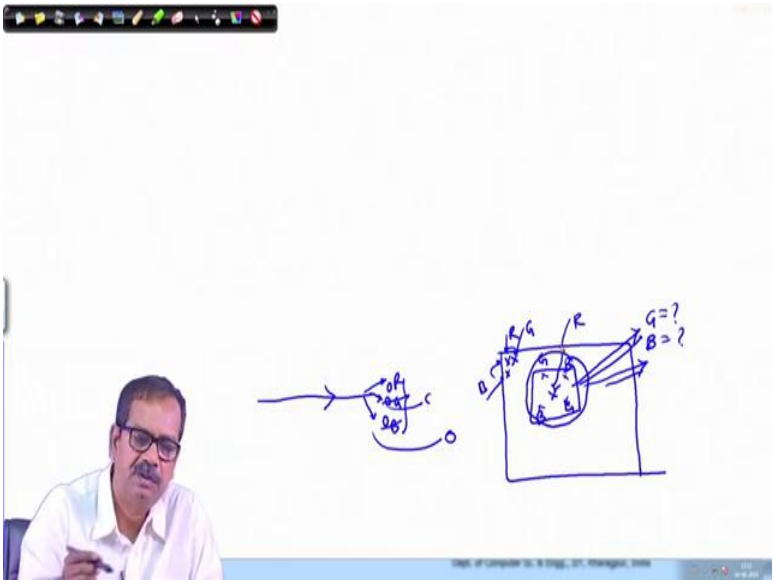
So, we can see actually the scene is perceived that it has been captured in some you know reddish under some reddish illumination which for natural scenery may considered that it is bit unnatural in that sense it should not be, so much of reddish. But that is what the color transfer operation did, but it would be near you can say sunset kind of operations. So, this is just the imaginations through which you can perceive this kind of image.

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So, next type of processing that we are going to discuss that is the color demosaicing operation, so in this problem we use a color filter array. So, first let me explain the principle of the single chip camera where actually we know that for a camera we required three types of filters. For a normal camera when we have full color reception which means you have a color signal, suppose there is a color signal and then you have to receive the signal in the form of a sensor.

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So, same signal should be received by three different sensors, say there are red, green, and blue and there are different techniques by which the same optical signal is guided to this or they are divided into three different parts. And then they are sensed by three different sensors independently. But it has to be the same coherent source and that coherency has to be maintained to represent the color of a particular object which should be received by the same signal.

Now, this kind of technology is quite complex and there are different ways people do it where they have that optical divider and like using prisms etc.. And the manufacturing of this kind of camera is requires high technological innovations and it is quite costly. And since you are using for each pixel three different sensors, we call it three chip colored camera or three chip sensors.

But there is a cost effective technology for sensing color images where instead of three sensors, I will be using only one of them. That means, I can use either red, I will be only sensing red or I will be sensing sometimes green or sometimes blue. But whenever I am sensing the signal whenever the camera is sensing this any signal of a coming from a particular point, it only captures only one component, but what it does, it does in an interleaved fashion. So, which means that in an image suppose from the image perspective if I record the corresponding received energies from different color channels.

At this point I will have only red whereas, next point I will have say green and say this point is blue. So, you can see that I will be losing my spatial resolutions, but if I assume my spectral resolution is much course; that means, I can associate this blue also to this pixel and this green to this pixel. So, I can consider at all this pixels they have the same R G B values, so this is a very crude way of estimating the colors.

So; that means, assigning colors of each pixels all the full color information estimating full color information's along this pixels. As you can see it is a very crude, because I am simply transferring the values of missing components to the neighboring pixels by considering the spatial correlation of the spectral channels.

Now, this problem itself I have just given you one particular solution, but there could be many solution it is an interpolation problem which means I have the missing components and missing components at this point. So, for this pixels say for example, for this pixels

if it is I am recording only red, so my green and blue they are missing. but in the neighborhood I have some pixels where those values are there some are green, some are blue say some are green, some are blue.

So, even I have also other red also, so considering this neighborhood pattern or this neighborhood information how best I could estimate this missing components that is the problem of color interpolation. And when we sense the image in this interleaved fashion of one of the spectral components we call that kind of image as a color filter array. So, let me go back to my slides, so that to summarize this particular computation problem and this kind of computation is known as color demosaicing or color interpolation.

The term demosaicing came as if the pattern interleaved pattern is called as a mosaic pattern with different color components. So, here we can see that in this problem you are using color filter array of a single chip CCD camera, from there you are generating a dense pixel maps of all the color components. So, they have sparse data as if as you have only one component at every point, but you would like to get all the color components for every point.

And as the motivation as I mentioned the technologically it is simpler to design and manufacture this kind of cameras. Because you know the spectral coherency of the reception to ensuring that coherency is not required here. So, we have you can use simply the filters in front of the images in front of your lens color filters which are in patterns. And for each pixel you will be receiving only energy in that spectral band.

So, there are different patterns are proposed by which you can design this filters that is why the array also is called color filter array. Because those are sensed by incident light through this filters. So, this example of this patterns are shown here, you can see in this pattern which is called Bayer's pattern and which is a very popular pattern in most of the literatures you will find people have worked on this pattern. And you can see that in this pattern what happens that green, red green red, so they are interleaved in one row.

In another row blue, green, blue, green they are interleaved, so the fact is that green has been sampled in the color filter array twice the red than red and blue. So, this is one consideration is that green anywhere more sensitive to the green channel. So, lots of

information of green channel will affect more than the other two components that is why we have this kind of sampling pattern.

But this is not the only pattern we can have other kind of patterns for example, there is a pattern adopted by Kodak , famous Kodak cameras. So, there you have this patterned like in the same row you have green, blue and reds on the same row this patterns are interleaved. Whereas, Bayer's pattern now we can also identify as if this is the red row because green is sampled in every row, but in one row when red pixels are sampled then the next row the blues pixels are sample.

So, rows can be you know denoted by for the purpose of describing this pattern or computations we can save a pixel in the red row or blue row and whereas, columns could be green and non green, so either it is red or blue. So, in this way you can identify a location in the Bayer's pattern; you should note also the starting pixel locations and the starting row, starting column and starting rows state. If I can provide you that and given this particular rule then I can describe the all the patterns.

So, in the color demosaicing problem, it is the pattern which is which remains a fixed, so you know already that given a color filter array for each pixel which is the spectral component of that particular image. So, for any every pixel I know that which spectral component it is, whether it is green, red and blue. And by following this pattern if I can tell you one particular pixel the rest of the pixel's spectral components can be determined or described.

So, this is the advantage when you use this kind of regular pattern when you describe a color filter array. So, our objective here is to interpolate the pixel values in such a way that you get all the full color components; that means, all three components at every pixel and that is what is a computation of color demosaicing. So, let me stop here and we will continue this discussion in the next lecture we will discuss about different algorithms for performing this color demosaicing. So, let us give a break at this point.

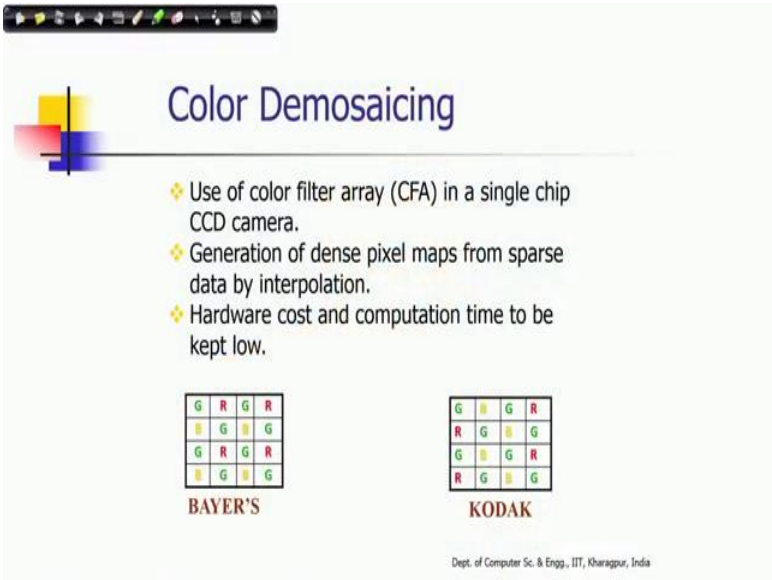
Thank you very much for your attention.

Computer Vision
Prof. Jayanta Mukhopadhyay
Department of Computer Science and Engineering
Indian Institute of Technology, Kharagpur

Lecture - 39
Color Fundamentals and Processing (Part VI)

We continue our discussion on color demosaicing and in the last lecture I have defined what is the problem of color interpolation.

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So, in this problem you are given a color filter array where there is a pattern by which you are sensing the colored image. So, in this pattern at every pixel only one spectral component is captured. So, the information of only one spectral component or spectral channel that is available. For example, say at this pixel only green value is available in its neighboring pixels red and blue are available.

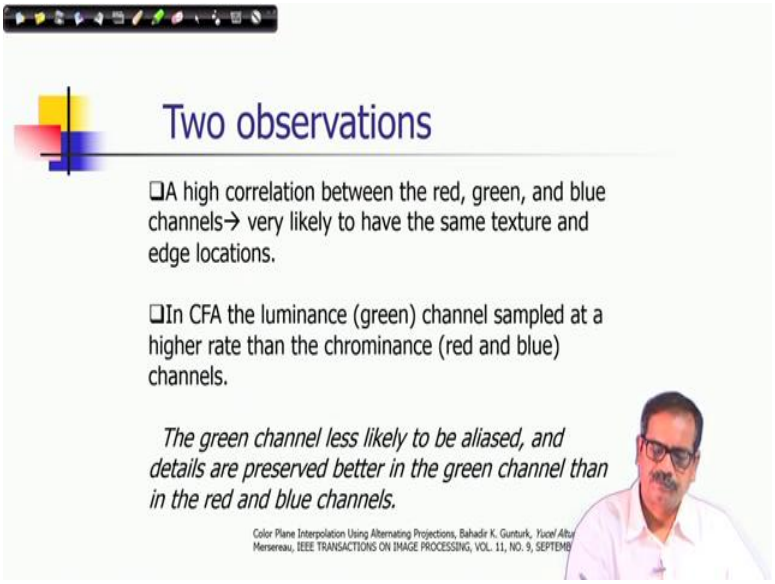
So, the problem of interpolation is to compute the missing components at every pixel by taking care of its neighboring distributions. So, we will be discussing different methodologies for that and in particular here in the given problem what is already provided that the interleaved pattern of channels that is standardized with respect to the pixel coordinates. So, at every pixel of CFA I know that, at that coordinate locations or at that array location which spectral component is available.

So, you should note that this could be this positions could be encoded by two different two indexing schemes; I mean, two index by this kind of indexing schemes you can observe that in this interleaved pattern one row has the alternations of green and red pixels, green and red values and in the next row alternately blue and greens are available. So, we can consider the row where red samples are available along with green that is red row and the row where blue samples are available along with green that is blue row where as the column can be denoted as either green or non green. then the column positions are either green or non green.

So, in this way in an alternate in a periodic fashion we can determine for any pixel locations in the CFA in which type of row it is lying and which type of column it is lying and from that information itself we can find out that which spectral component is available there.

So, for example, if the position is a red row and green column, then it is the green pixel location. So, any column which is green in a position, there will be a green pixel locations will be available where as if it is blue row and non green column then say this is one example then the blue component is available. So, with this kind of background now let me discuss different methodologies of interpolation.

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So, there are two key observations that we should make here; one thing is that there is a high correlation between the red green and blue channels. And which means that if you

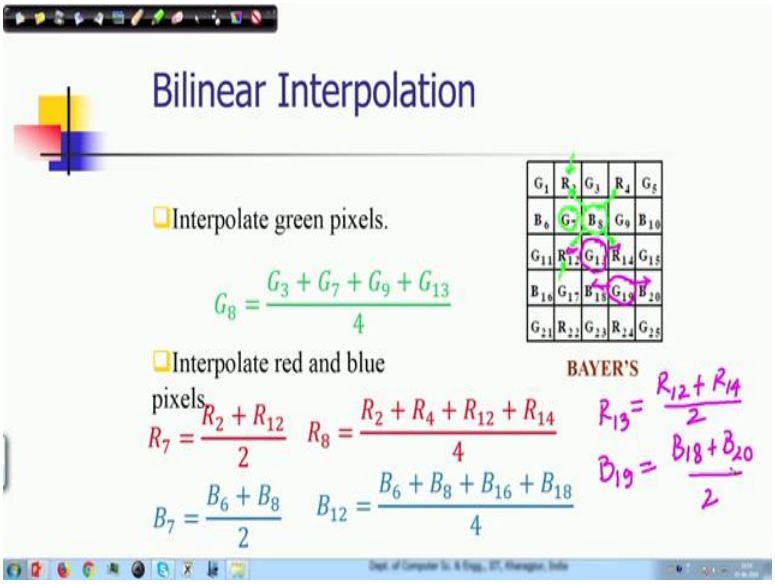
can if you have the informations of any of these channels, the other channel information also you can exploit this correlation to derive that information. They should very likely to have the same texture and edge locations particularly.

So, particularly the gradient estimations, cross channel gradient estimation is useful in this respect we will elaborate when we discuss different methodologies. The other observations from the CFA arrangement I mentioned these things in the previous lecture also that the green channels they are sampled at a higher rate and particularly in Bayer's pattern their sampled at twice the rate of red and blue channels.

So, usually green channel represents the intensity values they are more akin to representing intensity of values and the chromatic information is more embedded on red and blue occurs green also has its part there. So, that is why you know sometimes in the in the jargon of color interpolation green is considered synonymous to luminance channels. So, if you want to estimate luminance first we interpolate green and use that value kind of a luminance value in subsequent processing.

So, these are some of the advantages since you are having higher sampling rate in green channel because we are very sensitive to the green component of the color our vision system is very sensitive to that component. So, you would like to have higher quality of reconstruction of green channel and that is why there is it sampling rate is high. Andthat is why it should be less likely to be aliased and its details are preserved better in this channel than the other two chroma channels.

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So, first we will be discussing about very simple technique of bilinear interpolation since this is a two dimensional distribution. So, we will be considering interpolations in both the directions. So, that is why linear interpolation in both the directions and this is quite simple and direct as we can see here.

For example if you want to interpolate the green pixels let us observe that what kind of arrangement we have. So, when you are going to interpolate green pixels means, these information we need to find out in a location where green samples are not available. For example, a location like this which is shown here as say B_8 or a location say like this which is shown here as R_{14}. So, these are the locations where you need to compute the missing green component.

Now, you can observe that the neighboring information of green samples are available in the form of four neighbors. So, in the discrete rate we call this neighborhood definition; 4 neighborhood definition is quite straight forward. Suppose you have a pixel location and in a discrete rate say this is a pixel location and if I denote this location as say (x, y) then this neighboring pixels, these are the 4 neighbors of this grid or this pixel.

So, which means this co ordinate is say array indexing should be $(x+1, y)$ this should $(x, y+1)$ this is $(x-1, y)$ and this one is $(x, y-1)$. So, if I considered for example, computation of green at this location I can see these are the 4 neighbors.

Say this one, this one, this pixel and this pixel is 4 neighbors. So, what I can do I can take simple average of this pixels. So, that is how we interpolate green because everywhere wherever green sample is missing you will always find that in the 4 neighbors of that sample there are two green values are available. So, you simply take the average.

So, you can see this is the example of the average of 4 neighbors in the pixel location B_8 where you have the green sample missing. So, this is the location and you are considering this then G_7 , then G_9 and G_{13} and you are dividing it by 4, the sum of them you divide by 4, then you get the average.

Similarly, when you are going to interpolate the red and blue pixels there also you should consider only the neighboring samples which contain true pixel values of the corresponding channel which means that suppose I have a pixel where say red component is missing. So, there are two types of pixels the red component is missing in a pixel while green sample is available and red component is missing in a pixel where blue sample is available. So, these are the two scenarios say this is also another case.

So, in this green sample particularly note this location. So, here you observe that these are the red samples, two red samples are available. there is no other red samples in its neighborhood. So, you simply take the average of this two. So, what you need to do? You need to identify the type of pixel in a CFA that be which type of row and which type of column it is.

For example, this is a blue row and green column that is how you designate this pixel positions and for that case you have to consider the top neighbor and bottom neighbor and then you take the average. Where as if it is in red row and green column then you have to consider this one and this one; that means the left neighbor and the right neighbor and take the average.

If it is a blue column and blue row; that means, this is a scenario then you should consider in addition all this diagonal neighbors of this pixel and you take the average. So, these are the three cases which are available for interpolating red similarly you carry out the same operations for blue. So, what we get let us see these expressions.

$$R_7 = \frac{R_2 + R_{12}}{2} \quad R_8 = \frac{R_2 + R_4 + R_{12} + R_{14}}{4}$$

So, as I mentioned that if I considered the location G_7 say this is the location G_7 . So, this is a blue row and green column location. So, you get the top neighbor and bottom neighbor R_2 and R_{12} and you divide it by 2 that is averaging of those two values.

Similarly, you consider this case. So, here you have R_8 it is just the location beside the previous one this is the location and now the red samples are available in its diagonal neighbors. So, you can see that R_2 , R_4 , R_{12} and R_{14} and then you take the average. So, this is a second condition for interpolating red second case for interpolating red.

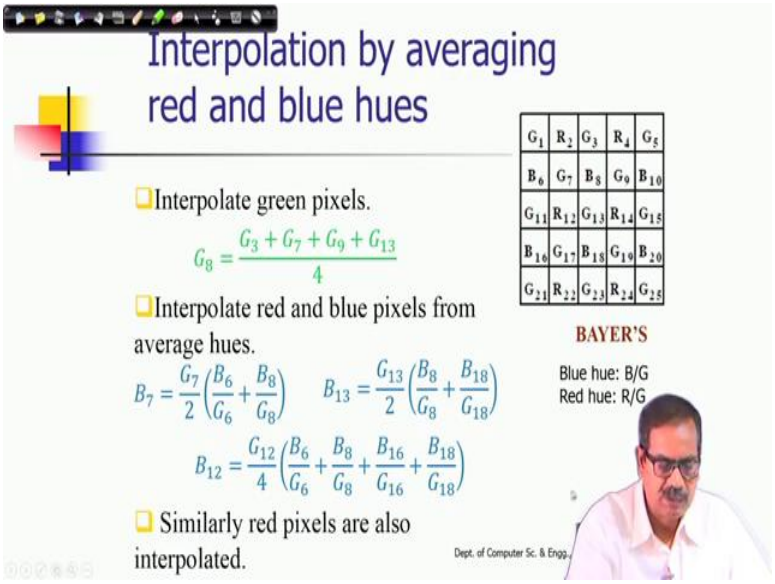
$$B_7 = \frac{B_6 + B_8}{2} \quad B_{12} = \frac{B_6 + B_8 + B_{16} + B_{18}}{4}$$

The third case as I mentioned, it could be blue row and blue column oh sorry this is in the same way you are doing B_7 and B_{12} , but no the other case could be as I was mentioning .these are the only two cases of red. No other case is missing here. So, which is not shown let me tell you. So, it is basically red row and green column. So, you consider this particular location. So, I should write missing component red and that location indexing the convention whatever we are following we call it R_{13} then I should take the average of left neighbor and right neighbor. $R_{13} = \frac{R_{12} + R_{14}}{2}$

And for the blue cases also, it should be the green column and blue row. So, which is this one, one example is this and I can write as say B_{19} that is equal to this is just for typical example we are use those using the indexes of those pixels. Anywhere where you satisfies this property of characterization of locations of the pattern what you mentioned that is the type of row and type of column you should apply the same technique.

So, in this case you have again left neighbor left blue neighbor and right blue neighbor. So, that should be $\frac{B_{18} + B_{20}}{2}$. So, this is how bilinear interpolation is carried out.

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So, next we will be discussing about a technique where we would like to exploit the correlation between green and blue and green and red samples. So, this technique what it is considering that at every point you can estimate hue and you are averaging over the corresponding hues.

first hue interpolate green pixels using the simple bilinear interpolation. So, in that case at every pixel you have all the green values. So, all the pixels wherever green samples were missing through bilinear interpolation you have computed it. So, once you have done it then actually you are able to compute this blue hue and red hue correspondingly in respective positions wherever blue sample is available, then you can compute blue hue wherever red sample is available you can compute red hue.

So, while performing interpolating blue sample at a pixel where blue sample is missing you consider its neighboring blue hues, then average it and then you multiply with the respective green value at that pixel that would give you the corresponding blue sample. So, this is a modifications it requires, similarly you do it for red hue. So, it is nothing, but bilinear interpolations of hues once you have computed green and then you convert it to the pixel value by multiplying the respective hue with the green channel, green component.

So, let me describe it further. So, as you can see that interpolation of green sample is the same as it is done in bilinear interpolation technique. You are simply averaging out the

the 4 neighbors of every non green pixel locations where 4 neighbors are always green neighbors that we have we can observe from this Bayer's pattern.

So, now we will be discussing about the interpolation of red and blue pixels. So, there you can find out this situation's; so, will be considering interpolation of blue pixels in this case. So, as you can see here that the location where you are interpolating that is this(G_7) location. Now, in this location you have the green sample that is available. So, its neighbors and it is a blue row; so it is left neighbor and right neighbors the two blue samples are available. Since you have already interpolated green, so we have the estimation of G_6 and estimation of G_8 these are available.

$$B_7 = \frac{G_7}{2}\left(\frac{B_6}{G_6} + \frac{B_8}{G_8}\right) \qquad B_{13} = \frac{G_{13}}{2}\left(\frac{B_8}{G_8} + \frac{B_{18}}{G_{18}}\right)$$

So, I can compute the blue hue at in those neighboring locations and you take the average of them. So, you divide it by two and then multiply with G_7 that is how you will get the blue value at that location. So, let us find out what you are doing in the other locations. So, in the location B_{13} as you can see here. So, this is the location where blue is missing and we need to compute the blue component, once again instead of left and right neighbor we have to consider top and bottom neighbors because this is in the red row and green column.

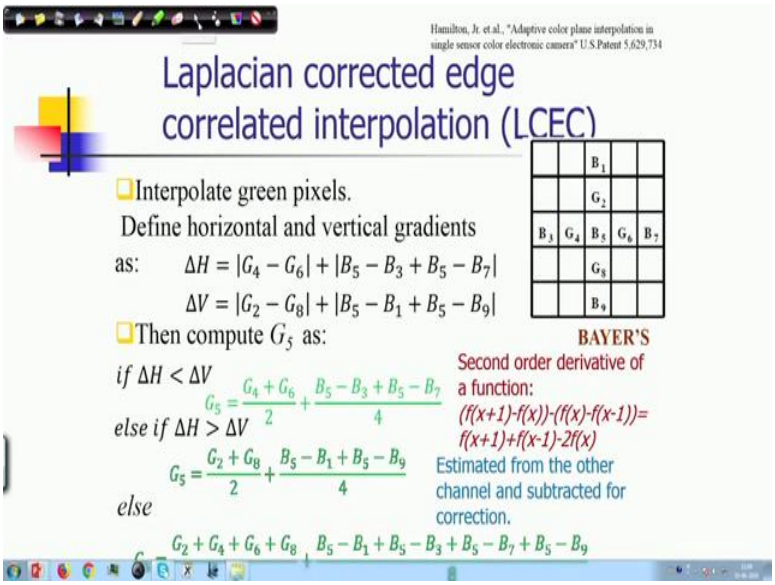
So, in its top and bottom neighbor two blue samples are available. So, you can estimate the blue hues take the average and multiply with the green value what is available at that location. So, this is the second case. And the third case is shown here where you have the blue values all in the diagonal locations. So, you can consider this particular guess say let me use a different colour just to say this is a location where blue sample is missing and you are going to estimate. So, once again the diagonal neighbors contains two blue pixels blue values and their greens are already estimated.

$$B_{12} = \frac{G_{12}}{4}\left(\frac{B_6}{G_6} + \frac{B_8}{G_8} + \frac{B_{16}}{G_{16}} + \frac{B_{18}}{G_{18}}\right)$$

So, you take the average of hues of those locations; that means, sum those hues and then divide it by 4 because there are 4 instances of estimation of blue hues and then you

multiply with the green sample. So, here green is also estimated because it was a red. So, the green value what is estimated by the green interpolation stage you can use that value to get the missing blue sample. So, this is how you compute the blue components wherever it is not there in the color filter array. In the same way you can do it for the red components. So, this is what you can perform these operations for red pixels.

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So, now I will be discussing another very effective technique where it is exploiting the information of gradient. So, let me discuss this particular property of interpolation this is very commonly used that when you interpolate the pixels you should interpolate along the directions of the least gradient. Because you can see particularly when you have edges let me give you an example through the sketches. Suppose you have a region where this side you have a brightness value say I_0 and this side you have a brightness value say I_1.

So, suppose this pixel value is not available and this pixel values are available. Now, if I am going to interpolate along these directions there is a direction of maximum gradient change. So, as you can see that these interpolated value will be in between I_0 and I_1 and there will be blurriness in the corresponding edge values edges. Where as if I interpolate along this direction we will be preserving this edges because those values they are similar and they are the least gradient changes there is the direction of least gradient change in this case.

So, we can consider that value should give you better interpolation. So, there is a principle by which you can perform this interpolation along the directions of low gradients. Since we have two directions natural directions in this pattern and by the definition of neighborhood there are horizontal and vertical gradient directions of course, you have also diagonal directions. So, other directions are also available, but depending upon the availability of your samples either they are horizontal neighbors or vertical neighbors or diagonal neighbors you can compute the corresponding gradients along those directions.

So, whenever there is a possibility of choosing samples among those directions. For example, in this particular case see we would like to compute green. Now, you can take the average of these two values that is in the horizontal directions or you can take the average of these two values that is in the vertical directions. So, which one you should choose? Now, if you find that vertical gradient is significantly smaller than the horizontal gradient which means there exists an edge then you should prefer the vertical direction.

So, that this is what and if you find then it is not much of difference then you can use all the neighbors. And if it is a diagonal directions where we coming to those situation there we have to consider again two directions diagonal directions which are also perpendicular and follow the same principle. So, estimation of gradient is important here, the other case what this algorithm does that estimation is further defined by considering the second order derivative of the distribution. So, I will be now discussing this algorithm in more details.

So, as I mentioned that first you have to compute say green interpolations. So, we are considering for example, this pixel location. So, the gradient value in the horizontal direction, this is an horizontal gradient direction is the difference between these two and you take the absolute amount because that is of your interest.

$$\Delta H = | G_4 - G_6 | + | B_5 - B_3 + B_5 - B_7 |$$

And then you can find that actually this is also added with it because what we are doing we are computing the difference of the difference which means the second order derivative; that means, if I take the difference of these two. Say $B_7 - B_5$ and then again

take the difference of these two and you subtract it say $B_7 - B_5$ and then again take it and subtract it $B_5 - B_3$, then you will get the, if you take the magnitude you will get the same value here. So, this is how you are estimating, but you are estimating the second order derivative from the blue channels as we mentioned in the very beginning of this today's lecture that there is a high correlation in the higher order derivatives across the channels.

So, we are estimating that same derivative should be also observed in the green channel when you can estimate it form the blue channel. So, this is how the gradient it estimate along the horizontal direction, similarly you can perform the estimation. So, this is just explaining that second order derivative of a function what I just have mentioned this is how you can compute.

$$\Delta V = | G_2 - G_8 | + | B_5 - B_1 + B_5 - B_9 |$$

And the vertical gradient can be estimated in this form you can see that the top neighbors of B_5 are considered for estimating the vertical gradient first order derivative and the second order derivatives are also computed from the vertical columns of blue components.

So, now the algorithm is that you have to find out in which direction the gradient is minimum. So, you have to compute G_5 like if ΔH is less than ΔV then if this is a scenario then you should consider the interpolation using this pixels. So, but what it does I could have taken $\frac{G_4 + G_6}{2}$ that is know 1 value, but it is further refined by a correction by a cross channel estimation of Laplacian derivative.

$$if \Delta H < \Delta V$$
$$G_5 = \frac{G_4 + G_6}{2} + \frac{B_5 - B_3 + B_5 - B_7}{4}$$

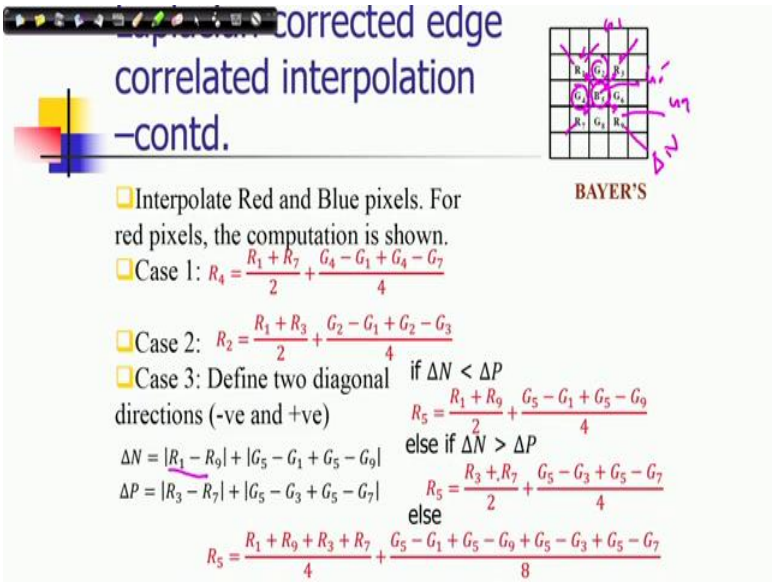
So, I will show you the final expression. So, this is the expression you can see that this estimate is refined by this second order derivative or you call it Laplacian value which is estimated from the blue channel because from the other channel which is available and it is subtracted from the value of $\frac{G_4 + G_6}{2}$.

So, this is one thing interesting because if you use the Taylor series expansion, then you will see that actually this value should be added during interpolation, but that performs poorly than this particular correction. And this is a technique which has been you proposed in this particular work and it is a very effective technique that what we have found out.

So, this is the work what is referred here and it has been observed that it gives very good interpolation quality of interpolation is very high good. So, this is further elaborated I mean used to repeated in other locations also same thing like if now the horizontal gradient is you know greater than vertical gradient, then you should use a vertical directions for interpolating green.

And otherwise if they are equal use all possible green values; that means, all the diagonal neighbors sorry not diagonal in this case they are all four neighbors in the non green locations. And then also use the corresponding estimates of derivatives using the both the directions vertical and horizontal directions and refine it.

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So, this is how the green pixel is interpolated and the blue and red pixel is also interpolated it in the same fashion on already you have green pixel. So, now, what you can do you since you have green values at every locations your cross channel estimation of Laplacian should be through green only. So, now, you have once again the condition of interpolation of red pixel. So, if it is at the location here you can see this is the location

where the missing red is computed and since there is there are no vertical this only the vertical neighbors are available for red true sample.

So, there is no question of comparing horizontal and vertical gradient in this case because only in one directions information is available and that is more reliable than the other values. So, we will be using them while doing interpolations and also we will be refining this estimate using their cross channel Laplacian estimate as we mentioned. And, but in other case also when it is the horizontal neighbors are available in the missing red location.

So, this is this particular location you see the only horizontal neighbors are available for red vertical there are blues. So, again you do not do any comparisons of horizontal and vertical gradient use similar kind of estimation process.

And the third case would be where actually the red neighbors, the neighboring red samples are in the diagonal directions. For example, this is the case where you have neighboring red samples in the CFA. So, now here you would like to apply the same technique of finding out the doing interpolations in the least gradient directions.

So, only thing is that your estimation of gradient should be along this direction, along these two perpendicular directions and that estimates once again can be refined using the Laplacian correction. So, what we are defining here as you can see that two directions; one is called ΔN. So, this direction ΔN you can see it is $R_1 - R_9$ and then you are also considering the Laplacian estimate along this. So, you are using G_5, so this is G_5, this is G_1 and this is G_9.

$$\Delta N = | R_1 - R_9 | + | G_5 - G_1 + G_5 - G_9 | \qquad \Delta P = | R_3 - R_7 | + | G_5 - G_3 + G_5 - G_7 |$$

So, using this value itself you can perform this Laplacian estimate here and in the other direction also you are doing this and then which one is smaller you are choosing that value. So, if ΔN is less than ΔP, then you are performing your interpolations using R_1, R_9 and those Laplacian corrections. And if ΔN is greater than ΔP, then you are doing

this and if they are equal, then in the same way you are using all the diagonal neighbors take the average and also perform the Laplacian corrections.

If $\Delta N < \Delta P$

$$R_5 = \frac{R_1 + R_9}{2} + \frac{G_5 - G_1 + G_5 - G_9}{4}$$

Else if $\Delta N > \Delta P$

$$R_5 = \frac{R_3 + R_7}{2} + \frac{G_5 - G_3 + G_5 - G_7}{4}$$

Else

$$R_5 = \frac{R_1 + R_9 + R_3 + R_7}{2} + \frac{G_5 - G_1 + G_5 - G_9 + G_5 - G_3 + G_5 - G_7}{8}$$

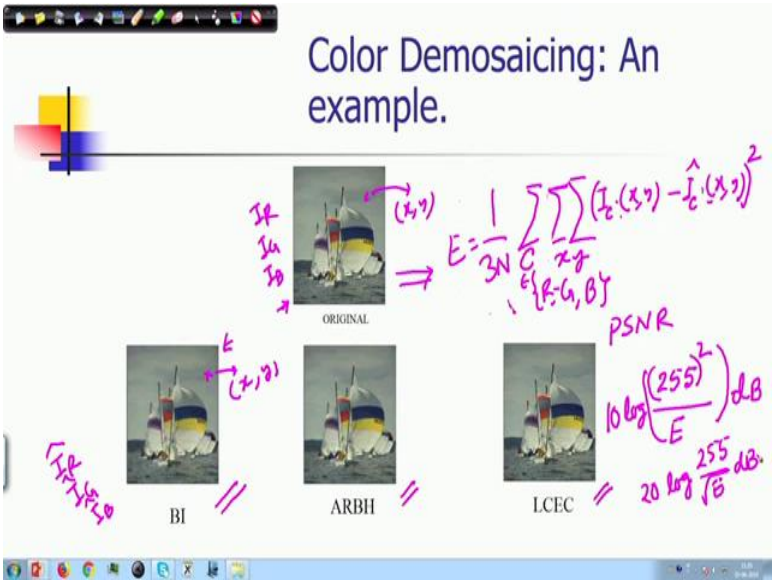
So, this is how this particular interpolation technique is carried out. So, let me give a break at this point, we will continue this discussion in the next lecture.

Thank you very much for your attention.

Computer Vision
Prof. Jayanta Mukhopadhyay
Department of Computer Science and Engineering
Indian Institute of Technology, Kharagpur

Lecture - 40
Color Fundamentals and Processing (Part VII)

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So, we continue our discussion on color interpolation. In the last lecture, I have discussed three particular techniques of interpolation; one is bilinear technique, then averaging of red and blue hue and then Laplacian correction of edge co related technique, which in acronyms are shown here ARBH, LCEC, for the this two and the bilinear is BI.

Now, here I have shown a few examples of demosaic patterns, just to explain how this algorithm has been applied. What we did from the original color image, we have extracted the CFA image using the Bayer's pattern as I have described. So, it is a simulated here color filter array from the original image. So, you we can compare and then you carried out interpolation say you apply bilinear interpolations, then you observe this particular result and this is average of red, blue and hue and this is Laplacian corrected edge correlated techniques LCEC.

So, you can see that visually, they look almost similar though, in the close examination you may find there are some color artifacts and blurriness in some cases and there is a measure by which this quality of this techniques could be evaluated. Suppose, you

consider this image that is the reconstructed image and this is your original image. So, you have a pixel value, consider at any location, the correspondingly same location of the original. Say, this is a location say let me write this location as xy and same location xy, this image let me consider this image has three component. So, this is reconstructed image so, we have $\hat{I}_R$, $\hat{I}_G$, $\hat{I}_B$

So, these are the three color components and this is the original color image of this component. So, what we can consider; we find out the corresponding error between this color channel. So, for the red we will be doing say $(I_R(x,y) - \hat{I}_R(x,y))$. So, if I perform the square of that error that would give me the red components error and overall red components error would be, if I sum over all the x and y values, that is for red.

That is a error of the red and you compute this error for all the channels for even blue and for green and the all this channel, if I take the average which means that if there are N samples I should compute it, so, if there are three channels. So, I will be writing this expression mathematically in this way that there are three channels. So, let me put those channel C and C could be either red or green or blue and if there are N pixels.

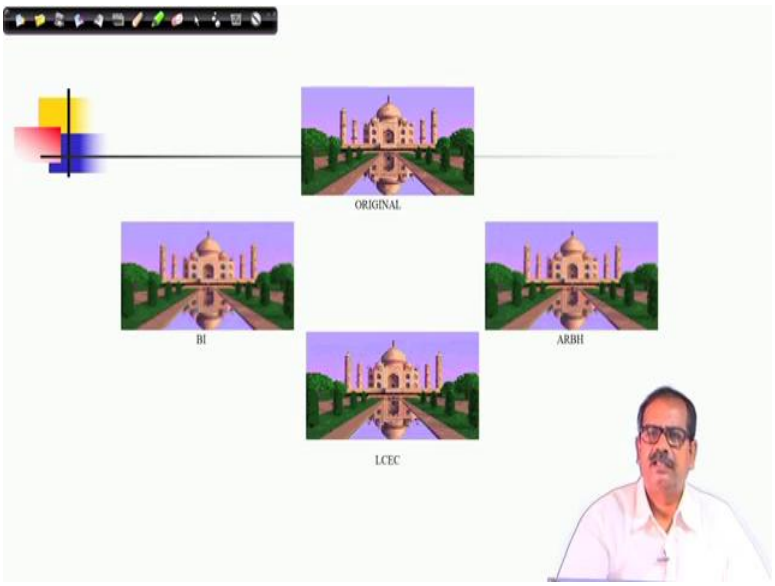
So, what I will do I will take the average of this error; that means, every sample, every instance is considered for squared deviations of the true values with the predicted value and count that many number of instances which is 3N in this case. N is the size of the image for every channel.

So, this is the error and when we define a peak signal to noise ratio, what we do? We perform the signal to noise ratio as $\frac{(255)^2}{E}$. So, this is this peak signal to noise ratio, in terms of ratio now, this value that you can see that signal strength has been taken as the constant that is the very convenient way of expressing.

It gives little, it quite inflated measure but it is very popular in image processing and then you express it in dB which is $10\log\left(\frac{255^2}{E}\right)$. When you express in this form, then in it becomes so you can see that, which means that I have to take root mean square sometimes it is written as $20\log\frac{255}{\sqrt{E}}$ dB that is ok.

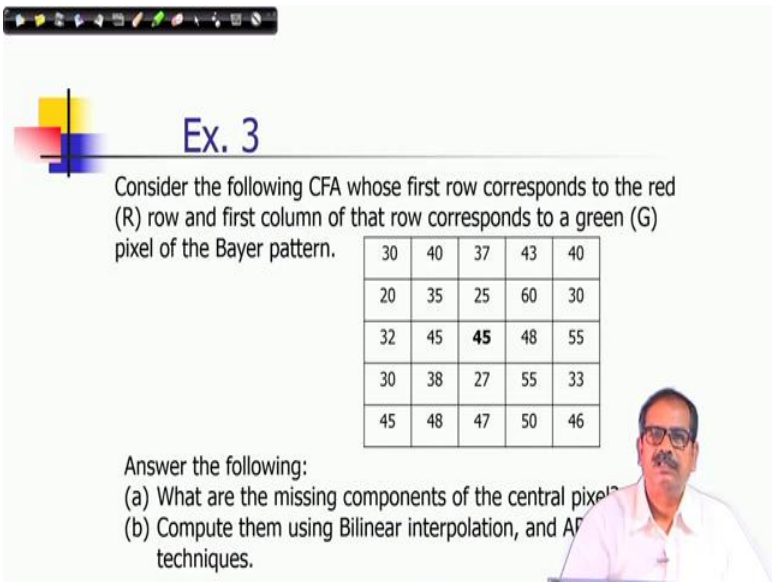
So, this is the PSNR value and you can compute this PSNR value it is not shown here in my slides and we can find that actually, LCEC will give you the highest PSNR and which shows that this LCEC technique is better than the other techniques in terms of PSNR .

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This is another example of reconstruction. Once again, the same experiments have been carried out only, we have shown another example.

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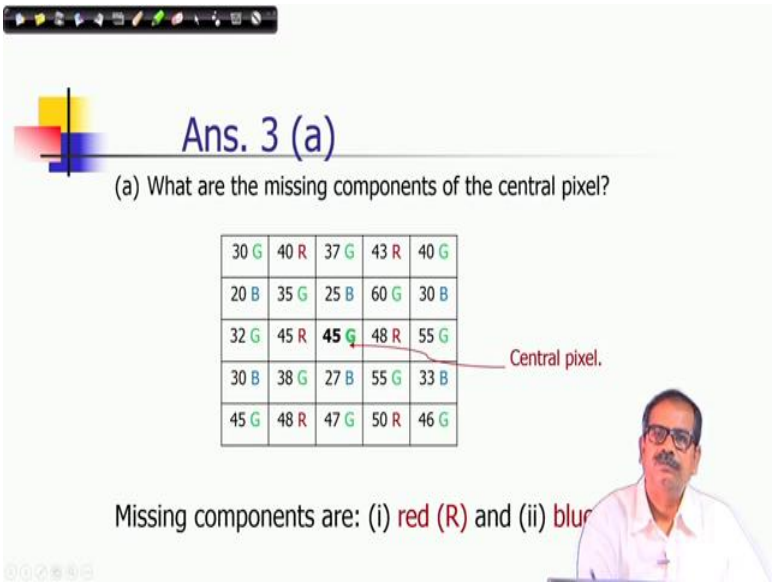


So, let me discuss one particular exercise to once again to make your understanding better for estimation of the missing spectral samples. So, let us do this exercise. So, it says that you consider the following color filter array, whose first row corresponds to the red row and first column of that row corresponds to a green pixel of the Bayer pattern, which means the leftmost and top most position of the array location, in this particular display that element corresponds to red row and green column in our convention of denoting any location.

And then what you are asked to answer the following that what are the missing components of the central pixel, which has been shown by the bold font and whose value is 45 here and then you should compute those missing components using bilinear interpolation and average of red and blue hue.

So, these are the two parts of this exercise. So, this is a central pixel as I mentioned.

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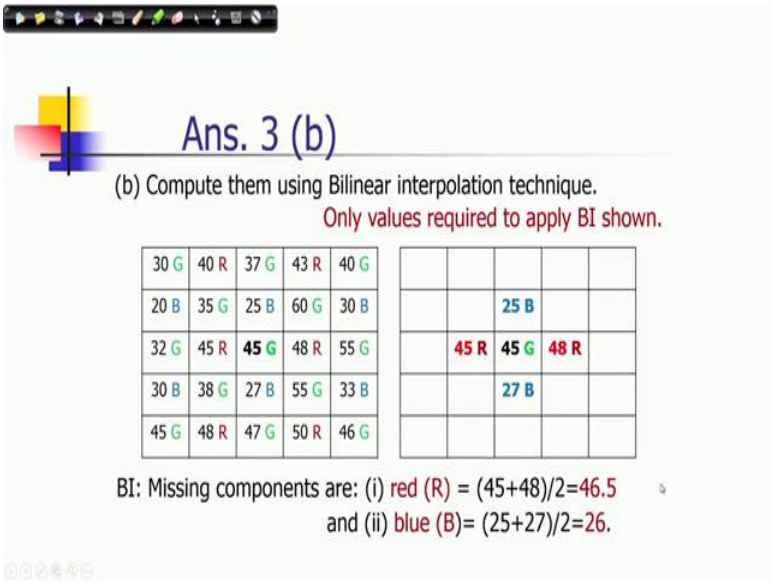
So, first let us consider what are the missing components of the central pixel . Now, in this particular diagram, I have just displayed the corresponding pixel types or which spectral samples are available in the locations corresponding locations to the pixel. As we have seen the first row we have the red row and green column.

So, which means the pixel value available at that location is green and then alternatively, or periodically, in that row the in an interleaved fashion green red green red in that way

the pixel samples are available. So, if I follow those Bayer pattern then the central pixel, we can say which we have highlighted that is also a green column pixel and it is also in the red row.

So, anyway since if the column if the type of column there it is green. So, we consider that is a only the green value is available. So, that missing components in those cases would be red and blue. So, this is a central pixel and missing components are red and blue.

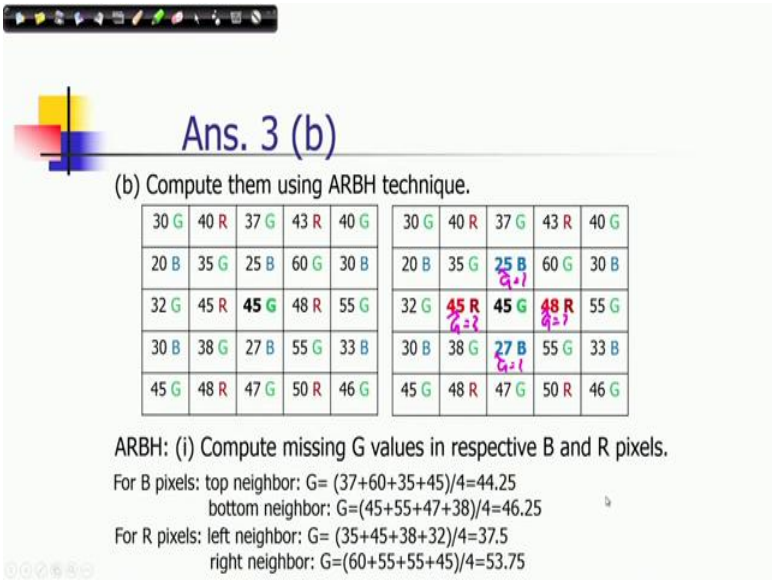
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Now, we have to compute the missing components red and blue. So, you can see; we have to apply the bilinear interpolation technique. So, we will be using bilinear interpolation. So, here I am showing part of these array the pixels samples which are used for our computation and by the color, I am showing those samples also the red is shown in red color and blue is shown in blue color.

So, for missing samples of red you simply you have to take average of 45 and 48 and for blue, you have to take average of 25 and 27. So, that is what you need to do to get red and blue for the bilinear interpolation technique.

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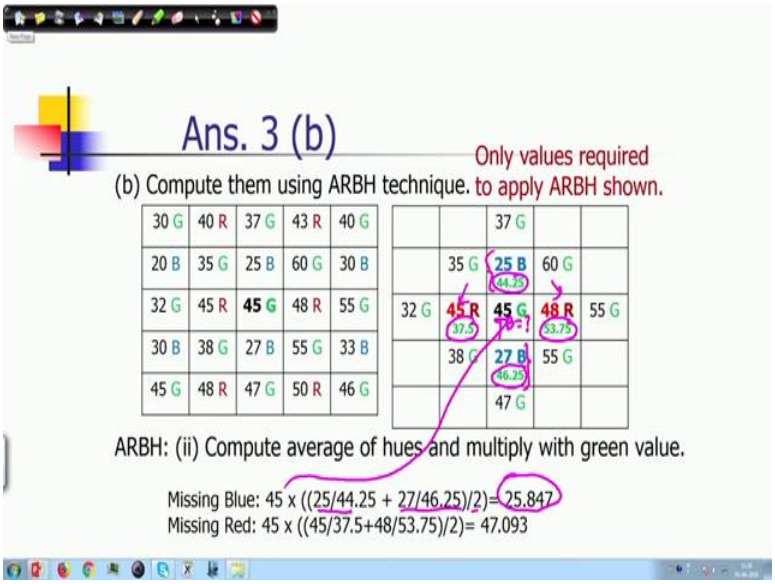


Now, let us consider the other technique that is averaging of red and blue hue. Here, once again I am showing those patterns, showing the highlighting those pixel values which are required for computing the corresponding know red and blue interpolations, but we need to compute the hue in those locations also; that means, in this location you need to compute the red hue, which means I should know I should also try to know what is the value of estimated green here.

Similarly, what is the value of estimated green here. So, I should know green value before computing hues that is what we require and for estimated green once again, you apply the same bilinear interpolation technique. So, the step would be first I will estimate this green values and then I will be applying the averaging of red and blue hues steps. So, computing missing G values; in respective B and R pixels.

So, for B pixels these values are computed and for bottom neighbor also these values are, I mean B pixels this is computed, the top neighbor it is 44.25 and bottom is 46.25 and red also is computed using left neighbor and right neighbor out here, it is corresponding left and right neighbors.

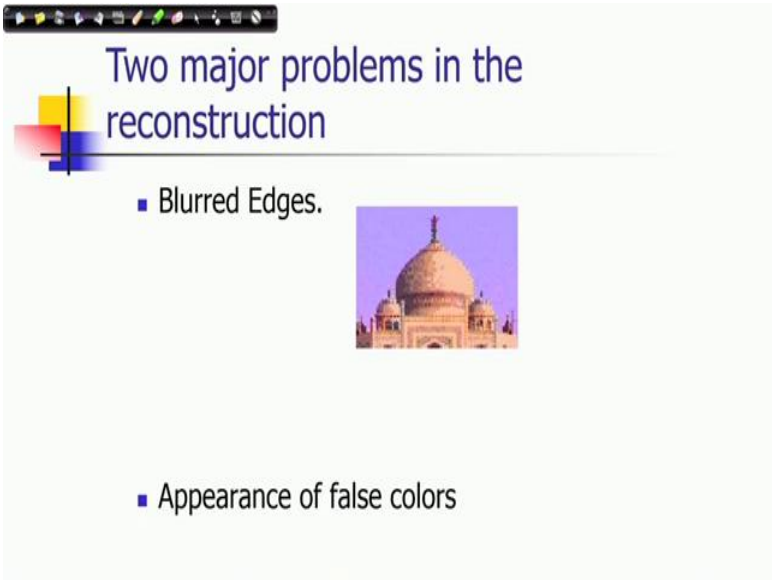
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So, (Refer Time: 11:03) just to show you what are those values. So, you can find out here, these values are shown here. So, this is the estimated green in those locations, following those rules. So, now, you can compute the corresponding hues. So, you can compute average of hues and multiply with green value. So, this is the missing blue, you can see that these are the hues we are computing. So, this is 25 by 44.25. So, corresponding to this location and 27 by 46.25 and then you have taken the average by dividing it by 2 and then you are multiplying the corresponding green pixel value and finally, you obtain the missing blue component at this position. So, this value is 25.847. In the same way you perform red by considering these two neighbors that value is coming as 47.

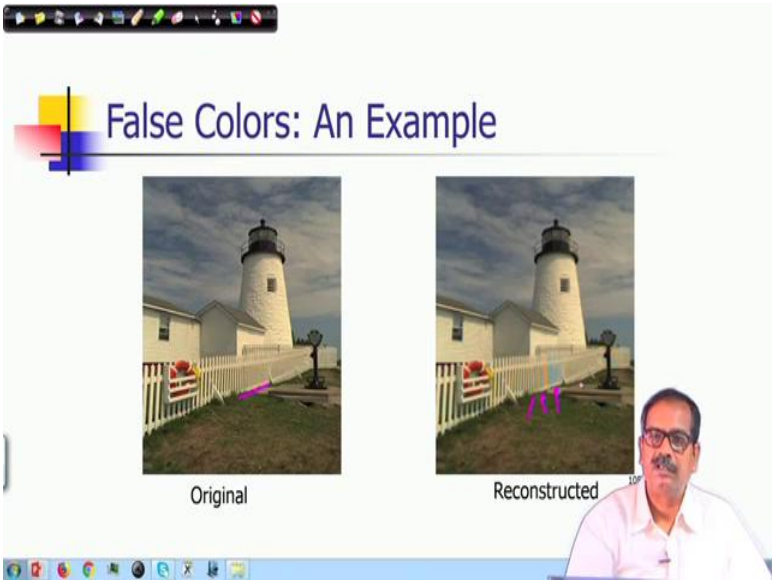
So, in this way you can get this result.

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So, there are two major problems of the reconstruction. I talked about blurriness of edges even though, we can use the less gradiant directions still, you can have the edges blurred. So, I have shown by know zooming the interpolated image what I have displayed earlier, the dome of the Taj Mahal in that picture and you can see that there the edges are a quite blurred in the zoomed portion, you can see and also there are appearance of false colors.

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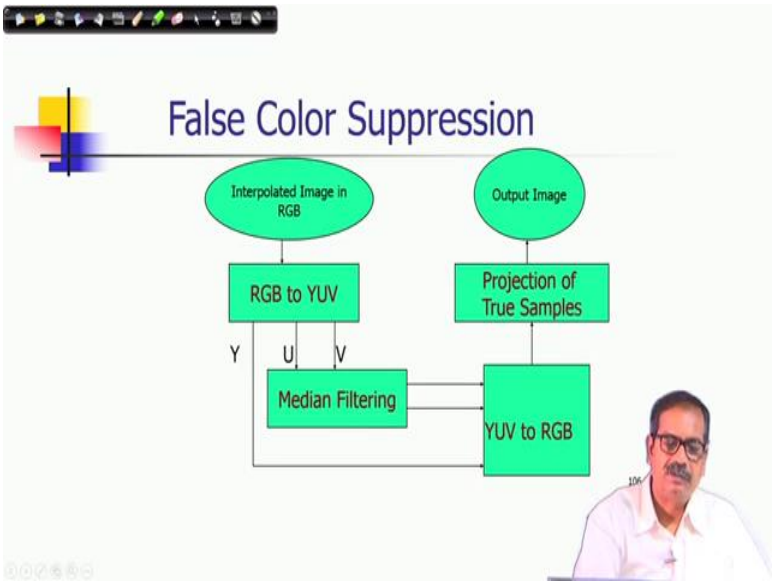


So, particularly it is severe, when you have a very high transitions of the paints here, very high frequency transitions and the color is white.

Since, the color is white, no all red and red, green, blue, they are to be estimated very accurately representing white. Even when a slight change in those estimations, slight error will cause appearance of know different kinds of colors. So, this is called the false color artifacts and those artifacts are quite visible, when you have this kind of images and this particular image is known for showing these artifacts and this image is also heavily used for testing algorithms for evaluating the quality of interpolation, quality of results.

So, these are the two particular concerns, but the thing is that for removing blurred image there a different filtering techniques and there are different other post processing techniques, which I am not going to discuss which is quite, which requires a bit more complex. You know concepts to be discussed, but whereas, for removing false color, I can describe a very simple technique and effective technique that is by using the median filtering.

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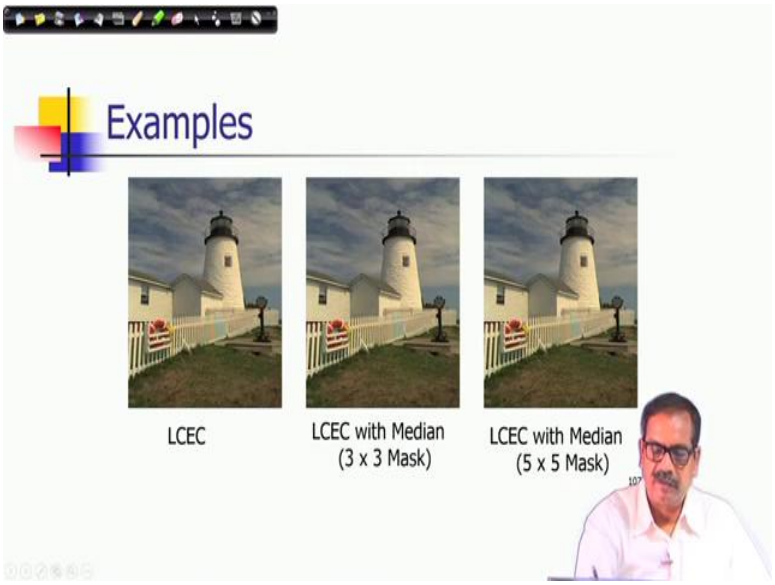


But what you are doing you know, in this case? you are performing median filtering in a different color space. You are separating out, the luminance and chroma components. You could have done in other color spaces also, where you can, you could have separated out, but I have used the U and V it is U and V is nothing but modified complementary blue and complementary of red of *YCbCr* transformation. So, which is linear transformation and there you perform median filter in each component and after

that you again, you project back to then you again transform it back to RGB and interesting is that while since, you are doing filtering some of the true pixel values are also getting modified.

So, you should project those true samples in this locations. that is another thing, you can you should do and that that is how you get the output of the image.

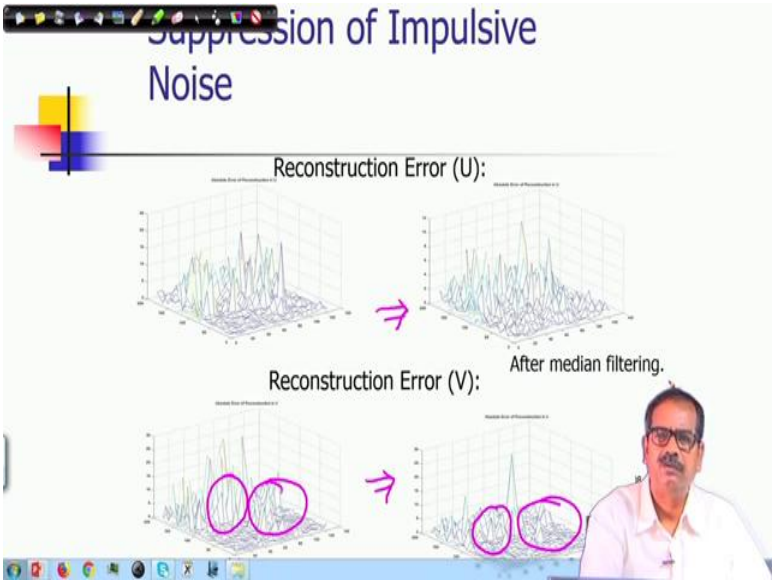
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So, if you perform these operation, I can show you the effect here. So, you can see that this was the original reconstruction image, reconstructed image using this technique LCEC which we have discussed. If I apply a 3×3 three cross three mask of median then you can find out this artifacts are getting reduced and 5×5 is reduced but the there is you know some disadvantage of having larger mask, it will blur the edge also. So, there is a trade-off between these.

So, this is one example. In fact, we can show also the, how the noise gets reduced in the corresponding space of U and V.

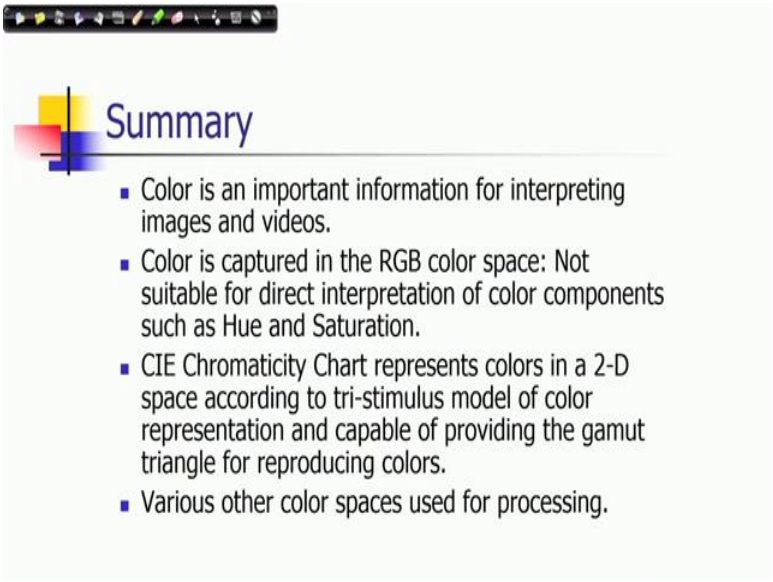
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So, this is typical plot, this plot corresponds to the reconstruction of the lighthouse image, what I have shown you before and in the error of reconstruction of U and V samples using the original interpolation technique LCEC, those errors are plotted against the locations, against the pixel locations. So, I have given you the surface plot. The top, it is showing the surface plot of error for the U component and the bottom, it is showing the surface plot of error of the V component. So, when we apply the median filter, then we can see the magnitude of this error is getting reduced.

So, this is the plot of U. So, this corresponds to U component and this is for V, you can see that it is significantly reduced. This part, if you are observant this part it is significantly reduced even this part it is significantly reduced, for U component it is not so prominent still it is reduced and there is a remarkable improvement of PSNR value which I am not quoting here. So, around 3 to 4 UV improvement those are reported.

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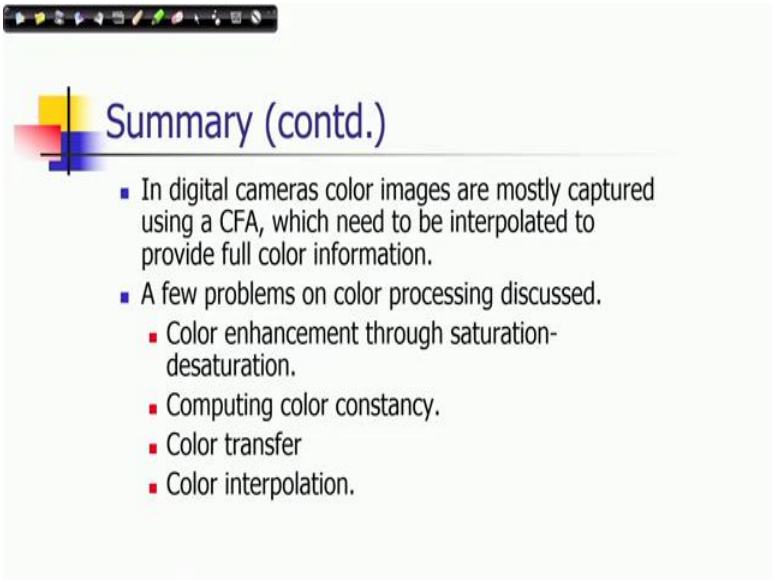


So, with this I will be concluding our topics in color fundamentals and color processing. So, just to summarize that what are the issues we have covered in this topic and what are the what is the take home, information regarding this particular topic that color is an important information for interpreting images and videos. you know that and it is understanding of it is representation is very important.

And we have seen how color could be represented and so, it is captured in the RGB color space, but it is not suitable for direct interpretation of color components such as hue and saturation and that is why there are spaces where you can separate them out. In particularly CIE, which is the international body of colors and they have recommended a chart which is called chromaticity chart and it represents colors in a 2-D space according to tri stimulus model of color representation. So, they have standardized the color representation and it is capable of providing the gamut triangle for reproducing colors.

There are various other colors spaces that we discussed that is used for processing in the just in the previous example; I have shown you processing in YUV color space, but there are others spaces also; we have mentioned using chromaticity chart itself, we have given some examples of processing; that means, you can convert from RGB to xyz, from xyz to normalized xy chromaticity space and there you can process in the hues and saturation to get different kind of information.

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So, the other topic, the last topic, what we discussed that is on color demosaicing. So, this is required when color images are captured using a color filtered array and mostly in present day widely used all the digital cameras, which are not expensive to that extent there it follows this principle of imaging and there is an interpolation process in built with that system itself.

So, it is needed to be interpolated to provide full color information, if you take get the image in the form of color filter array and we have discussed also a few problems on color processing. So, those are the some of this problem that is we discussed in color enhancement through saturation desaturation operation and we have seen how it could be done with the help of CIE chromaticity chart, that is one operation. Then we discussed also about computing color constancy there are two particular steps estimation of color of the illuminants and also correcting colors using the estimated color.

Then color transfer is more general competitions of transferring colors, which is the what the illumination is seemingly coming from a different target image from a target image or from a source image and that is the target illumination, which is described by target imageand the color of the source image to be transferred as if it has been illuminated by the illuminated of the target image.

So, there is a problem on color transfer then in the last examples of processing, we discussed about color interpolation. So, these are the four typical color processing of

color images that we discussed in this topic. With this let me conclude my lecture on this color processing and color fundamentals; we will go to the next topic of range image analysis, in my next lecture.

Thank you very much for your attention.